Notes on Functional Analysis*

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Contents

Metric Spaces and Normed Vector Spaces 2
Main Theorems of Normed Vector Spaces 7
Hilbert Spaces 14
L^p Spaces 22
Convolution and Smoothing 28
Duals of Continuous Function Spaces 35
Representations of Positive Linear Functionals 38
Schwartz Space of Functions with Rapid Decrease 48
Simple Properties of Fourier Transform 54
Inverse of $\mathcal{F}$ 57
Plancherel's Equality 62
Some Applications of the Fourier Transform 64
Tempered Distributions 66
Main Operations on $S' = S'(\mathbb{R}^d)$ 70
Some Applications of Tempered Distributions 74
Weak Convergence 76

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Metric Spaces and Normed Vector Spaces

Metric Space

A metric on a set M is a function $\rho : M \times M \to [0, \infty)$ such that

1. $\rho(x, y) = \rho(y, x)$, $\forall x, y \in M$.
2. $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$, $\forall x, y, z \in M$.
3. $\rho(x, y) = 0$ if and only if $x = y$.

Example

Let $M = \mathbb{R}^p$. Then, for $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and $y = (y_1, \dots, y_p) \in \mathbb{R}^p$,

$$\rho_2(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_p - y_p)^2},$$

$$\rho_1(x, y) = |x_1 - y_1| + \cdots + |x_p - y_p|, \quad \rho_\infty(x, y) = \max_{1 \leq i \leq p} |x_i - y_i|.$$

Convergence

$x_n \in M$ converges to $x \in M$ if $\rho(x_n, x) \xrightarrow{n \to \infty} 0$. We write $\lim_n x_n = x$ or $x_n \to x$ as $n \to \infty$.

Note. For the standard metrics on $\mathbb{R}^p$ $(\rho_1, \rho_2, \rho_\infty)$, $x^n = (x_1^n, \dots, x_p^n) \to x = (x_1, \dots, x_p)$ if and only if $x_i^n \to x_i$ for all $1 \leq i \leq p$.

Example

Let $M = C([0, 1])$, the space of continuous functions on $[0, 1]$. Then,

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| \, dt, \quad \rho_\infty(x, y) = \sup_{0 \leq t \leq 1} |x(t) - y(t)|.$$

Let $K = \rho_\infty(x, y)$.

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| \, dt \leq \int_0^1 K \, dt = K = \rho_\infty(x, y).$$

Note that $\rho_1(x, y) \leq \rho_\infty(x, y)$, so ρ_∞-convergence implies ρ_1-convergence, but not necessarily vice versa.

Cauchy Sequence

A sequence $x_n \in M$ is Cauchy if $\rho(x_n, x_m) \to 0$ as $n, m \to \infty$.

Proposition 1

If $x_n \to x$ in M, then (x_n) is Cauchy.

Proof. $\rho(x_n, x_m) \leq \rho(x_n, x) + \rho(x_m, x) \to 0$ as $n, m \to \infty$.

Completeness of Metric Space

(M, ρ) is complete if every Cauchy sequence has a limit. That is, if $\rho(x_n, x_m) \to 0$ as $n, m \to \infty$, then there exists $x \in M$ such that $x_n \to x$.

Example

a. $M = \mathbb{R}$ is complete, but $\mathbb{Q} \subset \mathbb{R}$ is not, with the subspace metric.

b. $M = L^1([0,1])$ with $\rho_1(x, y) = \int_0^1 |x(t) - y(t)|\, dt$ is complete.

c. $C([0,1]) \subset L^1([0,1])$ with $\rho_1(x, y)$ is not complete.

Remark. For example c, recall every $x \in L^1$ can be approximated by a sequence $x_n \in C([0,1])$. $\int_0^1 |x_n(t) - x(t)|\, dt \to 0$. Consider the specific sequence defined by the following step function:

$$x(t) = \begin{cases} 0 & \text{if } 0 \le t < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \le t \le 1 \end{cases} \in L^1([0,1]), \quad x_n(t) = \begin{cases} 0 & \text{if } 0 \le t < \frac{1}{2} \\ n(t - \frac{1}{2}) & \text{if } \frac{1}{2} \le t < \frac{1}{2} + \frac{1}{n} \\ 1 & \text{if } \frac{1}{2} + \frac{1}{n} \le t \le 1 \end{cases}.$$

Then,

$$\int_0^1 |x_n(t) - x(t)|\, dt = \frac{1}{2n} \to 0 \text{ as } n \to \infty.$$

x_n is Cauchy with respect to ρ_1 but $x \notin C([0,1])$.

Dense

$B \subseteq M$ is dense if every $x \in M$ can be approximated by $x_n \in B$: $x_n \in B \to x \in M$.

Theorem 1

For any (M, ρ) there exists a complete $(\tilde{M}, \tilde{\rho})$ such that M is isometric to a dense subset $B \subseteq \tilde{M}$. That is, there exists a bijection $h : M \to B$ such that

$$\rho(x, y) = \tilde{\rho}(h(x), h(y)), \quad x, y \in M.$$

$(\tilde{M}, \tilde{\rho})$ is called the completion of (M, ρ).

Remark. Isometric means distance-preserving. When we say that a map $h : M \to B$ is an isometry, it means that the distance between any two points in M is exactly equal to the distance between their corresponding images in B under the mapping h. In Theorem 1 above, h and its inverse $h^{-1} : B \to M$ are continuous.

Example

1. $\mathbb{R}$ is the completion of $\mathbb{Q}$.
2. $L^1([0,1])$ is the completion of $C([0,1])$ with respect to $\rho_1(x, y) = \int_0^1 |x(t) - y(t)|\, dt$.

Equivalence relation

If (x_n), (y_n) are Cauchy sequences, $(x_n) \sim (y_n)$ if $\rho(x_n, y_n) \xrightarrow{n\to\infty} 0$.

Continuous Functions

Let $f : (M, \rho) \to (Y, \nu)$ be a function.

1. f is continuous at x_0 if $x_n \to x_0$ implies $f(x_n) \to f(x_0)$. Equivalently, for any $\epsilon > 0$, there exists $\delta > 0$ such that

$$\nu(f(x), f(x_0)) \leq \epsilon \text{ if } \rho(x_0, x) \leq \delta.$$

   We write $\lim\limits_{x\to x_0} f(x) = f(x_0)$.

2. f is continuous if f is continuous at all $x \in M$.

Open Balls and Closed Ball

$$B_\delta(x) = \{y \in M \mid \rho(x, y) < \delta\}, \quad B_\delta[x] = \{y \in M \mid \rho(x, y) \leq \delta\}.$$

Open Set

$A \subseteq M$ is open if $\forall x \in A$, $\exists B_\delta(x) \subseteq A$ for some $\delta > 0$.

Closed Set

$F \subseteq M$ is closed if $F^c = M \setminus F$ is open.

Theorem 2

Let $\mathcal{T}$ be the collection of all open sets (all open subsets of M). Note that $M \in \mathcal{T}$, $\emptyset \in \mathcal{T}$. $\mathcal{T}$ is closed under arbitrary unions and finite intersections.

Remark. $\emptyset$ and M are open and closed. $\mathcal{T}$ is called the topology of M.

Corollary 1

Let $\mathcal{F}$ be the set of all closed sets. $\mathcal{F}$ is closed under arbitrary intersections and finite unions.

Neighborhood

$A \subseteq M$ is a neighborhood of x if $x \in A$ and there exists $B_\delta(x) \subseteq A$ with $\delta > 0$.

Inner Point

$x \in A$ is an inner point of A if there exists $B_\delta(x) \subseteq A$ with $\delta > 0$.

Fact. The set A° of all inner points of A is open.

Closure

Given $A \subseteq M$, its closure $\overline{A}$ is the smallest closed set that contains A.

Theorem 3

$\overline{A} = \{x \in M \mid \text{ there exists } \{x_n\} \subseteq A \text{ with } \lim_{n\to\infty} x_n = x\}$.

Corollary 2

$F \subseteq M$ is closed iff $\overline{F} = F$ iff $\{x_n\} \subseteq F$ with $\lim_{n\to\infty} x_n = x$ implies $x \in F$.

Normed Vector Spaces (nvs)

A normed vector space E is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) with a function, which we call a norm, $\|\cdot\|_E \colon E \to [0,\infty)$ such that

1. $\|\alpha x\|_E = |\alpha|\|x\|_E$.
2. $\|x+y\|_E \le \|x\|_E + \|y\|_E$.
3. $\|x\|_E = 0$ iff $x = 0$.

Note. $\|\cdot\|_E$ defines the distance $\rho(x,y) = \|x-y\|_E$, $x, y \in E$.

Triangle inequality

$$\|x-y\|_E = \|(x-z)+(z-y)\|_E \le \|x-z\|_E + \|z-y\|_E.$$

Example

1. Examples of norms on $\mathbb{R}^d$ from which the previous metrics arise: for $x = (x_1,\dots,x_d) \in \mathbb{R}^d$,
$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2},$$
$$\|x\|_1 = |x_1| + \cdots + |x_d|,$$
$$\|x\|_\infty = \max_i |x_i|.$$

2. $E = C([0,1])$ with ρ_1, ρ_∞:
$$\|x\|_1 = \int_0^1 |x(t)|\,dt, \quad \|x\|_\infty = \sup_{0\le t\le 1} |x(t)|.$$

Completeness of Normed Vector Space

If a normed vector space E is complete, it is called Banach.

Example

1. $\mathbb{R}^d$ with $\|x\|_1, \|x\|_2, \|x\|_\infty$ is Banach.
2. $C[0,1]$ is Banach with sup norm $\|x\|_\infty$ but not Banach with $\|x\|_1$-norm.
3. $L^1[0,1]$ is Banach $\to \|x\|_1 = \int_0^1 |x(t)|\, dt$. $C[0,1]$ is dense in $L^1[0,1]$.
4. $\mathbb{Q}^d$ is not Banach.

Linear Operators

Let E, F be normed vector spaces, $T : E \to F$ be linear. The following claims are equivalent:

(a) T is continuous at 0.

(b) T is continuous on E.

(c) T is bounded: There exists $C > 0$ so that $\|T(x)\|_F \leq C\|x\|_E$, $\forall x \in E$.

Notation. We denote $\mathcal{L}(E, F)$ the set of all bounded linear $T : E \to F$

Proof. (b) $\implies$ (a) is obvious. For (a) $\implies$ (b), let $x_n \to x$ in E. Then, by linearity, $x_n - x \to 0$ in E and $T(x_n - x) = T(x_n) - T(x) \to 0$ as $n \to \infty$. Thus, (a) $\implies$ (b).

For (a) $\implies$ (c), since T is continuous at 0, there exists $\delta > 0$ such that

$$\|T(x)\|_F = \|T(x) - T(0)\|_F \leq 1 \text{ if } \|x\|_E = \|x - 0\|_E \leq \delta.$$

For $y \in E$ and $y \neq 0$,

$$\|T(y)\|_F = \left\| T\left(\frac{y}{\|y\|_E} \cdot \delta\right) \right\|_F \cdot \frac{\|y\|_E}{\delta} \leq 1 \cdot \frac{\|y\|_E}{\delta} = \frac{1}{\delta} \cdot \|y\|_E = C\|y\|_E.$$

where $C = 1/\delta$. For (c) $\implies$ (a), if $x_n \to 0$ then,

$$\|T(x_n)\|_F \leq C\|x_n\|_E \to 0 \text{ as } n \to \infty.$$

Exercise. T is bounded iff $\sup \|T(x)\|_F < \infty$ for $\|x\|_E \leq R$, $\forall R > 0$.

Operator Norm

Given $T \in \mathcal{L}(E, F)$, its operator norm is the number:

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \sup_{\|x\|_E = 1} \|T(x)\|_F.$$

Note.

(i) $\|T\| \leq C$ in (c).

(ii) $\|T\|$ is the smallest $C > 0$ for which (c) holds.

(iii) **Important:** $\|\mathbf{T}(\mathbf{x})\|_{\mathbf{F}} \leq \|\mathbf{T}\|\|\mathbf{x}\|_{\mathbf{E}}$.

Exercise. $\mathcal{L}(E,F)$ with $\|\cdot\|$ is nvs.

Proof. We provide the proof for triangle inequality proof. Let $T, G \in \mathcal{L}(E,F)$, $\|x\|_E \le 1$.

$$\sup_{\|x\|_E \le 1} \|(T+G)(x)\|_F = \sup_{\|x\|_E \le 1} \|T(x) + G(x)\|_F \le \|T\| + \|G\|.$$

Dual Space

Given nvs E, its **dual space** is $E^* = \mathcal{L}(E,\mathbb{R})$ (or $E^* = \mathcal{L}(E,\mathbb{C})$). Note that E^* is Banach. For $\ell \in E^*, x \in E$, we denote

$$\ell(x) = \langle \ell, x\rangle_{E^*,E} = \langle \ell, x\rangle.$$

Example

1. $E = \mathbb{R}^d, \ell \in E^* = \mathcal{L}(\mathbb{R}^d,\mathbb{R})$ is $\ell(x) = a\cdot x = \langle a, x\rangle = \sum_{i=1}^d a_i x_i$ where $a = (a_1, \ldots, a_d) \in \mathbb{R}^d$, $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$.
2. Let E, F be nvs. Then $E \times F$ is nvs with $\|(x,y)\|_2 = \sqrt{\|x\|_E^2 + \|y\|_F^2}$. Also, $\|(x,y)\|_1 = \|x\|_E + \|y\|_F$, and $\|(x,y)\|_\infty = \max\{\|x\|_E, \|y\|_F\}$. Moreover, if E, F are Banach, then $E \times F$ is Banach.

Main Theorems of Normed Vector Spaces

Hahn-Banach Theorem

Let E be a normed vector space and $M \subseteq E$ be a vector subspace. Let $f \in M^* = \mathcal{L}(M,\mathbb{R})$ (the dual space of M). Then there exists $\ell \in E^* = \mathcal{L}(E,\mathbb{R})$ extending f such that $\|\ell\|_{E^*} = \|f\|_{M^*}$.

Remark. The proof uses Zorn's Lemma, but Zorn's Lemma is not needed if E is separable (i.e., if E has a countable dense subset).

Corollary 1

Let E be a normed vector space and $F \subset E$ be a closed vector subspace such that $F \neq E$. Then, there exists a nonzero $\ell \in E^*$ so that $\ell|_F = 0$ (i.e., $\ell(y) = 0$ for all $y \in F$). Moreover, if $x_0 \notin F$, then there exists $\ell \in E^*$ so that $\ell(x_0) = 1$ and $\ell(y) = 0$ for any $y \in F$.

Proof. Let $x_0 \notin F$. We begin by constructing the subspace M:

$$M = \mathbb{R}x_0 + F = \{tx_0 + y : t \in \mathbb{R}, y \in F\}.$$

Before proceeding, we establish a crucial property: if $tx_0 + y = 0$, then $t = 0$ and $y = 0$. To prove this, suppose $tx_0 + y = 0$ with $t \neq 0$. Then we would have $x_0 = -\frac{y}{t} \in F$ since F is a subspace, contradicting our assumption that $x_0 \notin F$. Therefore, t must be 0, and consequently $y = 0$.

We now define a linear functional $f : M \to \mathbb{R}$ by $f(tx_0 + y) = t$. Note that this definition implies $f(x_0) = 1$ and $f(y) = 0$ for all $y \in F$. To show f is well-defined, suppose $tx_0 + y = t'x_0 + y'$. Then $(t - t')x_0 + (y - y') = 0$. By our previous observation, this implies $t = t'$ and $y = y'$, so $f(tx_0 + y) = f(t'x_0 + y')$. The linearity of f follows naturally:

$$f(\alpha_1(t_1x_0 + y_1) + \alpha_2(t_2x_0 + y_2)) = \alpha_1 t_1 + \alpha_2 t_2 = \alpha_1 f(t_1x_0 + y_1) + \alpha_2 f(t_2x_0 + y_2).$$

To prove f is bounded, let $d = \inf_{z \in F} \|x_0 - z\|_E$. We claim that $d > 0$. Since F is closed and $x_0 \notin F$, if we had $d = 0$, there would exist a sequence $\{z_n\} \subseteq F$ with $z_n \to x_0$, contradicting $x_0 \notin F$.

For any $tx_0 + y \in M$, we have

$$\|tx_0 + y\|_E = |t| \left\| x_0 + \frac{y}{t} \right\|_E = |t| \left\| x_0 - \left(-\frac{y}{t}\right) \right\|_E \geq |t| d.$$

Therefore, if $\|tx_0 + y\|_E \leq 1$, then $|t| \leq \frac{1}{d}$. This shows:

$$\|f\|_{M^*} = \sup_{\|tx_0 + y\|_E \leq 1} |f(tx_0 + y)| = \sup_{\|tx_0 + y\|_E \leq 1} |t| \leq \frac{1}{d} < \infty.$$

By the Hahn-Banach Theorem, f extends to a functional $\ell \in E^*$. This extended functional ℓ satisfies $\ell(x_0) = f(x_0) = 1$ and $\ell(y) = f(y) = 0$ for all $y \in F$, completing our proof.

Comment. Corollary 1 is a standard tool to prove by contradiction that a vector subspace $D \subseteq E$ is dense ($\overline{D} = E$).

1. Assume the opposite, $\overline{D} \neq E$. Note that $\overline{D}$ is a closed vector subspace.
2. By Corollary 1, there exists a nonzero $\ell \in E^*$ such that $\ell|_{\overline{D}} = 0$.
3. But if other arguments show that $\ell|_{\overline{D}} = 0 \implies \ell = 0$ on E, we reach a contradiction.

Note that if $\ell|_D = 0$ and ℓ is continuous, then for any sequence $\{x_n\} \subseteq D$ with $x_n \to x_0 \in \overline{D}$, we have $\ell(x_n) \to \ell(x_0) \implies \ell(x_0) = 0$.

Corollary 2

(a) For each nonzero $x \in E$, there exists a nonzero $\ell \in E^*$ such that

$$\ell(x) = \langle \ell, x \rangle = \|\ell\|_{E^*} \|x\|_E.$$

We say ℓ and x are aligned.

(b) For any $x \in E$,

$$\|x\|_E = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x)|.$$

Recall that $\|\ell\|_{E^*} = \sup_{\|x\|_E \leq 1} |\ell(x)|$.

Remark. It is always the case that $|\ell(x)| \leq \|\ell\|_{E^*} \|x\|_E$. For (a):

1. If $E = \mathbb{R}^d$ with $\|\cdot\|_2$, then $E^* = \mathbb{R}^d$ with $\|\cdot\|_2$. Two vectors $\ell = (a_1, \dots, a_d)$ and $x = (x_1, \dots, x_d)$ are aligned if

$$\ell(x) = \langle a, x \rangle = a_1 x_1 + \cdots + a_d x_d = \|a\|_2 \|x\|_2.$$

2. If $E = \mathbb{R}^d$ with $\|\cdot\|_1$, then $E^* = \mathbb{R}^d$ with $\|\cdot\|_\infty$, and we have:

$$\langle a, x \rangle = \|a\|_\infty \|x\|_1.$$

Proof. Assume the base field is $\mathbb{R}$.

(a) Let $x_0 \in E$ and $x_0 \neq 0$. Define

$$M = \mathbb{R}x_0 = \{tx_0 : t \in \mathbb{R}\}.$$

Define a linear map $f : M \to \mathbb{R}$ by $f(tx_0) = t\|x_0\|_E$, and then:

$$\|f\|_{M^*} = \sup_{\|tx_0\|_E \leq 1} |f(tx_0)| = \sup_{|t|\|x_0\|_E \leq 1} |t|\|x_0\|_E = 1.$$

Note that $f(x_0) = \|x_0\|_E = \|f\|_{M^*}\|x_0\|_E$. Thus, f is linear and bounded, so $f \in \mathcal{L}(M, \mathbb{R})$. By the Hahn-Banach Theorem, there exists $\ell \in E^*$ extending f with

$$\|\ell\|_{E^*} = \|f\|_{M^*} = 1,$$

$$\implies \ell(x_0) = f(x_0) = \|x_0\|_E = \|\ell\|_{E^*}\|x_0\|_E.$$

(b) Let $x_0 \in E$ and $\|\ell\|_{E^*} \leq 1$. Then,

$$|\ell(x_0)| \leq \|\ell\|_{E^*}\|x_0\|_E \leq \|x_0\|_E \implies \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| \leq \|x_0\|_E.$$

By part (a), there exists $\ell \in E^*$ such that $\|\ell\|_{E^*} = 1$ and $\ell(x_0) = \|x_0\|_E$.

Norm of Adjoint Operator

Let E, F be normed vector spaces and $T \in \mathcal{L}(E, F)$. We define the adjoint $T' : F^* \to E^*$ to be the linear map that assigns to each $\ell \in F^*$ the functional $T'\ell \in E^*$ defined by

$$(T'\ell)(x) := \ell(T(x)), \quad x \in E.$$

Proposition 1

$T' \in \mathcal{L}(F^*, E^*)$ and $\|T'\| = \|T\|$.

Proof.

$$\|T'\| = \sup_{\|\ell\|_{F^*} \leq 1} \|T'\ell\|_{E^*} = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |(T'\ell)(x)| = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |\ell(T(x))|$$

$$\implies \|T'\| = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |\ell(T(x))| = \sup_{\|x\|_E \leq 1} \sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))|.$$

By Corollary 2 (b),

$$\|T'\| = \sup_{\|x\|_E \leq 1} \sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \|T\|.$$

Embedding

Let E be a normed vector space and $E^{**} = (E^*)^* = \mathcal{L}(E^*, \mathbb{R})$. Every $x_0 \in E$ defines naturally a functional J_{x_0} defined by

$$J_{x_0}(\ell) = \ell(x_0), \quad \ell \in E^*.$$

Note that $|J_{x_0}(\ell)| = |\ell(x_0)| \leq \|\ell\|_{E^*}\|x_0\|_E$.

Proposition 2

The map $J : E \to E^{**}$ that assigns to $x_0 \in E$ the linear functional $J_{x_0} \in E^{**}$ is an isometry:

$$\|x_0\|_E = \|J_{x_0}\|_{E^{**}}$$

i.e., the space E is isometrically embedded into E^{**} and we write $E \subseteq E^{**}$.

Proof. For $x_0 \in E$, by Corollary 2 (b),

$$\|x_0\|_E = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| = \sup_{\|\ell\|_{E^*} \leq 1} |J_{x_0}(\ell)| = \|J_{x_0}\|_{E^{**}}.$$

Baire's Lemma

Let (M, ρ) be a complete metric space and $M = \bigcup_n M_n$ with all M_n closed. Then, there exists $n_0 \geq 1$ so that $\mathring{M}_{n_0} \neq \emptyset$, i.e., there exists $B_{r_0}(x_0) \subseteq M_{n_0}$. Recall $\mathring{M}_{n_0} = \mathrm{Int}(M_{n_0}) = \{x \in M_{n_0} : B_r(x) \subseteq M_{n_0} \text{ for some } r > 0\}$.

Proof. We prove by contradiction. Assume $\mathring{M}_n = \emptyset$, $\forall n \geq 1$. Then for each n,

$$O_n = M_n^c = M \setminus M_n \text{ is dense and open in } M.$$

Every $B_r(x) \subseteq M$ contains $y \in O_n$. We will show that $G = \bigcap_n O_n$ is dense, but

$$G = \bigcap_n O_n = \bigcap_n M_n^c = \left(\bigcup_n M_n\right)^c = M^c = \emptyset$$

and this is a contradiction.

To see G is dense, note G is dense if every $B_{r_0}(x_0)$ contains $x \in G$. Fix $B_{r_0}(x_0)$. Since O_1 is dense,

$$\exists \overline{B}_{r_1}(x_1) \subseteq B_{r_0}(x_0) \cap O_1, \text{ with } r_1 \leq \frac{r_0}{2}.$$

To get $\overline{B}_{r_1}(x_1)$, we can always find an open ball in a nonempty open set. Thus, we can take a closed ball to be the closure of an open ball with half the radius (or the closure of any other smaller ball).

Since O_2 is dense, $B_{r_1}(x_1) \cap O_2$ is nonempty and open.

$$\exists \overline{B}_{r_2}(x_2) \subseteq B_{r_1}(x_1) \cap O_2 \text{ with } r_2 \leq \frac{r_1}{2} \leq \frac{r_0}{2^2}.$$

Since O_{n+1} is dense,

$$\exists \overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1} \text{ with } r_{n+1} \leq \frac{r_0}{2^{n+1}}.$$

We claim the sequence $\{x_n\}$ is Cauchy. If $j, j' \geq n$, then $x_j, x_{j'} \in B_{r_n}(x_n)$, and

$$\rho(x_j, x_{j'}) \leq \rho(x_j, x_n) + \rho(x_{j'}, x_n) \leq 2 \cdot \frac{r_0}{2^n} \to 0 \text{ as } j, j' \to \infty.$$

Since M is complete, $x = \lim_{n\to\infty} x_n$ exists. Note that

$$x \in \bigcap_{n\geq 1} \overline{B}_{r_n}(x_n) \subseteq \bigcap_{n\geq 0} B_{r_n}(x_n) \subseteq O_m, \quad \forall m \geq 1$$

because $\overline{B}_{r_n}(x_n)$ is closed so the limit belongs to it. This means $x \in \bigcap_{m\geq 1} O_m = G$.

Banach-Steinhaus Theorem

Theorem 2

Let E, F be normed vector spaces and E be Banach. Let $T_i \in \mathcal{L}(E, F)$, $i \in I$, and for each $x \in E$,

$$\sup_{i\in I} \|T_i(x)\|_F < \infty \text{ (pointwise uniform boundedness).}$$

Then $\sup_{i\in I} \|T_i\| < \infty$, or equivalently, for every $R > 0$,

$$\sup_{i\in I} \sup_{\|x\|_E \leq R} \|T_i(x)\|_F = \sup_{\|x\|_E\leq R} \sup_{i\in I} \|T_i(x)\|_F < \infty.$$

Proof. By assumption, for each $n \geq 1$, the set

$$M_n = \{x \in E : \sup_{i\in I} \|T_i(x)\|_F \leq n\} = \bigcap_{i\in I} \{x \in E : \|T_i(x)\|_F \leq n\}$$

is closed because each T_i is continuous. Since every $x \in E$ is pointwise bounded, we have $E = \bigcup_{n=1}^{\infty} M_n$. By Baire's Lemma, there exists $n_0 \geq 1$ such that $\mathring{M}_{n_0} \neq \emptyset$, meaning:

$$\exists \overline{B}_{r_0}(x_0) \subseteq \{x \in E : \sup_{i\in I} \|T_i(x)\|_F \leq n_0\}.$$

We will show that

$$\|T_i\| = \frac{1}{r_0} \sup_{\|y\|_E \leq r_0} \|T_i(y)\|_F \leq \frac{1}{r_0} C_0$$

where C_0 is independent of i, by showing:

$$\sup_{\|y\|_E\leq r_0} \|T_i(y)\|_F \leq C_0.$$

Let $y \in E$ with $\|y\|_E \leq r_0$. Then,

$$\|T_i(y)\|_F = \|T_i(y + x_0 - x_0)\|_F \leq \|T_i(x_0 + y)\|_F + \|T_i(x_0)\|_F \leq n_0 + \sup_{i\in I} \|T_i(x_0)\|_F = 2n_0$$

because $x_0 + y \in \overline{B}_{r_0}(x_0)$ and $\sup_{i\in I} \|T_i(x_0)\|_F \le n_0$. Thus, we can set $C_0 = 2n_0$.

Remark. Let $T_i \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^d)$ be given by matrix components:

$$T_i = (a^i_{k\ell}), \quad 1 \le k \le d,\, 1 \le \ell \le d.$$

Then $\sup_i \|T_i(x)\| < \infty$ for all $x \in \mathbb{R}^d$ implies $\sup_i |a^i_{k\ell}| < \infty$ for all k, ℓ. Indeed, if e_k $(k = 1, \ldots, d)$ is the standard basis of $\mathbb{R}^d$, then

$$\sup_i |a^i_{k\ell}| = \sup_i |e_k \cdot (T_i(e_\ell))| \le \sup_i \|T_i(e_\ell)\| < \infty, \quad \ell = 1, \ldots, d.$$

As $\|e_k\| = 1$ and $|\cos\alpha| \le 1$, we have:

$$e_k \cdot (T_i(e_\ell)) = \|e_k\| \|T_i(e_\ell)\| \cos\alpha.$$

Open Mapping Theorem

Let M, Y be metric spaces and $f : M \to Y$. The following are equivalent:

(a) f is continuous.

(b) $f^{-1}(U) \subseteq M$ is open for any open $U \subseteq Y$.

(c) $f^{-1}(F) \subseteq M$ is closed for any closed $F \subseteq Y$.

Let E, F be Banach spaces and $T \in \mathcal{L}(E, F)$ be surjective (onto). If $U \subseteq E$ is open, then $T(U) \subseteq F$ is open.

Theorem 3

Let E, F be Banach spaces and $T \in \mathcal{L}(E, F)$ be surjective. Then there exists $c > 0$ so that $B^F_c \subseteq T(B^E_1)$, where

$$B^F_c = \{y \in F : \|y\|_F < c\}, \quad B^E_1 = \{x \in E : \|x\|_E < 1\}.$$

Here, $B^H_r = \{x \in H : \|x\|_H < r\}$ denotes a general ball of radius r in space H.

Proof. Since T is onto,

$$E = \bigcup_{n=1}^{\infty} B^E_n, \quad F = \bigcup_{n=1}^{\infty} \overline{T(B^E_n)}.$$

We take the closure here to apply Baire's Lemma. By Baire's Lemma, there exists $n_0 \ge 1$ such that $B^F_{r_0}(x_0) \subseteq \overline{T(B^E_{n_0})}$. Since T is linear, we have $y = T(v) \implies -y = T(-v)$, meaning $y \in T(B^E_{n_0}) \implies -y \in T(B^E_{n_0})$. Thus, $B^F_{r_0}(-x_0) \subseteq \overline{T(B^E_{n_0})}$.

For $y \in B^F_{r_0}$,

$$y = \frac{1}{2}(x_0 + y) + \frac{1}{2}(-x_0 + y) \in \overline{T(B^E_{n_0})},$$

where $x_0 + y \in B^F_{r_0}(x_0)$ and $-x_0 + y \in B^F_{r_0}(-x_0)$. By convexity, $B^F_{r_0} \subseteq \overline{T(B^E_{n_0})}$. Now we claim:

$$B^F_{r_0/n_0} \subseteq \overline{T(B^E_1)} \subseteq T(B^E_2).$$

Let $\delta = r_0/n_0$. If this holds, then $B^F_\delta \subseteq T(B^E_2) \implies B^F_{\delta/2} \subseteq T(B^E_1)$.

We first show $\overline{T(B^E_1)} \subseteq T(B^E_2)$. Let $z \in \overline{T(B^E_1)}$. Then there exists $x_1 \in B^E_1$ such that

$$z - T(x_1) \in B^F_{\delta/2} \subseteq \overline{T(B^E_{1/2})}.$$

This is because z lies in the closure of the image of B^E_1, allowing us to approximate z closely by elements in the image. Similarly, there exists $x_2 \in B^E_{1/2}$ such that:

$$z - T(x_1) - T(x_2) \in B^F_{\delta/2^2} \subseteq \overline{T(B^E_{1/2^2})},$$

Inductively, there exists $x_n \in B^E_{1/2^{n-1}}$ such that:

$$z - T(x_1 + \cdots + x_n) \in B^F_{\delta/2^n} \subseteq \overline{T(B^E_{1/2^n})},$$

Note that:

$$\|z - T(x_1 + \cdots + x_n)\|_F \leq \frac{\delta}{2^n} \xrightarrow{n\to\infty} 0.$$

The series $\sum_{k=1}^\infty x_k$ converges absolutely since:

$$\sum_{k=1}^\infty \|x_k\|_E \leq 1 + \sum_{k=1}^\infty \frac{1}{2^k} = 2.$$

Since E is Banach, $x = \sum_{k=1}^\infty x_k = \lim_{n\to\infty} \sum_{k=1}^n x_k$ exists, and:

$$\|x\|_E \leq \sum_{k=1}^\infty \|x_k\|_E < 2.$$

Therefore,

$$z = \lim_{n\to\infty} T(x_1 + \cdots + x_n) = T(x) \quad \text{and} \quad \|x\|_E < 2.$$

Hence, $z \in T(B^E_2)$, which proves $\overline{T(B^E_1)} \subseteq T(B^E_2)$.

Corollary 3

Let E, F be Banach spaces and $T \in \mathcal{L}(E, F)$ be onto. Then, $T(U) \subseteq F$ is open for any open $U \subseteq E$.

Proof. Let $U \subseteq E$ be open and $y_0 \in T(U)$. Then, there exists $x_0 \in U$ and $\delta > 0$ so that $B^E_\delta(x_0) \subseteq U$ and $y_0 = T(x_0)$. By Theorem 3, there exists $c > 0$ such that $B^F_c \subseteq T(B^E_1)$, implying $B^F_{c\delta} \subseteq T(B^E_\delta)$. We have:

$$y_0 + B^F_{c\delta} \subseteq T(x_0 + B^E_\delta) = T(B^E_\delta(x_0)) \subseteq T(U).$$

Therefore, $T(U)$ is an open set.

Inverse Mapping Theorem

Let E, F be Banach spaces and $T \in \mathcal{L}(E, F)$ be bijective. Then, $T^{-1} \in \mathcal{L}(F, E)$.

Proof. Let $U \subseteq E$ be open. Then $(T^{-1})^{-1}(U) = T(U)$ is open by the Open Mapping Theorem. Thus, T^{-1} is continuous, so it is bounded.

Closed Graph Theorem

Let E, F be Banach spaces and $T : E \to F$ be linear. Assume its graph:

$$\Gamma = \{(x, T(x)) \in E \times F : x \in E\} \subseteq E \times F \text{ is closed.}$$

Then, $T \in \mathcal{L}(E, F)$.

Proof. Since Γ is a closed vector subspace of $E \times F$, Γ is a Banach space as well. Both projections of Γ are continuous:

$$(x, T(x)) \in \Gamma \xrightarrow{\pi_1} x \in E,$$

$$(x, T(x)) \in \Gamma \xrightarrow{\pi_2} T(x) \in F.$$

Since $\|\pi_1(x, T(x))\|_E = \|x\|_E \leq \sqrt{\|x\|_E^2 + \|T(x)\|_F^2} = \|(x, T(x))\|_{E \times F}$, π_1 is bounded and clearly bijective. By the Inverse Mapping Theorem, $\pi_1^{-1} : E \to \Gamma$ is continuous. Since $T = \pi_2 \circ \pi_1^{-1}$ is the composition of two continuous functions, T is continuous and thus bounded.

Comment. Proving continuity of $T : E \to F$:

1. In a standard way, given $x_n \to x$ in E:
   (i) Prove $\lim_n T(x_n) = y$ exists.
   (ii) Prove $y = T(x)$.
2. Using the Closed Graph Theorem: Given $x_n \to x$ in E and $T(x_n) \to y$ in F, prove $y = T(x)$.

Hilbert Spaces

Inner Product

Let H be a complex vector space. An inner product on H is a $\mathbb{C}$-valued function denoted by $\langle x, y\rangle$, for $x, y \in H$, such that:

i. $\langle x, x\rangle \geq 0\ \forall x \in H$; $\langle x, x\rangle = 0$ if and only if $x = 0$

ii. $\forall x, y, z \in H$, and $\alpha, \beta \in \mathbb{C}$:

$$\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle,$$

$$\langle z, \alpha x + \beta y\rangle = \overline{\alpha}\langle z, x\rangle + \overline{\beta}\langle z, y\rangle,$$

$$\langle x, y\rangle = \overline{\langle y, x\rangle}.$$

Examples

1. $H = \mathbb{C}^d$, $x = (x_1, \ldots, x_d)$, $y = (y_1, \ldots, y_d) \in \mathbb{C}^d$:

$$\langle x, y \rangle = \sum_{j=1}^{d} x_j \overline{y}_j.$$

   Then, $\langle x, x \rangle = \sum_{j=1}^{d} |x_j|^2$.

2. ℓ^2 is the space of all sequences $(x_1, x_2, \ldots)$ such that

$$\sum_{n=1}^{\infty} |x_n|^2 < \infty,$$

   with the inner product $\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \overline{y}_n$.

3. $\mathcal{C} = C([a, b])$ is the space of continuous complex-valued functions on $[a, b]$ with

$$\langle f, g \rangle = \int_a^b f(t)\overline{g(t)}\, dt.$$

   Then, $\langle f, f \rangle = \int_a^b |f(t)|^2 dt$.

4. $L^2(X, \mu)$ is the space of complex-valued square-integrable functions (equivalence classes) on a measure space $(X, \mathcal{X}, \mu)$ with

$$\langle f, g \rangle = \int f(x)\overline{g(x)}\, d\mu.$$

   In fact, Examples 1 and 2 are just specific cases of Example 4 using the counting measure.

Inner Product Space (Pre-Hilbert Space)

A complex vector space equipped with an inner product is called an inner product space or pre-Hilbert space.

Orthogonality

Let H be a pre-Hilbert space. We say $x \perp y$ in H if $\langle x, y \rangle = 0$. The induced norm is defined as:

$$\|x\|_H := \sqrt{\langle x, x \rangle}, \quad x \in H.$$

Pythagorean Theorem

If $x \perp y$, then

$$\|x + y\|_H^2 = \|x\|_H^2 + \|y\|_H^2.$$

Proof. Let $\langle x, y\rangle = \langle y, x\rangle = 0$. Then,

$$\begin{aligned}\|x+y\|_H^2 &= \langle x+y, x+y\rangle \\ &= \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle \\ &= \|x\|_H^2 + \|y\|_H^2\end{aligned}$$

Orthonormal System

A subset $\{e_j : j = 1, \ldots, N\}$ is an orthonormal system (ONS) if

$$\langle e_k, e_\ell\rangle = \begin{cases} 1 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases}.$$

Note that $\|e_k\|_H^2 = \langle e_k, e_k\rangle = 1$.

Theorem 2

Let $\{e_j : j = 1, \ldots, N\}$ be an ONS. Then for any $x \in H$:

i. $\sum_{j=1}^N \langle x, e_j\rangle e_j$ and $x - \sum_{j=1}^N \langle x, e_j\rangle e_j$ are orthogonal.

ii. $\|x\|_H^2 = \|x - \sum_{j=1}^N \langle x, e_j\rangle e_j\|_H^2 + \|\sum_{j=1}^N \langle x, e_j\rangle e_j\|_H^2$.

iii. **Bessel's Inequality:**

$$\sum_{j=1}^N |\langle x, e_j\rangle|^2 \leq \|x\|_H^2.$$

Proof. We first prove (i) by showing that $x - \sum_{j=1}^N \langle x, e_j\rangle e_j \perp e_k$ for all $k = 1, \ldots, N$:

$$\langle x - \sum_{j=1}^N \langle x, e_j\rangle e_j, e_k\rangle = \langle x, e_k\rangle - \langle x, e_k\rangle\langle e_k, e_k\rangle = \langle x, e_k\rangle - \langle x, e_k\rangle = 0.$$

Next, (ii) follows directly from (i) and the Pythagorean Theorem, and (iii) clearly follows by dropping the non-negative first term from the right-hand side of (ii).

Remark. If $\{e_j : j = 1, \ldots, N\}$ is an ONS, then $\{e_1, \ldots, e_N\}$ is linearly independent. Indeed,

$$0 = \sum_{j=1}^N \alpha_j e_j \implies \alpha_k = 0 \quad \forall k$$

which is shown by taking the inner product of both sides with each e_k.

Cauchy-Schwarz Inequality

For any $x, y \in H$,

$$|\langle x, y\rangle| \leq \|x\|_H \|y\|_H.$$

Proof. Let $y \neq 0$. Consider the single-element ONS $\{\frac{y}{\|y\|_H}\}$. By Bessel's inequality:

$$\sum_{j=1}^{1} |\langle x, e_j\rangle|^2 \leq \|x\|_H^2 \implies \left|\left\langle x, \frac{y}{\|y\|_H}\right\rangle\right|^2 \leq \|x\|_H^2 \implies |\langle x, y\rangle|^2 \leq \|x\|_H^2 \|y\|_H^2.$$

Corollary 1

$\|x\|_H = \sqrt{\langle x, x\rangle}$, $x \in H$, is a well-defined norm.

Proof. We prove the triangle inequality. For any $x, y \in H$:

$$\begin{aligned}
\|x+y\|_H^2 &= \langle x+y, x+y\rangle \\
&= \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle \\
&\leq \langle x, x\rangle + |\langle x, y\rangle| + |\langle y, x\rangle| + \langle y, y\rangle \\
&\leq \|x\|_H^2 + \|y\|_H^2 + 2\|x\|_H\|y\|_H \quad \text{(by Cauchy-Schwarz)} \\
&= (\|x\|_H + \|y\|_H)^2.
\end{aligned}$$

Hilbert Space

A pre-Hilbert space H is called a **Hilbert space** if it is complete with respect to the induced norm $\|\cdot\|_H$.

Projection Theorem

Let H be a pre-Hilbert space. Given a vector subspace $M \subseteq H$, its orthogonal complement is defined as:

$$M^\perp = \{x \in H : \langle x, m\rangle = 0 \quad \forall m \in M\}.$$

We write $x \perp M$ if $x \in M^\perp$. Note that $M^\perp$ is always a closed vector subspace of H due to the continuity of the inner product.

Theorem 3

Let H be a Hilbert space and $M \subseteq H$ be a closed vector subspace.

i. $\forall x \in H$, there exists a unique $m_0 \in M$ such that

$$d := \inf_{m \in M} \|x - m\|_H = \|x - m_0\|_H$$

Moreover, $(x - m_0) \perp M$. The vector m_0 is called the **orthogonal projection** of x onto M, denoted by $P_M x = m_0$.

ii. Let $m_1 \in M$ be such that $(x - m_1) \perp M$. Then $m_1 = m_0$.

Remark. Theorem 3 implies that $\forall x \in H$, there exists a unique decomposition $x = m_0 + w$ where $m_0 \in M$ and $w \in M^\perp$. Thus, $H = M \oplus M^\perp$.

Proof of i. Let $x \in H$. By the definition of infimum, there exists a sequence $\{m_k\} \subseteq M$ such that $\|x - m_k\|_H \to d$ as $k \to \infty$. We show that $\{m_k\}$ is a Cauchy sequence using the Parallelogram Law:

$$\|m_k - m_\ell\|_H^2 = \|(x - m_\ell) - (x - m_k)\|_H^2 = 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - \|2x - (m_\ell + m_k)\|_H^2.$$

Since M is a vector subspace, $\frac{m_\ell + m_k}{2} \in M$, which implies $\|x - (\frac{m_\ell + m_k}{2})\|_H \geq d$. Thus,

$$\|2x - (m_\ell + m_k)\|_H^2 = 4\left\|x - \left(\frac{m_\ell + m_k}{2}\right)\right\|_H^2 \geq 4d^2.$$

Substituting this back gives:

$$\|m_k - m_\ell\|_H^2 \leq 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - 4d^2 \xrightarrow{k,\ell\to\infty} 2d^2 + 2d^2 - 4d^2 = 0.$$

Since H is complete, $m_0 := \lim_{k\to\infty} m_k$ exists, and by closure of M, $m_0 \in M$ with $\|x - m_0\|_H = d$.

For uniqueness, if $\tilde{m}_0 \in M$ also achieves the distance d, applying the same inequality with $m_k = m_0$ and $m_\ell = \tilde{m}_0$ yields:

$$\|m_0 - \tilde{m}_0\|_H^2 \leq 2d^2 + 2d^2 - 4d^2 = 0 \implies m_0 = \tilde{m}_0.$$

To prove that $(x - m_0) \perp M$, let $m \in M$ and $t \in \mathbb{R}$. Expanding the norm using inner product properties:

$$\|(x - m_0) - tm\|_H^2 = \|x - m_0\|_H^2 - 2t\mathrm{Re}\{\langle x - m_0, m\rangle\} + t^2\|m\|_H^2.$$

Since m_0 minimizes the distance, we have $d^2 \leq \|(x - m_0) - tm\|_H^2$, which simplifies to:

$$0 \leq t^2\|m\|_H^2 - 2t\mathrm{Re}\{\langle x - m_0, m\rangle\} \quad \forall t \in \mathbb{R}.$$

For this quadratic function to remain non-negative for all t, the linear coefficient must vanish:

$$\mathrm{Re}\{\langle x - m_0, m\rangle\} = 0.$$

Replacing t by it in the original formulation yields $\mathrm{Im}\{\langle x - m_0, m\rangle\} = 0$. Since both parts are zero, $\langle x - m_0, m\rangle = 0$, proving orthogonality.

Proof of ii. Let $m_1 \in M$ satisfy $(x - m_1) \perp M$. For any $m \in M$, since $(m_1 - m) \in M$, we have $(x - m_1) \perp (m_1 - m)$. By the Pythagorean Theorem:

$$\|x - m\|_H^2 = \|(x - m_1) + (m_1 - m)\|_H^2 = \|x - m_1\|_H^2 + \|m_1 - m\|_H^2 \geq \|x - m_1\|_H^2.$$

Taking the infimum over all $m \in M$ implies $\|x - m_1\|_H^2 = d^2$. By the uniqueness proven in part (i), $m_1 = m_0$.

Corollary 2

Let $\{e_1, \ldots, e_N\}$ be an ONS in a Hilbert space H, and let $M = \text{span}\{e_1, \ldots, e_N\}$. Then for any $x \in H$:

$$P_M(x) = \sum_{j=1}^{N} \langle x, e_j \rangle e_j.$$

Proof. Theorem 2 established that $(x - \sum_{j=1}^{N} \langle x, e_j \rangle e_j) \perp e_k$ for all k, which implies it is orthogonal to the entire span M. Applying Theorem 3 (ii) completes the proof.

Hilbert Space Basis

Let H be a Hilbert space. An orthonormal system $S \subseteq H$ is called a **complete orthonormal system (CONS)** or a **Hilbert basis** if for any $z \in H$:

$$\langle z, e_\alpha \rangle = 0 \quad \forall e_\alpha \in S \implies z = 0.$$

Example

Let μ be the counting measure on $[0,1]$. Consider $H = L^2([0,1], \mu)$, the space of functions $f : [0,1] \to \mathbb{R}$ such that

$$\sum_{t \in [0,1]} [f(t)]^2 = \int_{[0,1]} [f(t)]^2 d\mu < \infty.$$

The inner product is given by $\langle f, g \rangle = \sum_{t \in [0,1]} f(t)g(t)$. For each $\alpha \in [0,1]$, defining the standard indicator functions $e_\alpha(t) = 1$ if $\alpha = t$ and 0 otherwise forms a non-countable CONS where $f(t) = \sum_{\alpha \in [0,1]} \langle f, e_\alpha \rangle e_\alpha(t)$.

Theorem 4

Let H be a Hilbert space with a CONS $\{e_\alpha : \alpha \in A\}$. For any $x \in H$:

i. The set $\mathcal{A} := \{\alpha \in A : \langle x, e_\alpha \rangle \neq 0\}$ is at most countable: $\mathcal{A} = \{\alpha_1, \alpha_2, \ldots\}$.

ii. $x = \sum_{\alpha \in A} \langle x, e_\alpha \rangle e_\alpha = \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} = \lim_{n \to \infty} \sum_{k=1}^{n} \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$.

iii. **Parseval's Identity:**

$$\|x\|_H^2 = \sum_{\alpha \in A} |\langle x, e_\alpha \rangle|^2 = \sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2.$$

Proof.

i. For any finite subset $\{\beta_1, \ldots, \beta_n\} \subseteq A$, Bessel's inequality states $\sum_{k=1}^{n} |\langle x, e_{\beta_k} \rangle|^2 \leq \|x\|_H^2 < \infty$. For each $n \geq 1$, define $A_n = \{\alpha \in A : |\langle x, e_\alpha \rangle| > \frac{1}{n}\}$. If any A_n were infinite, it would contain a countable sequence contradicting Bessel's bounded sum. Hence, each A_n is finite, making $\mathcal{A} = \bigcup_{n=1}^{\infty} A_n$ a countable union of finite sets, which is countable.

ii. Let $\{\alpha_k\}_{k=1}^{\infty}$ be an enumeration of $\mathcal{A}$ and define the partial sums $s_n = \sum_{k=1}^{n} \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$. For $m > n$, by the Pythagorean theorem:

$$\|s_m - s_n\|_H^2 = \sum_{k=n+1}^{m} |\langle x, e_{\alpha_k} \rangle|^2.$$

As $n, m \to \infty$, this goes to 0 because the total series $\sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2$ converges. Thus $\{s_n\}$ is Cauchy, and since H is complete, it converges to some $y \in H$. To show $y = x$, we look at the inner product for any $\alpha \in A$:

$$\langle x - y, e_\alpha \rangle = \langle x, e_\alpha \rangle - \lim_{n \to \infty} \sum_{k=1}^{n} \langle x, e_{\alpha_k} \rangle \langle e_{\alpha_k}, e_\alpha \rangle.$$

If $\alpha \notin \mathcal{A}$, both terms are 0. If $\alpha \in \mathcal{A}$, then $\alpha = \alpha_n$ for some n, yielding $\langle x, e_{\alpha_n} \rangle - \langle x, e_{\alpha_n} \rangle = 0$. Since $\langle x - y, e_\alpha \rangle = 0$ for all α, by the completeness of the system, $x - y = 0 \implies x = y$.

iii. Applying Theorem 2 (ii) to the finite partial sums yields:

$$\|x\|_H^2 = \left\| x - \sum_{k=1}^{n} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \right\|_H^2 + \sum_{k=1}^{n} |\langle x, e_{\alpha_k} \rangle|^2.$$

Taking $n \to \infty$, the first norm term vanishes due to part (ii), leaving Parseval's Identity.

Theorem 5

A Hilbert space H has a countable CONS if and only if H is separable (contains a countable dense subset).

Proof. ($\Longrightarrow$) Let $\{e_n\}_{n\geq 1}$ be a countable CONS. The set of all finite linear combinations $\sum_{k=1}^{n} r_k e_k$ where $r_k \in \mathbb{Q} + i\mathbb{Q}$ forms a countable dense subset in H. ($\Longleftarrow$) Let $\{y_1, y_2, \ldots\}$ be a countable dense subset. We can filter out linearly dependent vectors to get a linearly independent sequence $\{x_n\}$ spanning the same subspace. Applying the standard Gram-Schmidt procedure to $\{x_n\}$ constructs a countable orthonormal system $\{e_n\}$ whose span is dense in H, hence forming a CONS.

Proposition 2

Let H be a pre-Hilbert space. For a fixed $z \in H$, define $\ell_z(x) = \langle x, z \rangle$. Then $\ell_z \in H^*$ and $\|\ell_z\|_{H^*} = \|z\|_H$.

Proof. Boundedness follows from Cauchy-Schwarz: $|\ell_z(x)| = |\langle x, z \rangle| \leq \|z\|_H \|x\|_H$, showing $\|\ell_z\|_{H^*} \leq \|z\|_H$. Testing the functional on the unit vector $x = \frac{z}{\|z\|_H}$ (for $z \neq 0$) yields exactly $\|z\|_H$, establishing equality.

Riesz Representation Theorem

Let H be a Hilbert space. For every bounded linear functional $\ell \in H^*$, there exists a unique $z_\ell \in H$ such that

$$\ell(x) = \langle x, z_\ell \rangle, \quad \forall x \in H.$$

Moreover, $\|\ell\|_{H^*} = \|z_\ell\|_H$.

Proof. If $\ell = 0$, choosing $z_\ell = 0$ works. Assume $\ell \neq 0$, and let $M = \ker(\ell)$. Since ℓ is continuous, M is a closed subspace, and $M \neq H$. By the Projection Theorem, $M^\perp \neq \{0\}$. Pick a nonzero $x_0 \in M^\perp$ scaled such that $\ell(x_0) = 1$. For any $x \in H$, the vector $x - \ell(x)x_0$ belongs to M because $\ell(x - \ell(x)x_0) = \ell(x) - \ell(x) = 0$. Since $x_0 \in M^\perp$, we have:

$$\langle x - \ell(x)x_0, x_0 \rangle = 0 \implies \langle x, x_0 \rangle = \ell(x)\|x_0\|_H^2 \implies \ell(x) = \left\langle x, \frac{x_0}{\|x_0\|_H^2} \right\rangle.$$

Setting $z_\ell = \frac{x_0}{\|x_0\|_H^2}$ proves existence. Uniqueness follows since if $\langle x, z_1 \rangle = \langle x, z_2 \rangle$ for all x, then $\langle x, z_1 - z_2 \rangle = 0 \implies z_1 - z_2 = 0$.

Adjoint Operator

Let H be a Hilbert space. For each $T \in \mathcal{L}(H)$, there exists a unique $T^* \in \mathcal{L}(H)$ such that

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \forall x, y \in H.$$

Moreover, $\|T^*\| = \|T\|$.

Proof. For a fixed $y \in H$, $x \mapsto \langle Tx, y \rangle$ is a bounded linear functional on H. By the Riesz Representation Theorem, there exists a unique vector, which we denote as T^*y, such that $\langle Tx, y \rangle = \langle x, T^*y \rangle$. To verify that T^* is linear on a complex Hilbert space, consider $y_1, y_2 \in H$ and $\alpha, \beta \in \mathbb{C}$. For any $x \in H$:

$$\begin{aligned}
\langle x, T^*(\alpha y_1 + \beta y_2) \rangle &= \langle Tx, \alpha y_1 + \beta y_2 \rangle \\
&= \overline{\alpha}\langle Tx, y_1 \rangle + \overline{\beta}\langle Tx, y_2 \rangle \quad \text{(conjugate-linear in 2nd slot)} \\
&= \overline{\alpha}\langle x, T^*y_1 \rangle + \overline{\beta}\langle x, T^*y_2 \rangle \\
&= \langle x, \alpha T^*y_1 + \beta T^*y_2 \rangle \quad \text{(bringing scalars back inside)}
\end{aligned}$$

Since this holds for all $x \in H$, $T^*(\alpha y_1 + \beta y_2) = \alpha T^*y_1 + \beta T^*y_2$. The norm equality follows from:

$$\|T^*\| = \sup_{\|y\| \le 1} \sup_{\|x\| \le 1} |\langle x, T^*y \rangle| = \sup_{\|y\| \le 1} \sup_{\|x\| \le 1} |\langle Tx, y \rangle| = \sup_{\|x\| \le 1} \|Tx\|_H = \|T\|.$$

Self-Adjoint Operator

An operator $T \in \mathcal{L}(H)$ is called **self-adjoint** if $T^* = T$.

Lemma

Let H be a Hilbert space and $T \in \mathcal{L}(H)$. Then,

$$\|T^*T\| = \|T\|^2 = \|T^*\|^2.$$

Proof. For any $x \in H$ with $\|x\|_H \leq 1$:

$$\|Tx\|_H^2 = \langle Tx, Tx \rangle = \langle x, T^*Tx \rangle \leq \|x\|_H \|T^*Tx\|_H \leq \|T^*T\|.$$

Taking the supremum over all such x yields $\|T\|^2 \leq \|T^*T\|$. Combined with the standard norm inequality $\|T^*T\| \leq \|T^*\|\|T\| = \|T\|^2$, we get equality.

Remark

Let H be a Hilbert space. Then:

1. $\mathcal{A} := \mathcal{L}(H)$ is a C^*-algebra.
2. $\mathcal{A}$ is a Banach algebra satisfying $\|ST\| \leq \|S\|\|T\|$.
3. The adjoint operation $T \mapsto T^*$ is an involution satisfying:
   - $(\alpha T + \beta S)^* = \overline{\alpha} T^* + \overline{\beta} S^*$.
   - $(ST)^* = T^* S^*$.
   - $\|T^*T\| = \|T\|^2$.

L^p Spaces

Let $(X, \mathcal{X}, \mu)$ be a measure space. For $0 < p \leq \infty$, we define the space $L^p(X, \mathcal{X}, \mu)$ (abbreviated as $L^p(\mu)$ or L^p) as:

$$L^p = \left\{ f : X \to \mathbb{C} \text{ is measurable} : \int_X |f|^p \, d\mu < \infty \right\}.$$

We identify functions that are equal almost everywhere (a.e.) to form equivalence classes. Additionally, we define L^∞ to be the space of equivalence classes of measurable functions $f : X \to \mathbb{C}$ that are essentially bounded:

$$|f| \leq C \text{ a.e. for some } C > 0.$$

Remark. L^p is a vector space. For instance, if $p \in (0, \infty)$ and $a, b > 0$, the inequality $(a+b)^p \leq 2^p(a^p + b^p)$ holds because:

$$(a+b)^p \leq (2 \max\{a, b\})^p = 2^p \max\{a^p, b^p\} \leq 2^p(a^p + b^p).$$

For $f \in L^p$ where $p \in (0, \infty)$, we define the L^p norm (or quasi-norm if $p < 1$) as:

$$\|f\|_{L^p} = \|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

For $p = \infty$, the essential supremum norm is defined as:

$$\|f\|_{L^\infty} = \|f\|_\infty = \inf\{C \geq 0 : |f| \leq C \text{ a.e.}\}.$$

Note. By definition, $|f| \leq \|f\|_\infty$ a.e.

Young's Inequality

For $a, b \geq 0$, $p \in (1, \infty)$, and $\frac{1}{p} + \frac{1}{q} = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. Let $a, b > 0$ (the case where $a = 0$ or $b = 0$ is trivial). Since the natural logarithm function ln is concave, we can apply Jensen's inequality:

$$\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q) = \ln(a) + \ln(b) = \ln(ab).$$

Taking the exponential (exp) of both sides yields the desired result.

Hölder's Inequality

Let $p \in [1, \infty]$ and let $\frac{1}{p} + \frac{1}{q} = 1$ (with $\frac{1}{\infty} = 0$). If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and:

$$\int_X |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

Proof. For $p = 1$ or $p = \infty$, the proof is straightforward and left as an exercise. Let $p \in (1, \infty)$, $f \neq 0$, and $g \neq 0$ a.e. By Young's inequality, for any parameter $t > 0$, we have:

$$\int_X |fg| \, d\mu = \int_X |(tf)(t^{-1}g)| \, d\mu \leq \frac{t^p}{p} \int_X |f|^p \, d\mu + \frac{t^{-q}}{q} \int_X |g|^q \, d\mu.$$

Let $A = \int_X |f|^p \, d\mu$ and $B = \int_X |g|^q \, d\mu$. Define the auxiliary function $L(t)$ for $t > 0$:

$$L(t) = \frac{t^p}{p} A + \frac{t^{-q}}{q} B.$$

To minimize $L(t)$, we find its derivative and set it to zero:

$$L'(t) = t^{p-1}A - t^{-q-1}B = 0 \implies t^{p+q} = \frac{B}{A} \implies t = \left(\frac{B}{A}\right)^{\frac{1}{p+q}}.$$

Evaluating $L(t)$ at this critical point gives:

$$\inf_{t>0} L(t) = \frac{1}{p}\left(\frac{B}{A}\right)^{1/q} A + \frac{1}{q}\left(\frac{B}{A}\right)^{-1/p} B = \frac{1}{p}A^{1/p}B^{1/q} + \frac{1}{q}A^{1/p}B^{1/q} = A^{1/p}B^{1/q}.$$

Substituting $A^{1/p} = \|f\|_p$ and $B^{1/q} = \|g\|_q$ completes the proof.

Remark. If $f \in L^p$ and $\frac{1}{p} + \frac{1}{q} = 1$, then $|f|^{p-1} \in L^q$.

Proposition 1 (Minkowski's Inequality)

For $p \in [1, \infty]$, $\| \cdot \|_p$ satisfies the triangle inequality and forms a norm on L^p.

Proof. For $p = 1$ or $p = \infty$, the proof is left as an exercise. Let $p \in (1, \infty)$ and $f, g \in L^p$. We assume $f + g \neq 0$ dynamically. Splitting the integrand yields:

$$\int_X |f+g|^p \, d\mu = \int_X |f+g|^{p-1}|f+g| \, d\mu \leq \int_X |f+g|^{p-1}|f| \, d\mu + \int_X |f+g|^{p-1}|g| \, d\mu.$$

Applying Hölder's inequality to both integrals on the right-hand side using conjugate exponents p and $q = \frac{p}{p-1}$, we get:

$$\int_X |f+g|^p \, d\mu \leq \left(\int_X |f+g|^{(p-1)q} \, d\mu \right)^{1/q} \|f\|_p + \left(\int_X |f+g|^{(p-1)q} \, d\mu \right)^{1/q} \|g\|_p.$$

Since $(p-1)q = p$ and $1 - \frac{1}{q} = \frac{1}{p}$, this simplifies directly to:

$$\int_X |f+g|^p \, d\mu \leq \left(\int_X |f+g|^p \, d\mu \right)^{1-\frac{1}{p}} \left(\|f\|_p + \|g\|_p \right).$$

Dividing both sides by $\left(\int_X |f+g|^p \, d\mu \right)^{1-\frac{1}{p}}$ yields:

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p.$$

Remark. Let $f_n, f \in L^\infty$. Then $\|f_n - f\|_\infty \to 0$ if and only if there exist representatives $\tilde{f}_n, \tilde{f}$ in their respective equivalence classes such that $\sup_{x \in X} |\tilde{f}_n(x) - \tilde{f}(x)| \to 0$ as $n \to \infty$.

Chebyshev's Inequality

Let $p \in (0, \infty)$ and $f \in L^p$. Then, for any $\epsilon > 0$:

$$\mu(\{x \in X : |f(x)| > \epsilon\}) \leq \frac{\int_X |f|^p \, d\mu}{\epsilon^p}.$$

Proof. Let $\epsilon > 0$. By utilizing monotonic sets, we observe:

$$\int_X |f|^p \, d\mu \geq \int_{\{|f| > \epsilon\}} |f|^p \, d\mu \geq \epsilon^p \mu(\{x \in X : |f(x)| > \epsilon\}).$$

Theorem 2

L^p is a Banach space for any $p \in [1, \infty]$.

Proof. The case $p = \infty$ is left as an exercise. Let $p \in [1, \infty)$ and let (f_n) be a Cauchy sequence in L^p. By Chebyshev's Inequality, for any $\epsilon > 0$:

$$\mu(\{|f_n - f_m| > \epsilon\}) \leq \frac{\int_X |f_n - f_m|^p \, d\mu}{\epsilon^p} \xrightarrow{n,m \to \infty} 0.$$

Hence, (f_n) is Cauchy in measure. This implies there exists a measurable function f and a subsequence (f_{n_k}) such that $f_{n_k} \to f$ a.e. as $k \to \infty$.

Fix $\epsilon > 0$. Choose N such that $\int_X |f_n - f_m|^p \, d\mu \leq \epsilon$ for all $n, m \geq N$. Letting $m = n_l \to \infty$, Fatou's Lemma implies:

$$\int_X |f_n - f|^p \, d\mu = \int_X \lim_{l \to \infty} |f_n - f_{n_l}|^p \, d\mu \leq \liminf_{l \to \infty} \int_X |f_n - f_{n_l}|^p \, d\mu \leq \epsilon$$

for all $n \geq N$. Thus, $f - f_n \in L^p$, which means $f = (f - f_n) + f_n \in L^p$, and $f_n \to f$ in L^p.

Comparison of L^p Spaces

In general, spaces are not nested: neither $L^p \subseteq L^q$ nor $L^q \subseteq L^p$ if $p \neq q$ on arbitrary measure spaces.

Proof. Let $X = (0, \infty)$ equipped with Lebesgue measure μ. Consider the function:

$$f(x) = x^{-\alpha} \chi_{(0,1)}(x) + x^{-\beta} \chi_{[1,\infty)}(x) \quad (\alpha, \beta > 0).$$

Calculating the L^r integral gives $\int_X |f(x)|^r \, d\mu = \int_0^1 x^{-\alpha r} \, dx + \int_1^\infty x^{-\beta r} \, dx$. This converges if and only if $\alpha < \frac{1}{r} < \beta$. Thus, if we select exponents such that $\frac{1}{q} < \alpha < \frac{1}{p} < \beta$, then $f \in L^p$ but $f \notin L^q$. Conversely, choosing $\alpha < \frac{1}{q} < \beta < \frac{1}{p}$ yields $f \in L^q$ but $f \notin L^p$.

Proposition 3

Let $\mu(X) < \infty$ and $0 < p < q \leq \infty$. Then $L^q \subseteq L^p$ and:

$$\|f\|_p \leq \|f\|_q \, \mu(X)^{\frac{1}{p} - \frac{1}{q}}.$$

Proof. Applying Hölder's inequality with conjugate exponents $\frac{q}{p}$ and $\frac{q}{q-p}$ yields:

$$\int_X |f|^p \cdot 1 \, d\mu \leq \left(\int_X |f|^q \, d\mu \right)^{\frac{p}{q}} \left(\int_X 1 \, d\mu \right)^{1 - \frac{p}{q}} = \|f\|_q^p \, \mu(X)^{1 - \frac{p}{q}}.$$

Taking the p-th root of both sides proves the assertion.

Proposition 4

Let $0 < p < q < r \leq \infty$. Then $L^q \subseteq L^p + L^r$.

Proof. Let $f \in L^q$. We decompose $f = f\chi_{\{|f| \leq 1\}} + f\chi_{\{|f| > 1\}} =: k + h$. Then $|k| \leq |f|$ and $|k| \leq 1$, so $|k|^r \leq |f|^q \in L^1$, implying $k \in L^r$. Similarly, $|h| \leq |f|$ and on the support of h, $|f| > 1$, meaning $|h|^p \leq |f|^q \in L^1$, implying $h \in L^p$.

Proposition 5

Let $0 < p < q < r \leq \infty$. Then $L^p \cap L^r \subseteq L^q$ and $\|f\|_q \leq \|f\|_p^t \|f\|_r^{1-t}$ where $\frac{1}{q} = \frac{t}{p} + \frac{1-t}{r}$.

Proof. Let $f \in L^p \cap L^r$.

i. If $r = \infty$, then $\int_X |f|^q \, d\mu = \int_X |f|^{q-p} |f|^p \, d\mu \leq \|f\|_\infty^{q-p} \|f\|_p^p$. Taking the q-th root gives the result with $t = p/q$.

ii. If $r < \infty$, rewrite $|f|^q = |f|^{tq} |f|^{(1-t)q}$. Apply Hölder's inequality with conjugate exponents $\frac{p}{tq}$ and $\frac{r}{(1-t)q}$. The conditions require $\frac{tq}{p} + \frac{(1-t)q}{r} = 1$, which aligns perfectly with the definition of t.

Proposition 6 (Sequence Spaces)

If $X = \mathbb{N}$ and μ is the counting measure, we denote $\ell^p = L^p(\mathbb{N}, \mu)$. If $0 < p < q \leq \infty$, then $\ell^p \subseteq \ell^q$ and $\|x\|_q \leq \|x\|_p$.

Proof. Let $x \in \ell^p$. For $q = \infty$, $\|x\|_\infty^p = \sup_n |x_n|^p \leq \sum_n |x_n|^p = \|x\|_p^p \implies \|x\|_\infty \leq \|x\|_p$. For $q < \infty$, by interpolation (Proposition 5), $\|x\|_q \leq \|x\|_p^{p/q} \|x\|_\infty^{1-p/q} \leq \|x\|_p$.

Simple Function Spaces

Define $\Sigma = \{f \in \mathcal{E} : f \text{ vanishes outside a set of finite measure}\}$, where $\mathcal{E}$ is the space of all complex-valued simple functions:

$$\mathcal{E} = \left\{ f = \sum_{j=1}^{n} z_j \chi_{E_j} : n \geq 1,\, z_j \in \mathbb{C},\, E_j \in \mathcal{X} \text{ are disjoint} \right\}.$$

Remark. $\mathcal{E} \subseteq L^\infty$ and $\Sigma \subseteq L^p$ for any $p \in (0, \infty]$.

Theorem 2.10 [Folland]

Let $(X, \mathcal{X})$ be a measurable space. If $f : X \to \mathbb{C}$ is measurable, there exists a sequence $\{f_n\} \subseteq \mathcal{E}$ such that $0 \leq |f_1| \leq |f_2| \leq \cdots \leq |f|$, $f_n \to f$ pointwise, and $f_n \to f$ uniformly on any set where f is bounded.

Proposition 7

i. $\mathcal{E}$ is dense in L^∞.

ii. Σ is dense in L^p if $p \in [1, \infty)$.

Proof. Apply Theorem 2.10. For (i), since $f \in L^\infty$ is bounded outside a null set, $f_n \to f$ uniformly a.e., meaning $\|f_n - f\|_\infty \to 0$. For (ii), if $f \in L^p$, $|f_n - f|^p \leq 2^{p+1} |f|^p \in L^1$. By the Dominated Convergence Theorem, $\int_X |f_n - f|^p \, d\mu \to 0$.

Generalized Dominated Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions such that $f_n \to f$ a.e. Suppose there exists a sequence of nonnegative integrable functions $\{g_n\}$ such that $|f_n| \leq g_n$ a.e., $g_n \to g$ a.e., and $\int_X g_n \, d\mu \to \int_X g \, d\mu < \infty$. Then $f_n \to f$ in L^1.

Proof. Consider the sequence of nonnegative functions $h_n = g_n + g - |f_n - f|$. Since $|f_n - f| \leq g_n + g$ a.e., $h_n \geq 0$ a.e. Applying Fatou's Lemma to $\{h_n\}$ yields:

$$\int_X \liminf_{n\to\infty} h_n \, d\mu \leq \liminf_{n\to\infty} \int_X h_n \, d\mu.$$

Since $f_n \to f$ and $g_n \to g$ a.e., $\liminf h_n = 2g$ a.e. Thus:

$$\int_X 2g \, d\mu \leq \int_X g \, d\mu + \int_X g \, d\mu - \limsup_{n\to\infty} \int_X |f_n - f| \, d\mu.$$

Subtracting $2\int_X g \, d\mu < \infty$ from both sides results in $\limsup_{n\to\infty} \int_X |f_n - f| \, d\mu \leq 0$, which ensures $\lim_{n\to\infty} \|f_n - f\|_1 = 0$.

The Dual of L^p

Let $p \in (1, \infty]$ and $q \in [1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. For any $h \in L^q$, define the linear functional T_h on L^p by:

$$T_h(f) = \int_X fh \, d\mu, \quad f \in L^p.$$

Then $T_h \in (L^p)^*$ and $\|T_h\|_{(L^p)^*} = \|h\|_q = \sup\{|\int_X fh \, d\mu| : \|f\|_p \leq 1\}$.

Theorem 4 (Riesz Representation Theorem for L^p)

Let $p \in (1, \infty)$. For each bounded linear functional $T \in (L^p)^*$, there exists a unique $h \in L^q$ such that $T(f) = \int_X hf \, d\mu$ for all $f \in L^p$, and $\|T\|_{(L^p)^*} = \|h\|_q$. If μ is σ-finite, this also holds for $p = 1$ and $q = \infty$.

Weak L^p

Let $p \in (0, \infty)$. A measurable function f belongs to **weak** L^p if its distribution function $\lambda_f(\alpha) = \mu(\{x \in X : |f(x)| > \alpha\})$ satisfies:

$$[f]_p := \left(\sup_{\alpha>0} \alpha^p \lambda_f(\alpha)\right)^{1/p} < \infty.$$

1. By Chebyshev's Inequality, $L^p \subseteq$ weak L^p and $[f]_p \leq \|f\|_p$.
2. Using layer-cake integration by parts, if $f \in L^p$, its norm can be computed via:

$$\|f\|_p^p = \int_X |f|^p \, d\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) \, d\alpha.$$

Convolution and Smoothing

Let $X = \mathbb{R}^d$, $\mathcal{X} = \mathcal{L}$, and $\mu = m \implies d\mu = dm = dx$.

Lemma 1

Let f, g be measurable, $x_0 \in \mathbb{R}^d$. Then,

$$f(x_0 - \cdot)g \in L^1 \iff g(x_0 - \cdot)f \in L^1 \tag{1}$$

and if Eq.(1) holds,

$$\int f(x_0 - y)g(y)\,dy = \int g(x_0 - y)f(y)\,dy.$$

All functions involved in Eq.(1) are measurable including $f(x_0 - \cdot)$ and $g(x_0 - \cdot)$.

Proof. Let $G : \mathbb{R}^d \to \mathbb{R}^d$ be bijective with $G, G^{-1} \in C^1$ and $x = G(y) = (g_1(y), \dots, g_d(y))$. Note that C^1 denotes the class of continuously differentiable functions. Then, by the formula of changing variables,

$$\int f(x)\,dx = \int f(G(y))|\det \nabla G(y)|\,dy.$$

$$\nabla G(y) = \begin{bmatrix} \frac{\partial g_1(y)}{\partial y_1} & \cdots & \frac{\partial g_1(y)}{\partial y_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d(y)}{\partial y_1} & \cdots & \frac{\partial g_d(y)}{\partial y_d} \end{bmatrix}.$$

In our case, we have $G(y) = x_0 - y = z$. Then,

$$\nabla G(y) = \begin{bmatrix} -1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{bmatrix} = -I_d \implies |\det \nabla G(y)| = |(-1)^d| = 1.$$

$$\implies \int f(x_0 - y)g(y)\,dy = \int f(G(y))g(x_0 - G(y))|\det \nabla G(y)|\,dy = \int g(x_0 - z)f(z)\,dz.$$

Convolution

Let f, g be measurable and $f(x - \cdot)g \in L^1$ a.e. with respect to x. The following function is well-defined:

$$(f * g)(x) = \int f(x - y)g(y)\,dy = \int g(x - y)f(y)\,dy = (g * f)(x).$$

$f * g$ is called the convolution of f and g.

Well-defined Convolution

Let f, g be measurable. If

$$\int |f(x-y)||g(y)|\, dy < \infty \quad \text{a.e. with respect to } x,$$

then we say

$$(f * g)(x) = \int f(x-y)g(y)\, dy,$$

is well-defined. We say $(f * g)(x)$ is defined for the x such that

$$\int |f(x-y)||g(y)|\, dy < \infty.$$

Support

Given $f : \mathbb{R}^d \to \mathbb{C}$, the support of f, denoted $\operatorname{supp}(f)$, is the closure of $\{x : f(x) \neq 0\}$. $C_c(\mathbb{R}^d)$ is the space of continuous functions with compact support.

Lemma 2

If $f, g \in C_c(\mathbb{R}^d)$, then $f * g \in C_c(\mathbb{R}^d)$. Moreover,

$$\operatorname{supp}(f * g) \subseteq \overline{\operatorname{supp}(f) + \operatorname{supp}(g)},$$

where $A + B = \{x + y : x \in A, y \in B\}$.

Proof. Since $f, g \in C_c(\mathbb{R}^d)$, both f, g are bounded. Therefore,

$$\int |f(x-y)||g(y)|\, dy < \infty \text{ for any } x \in \mathbb{R}^d.$$

Hence, $f * g$ is defined for any $x \in \mathbb{R}^d$. Now, let $\{x_n\}$ be a sequence that converges to $x_0 \in \mathbb{R}^d$. Define $h_n(y) = [f(x_n - y) - f(x_0 - y)]g(y)$ and $h(y) = [f(x_0 - y) - f(x_0 - y)]g(y) = 0$. Next, we apply the Dominated Convergence Theorem to prove $f * g$ is continuous. DCT needs:

1. $h_n \to h$ a.e.
2. There exists a nonnegative $\phi \in L^1$ with $|h_n| \leq \phi$ a.e. for all n.

Since $f \in C_c(\mathbb{R}^d)$, f is continuous. Therefore, for each fixed y, we have $f(x_n - y) \to f(x_0 - y)$ as $x_n \to x_0$. This gives us pointwise convergence: $h_n(y) \to 0 = h(y)$ for each y. We need an integrable function that dominates $|h_n(y)|$. Note that since f has compact support and is continuous, it's bounded: $|f(z)| \leq M$ for some constant M. Then,

$$|h_n(y)| = |[f(x_n - y) - f(x_0 - y)]g(y)| \leq 2M|g(y)|.$$

Since $g \in C_c(\mathbb{R}^d)$, g is bounded, so $\phi = 2M|g(y)| \in L^1$. By DCT,

$$\lim_{n\to\infty} \left(\int f(x_n - y)g(y)\, dy - \int f(x_0 - y)g(y)\, dy \right) = \lim_{n\to\infty} \int h_n(y)\, dy = \int \lim_{n\to\infty} h_n(y)\, dy = 0.$$

Therefore, $f * g$ is continuous.

Now, we claim $\{x : (f * g)(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g)$.

$$\begin{aligned}
(f * g)(x) \neq 0 &\implies \text{There exists } y \text{ such that } f(x-y)g(y) \neq 0, \\
&\implies x - y \in \text{supp}(f), \quad y \in \text{supp}(g), \\
&\implies x = (x-y) + y \in \text{supp}(f) + \text{supp}(g), \\
&\implies \{x : (f * g)(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g).
\end{aligned}$$

Taking the closure of both sides,

$$\text{supp}(f * g) = \overline{\{x : (f * g)(x) \neq 0\}} \subseteq \overline{\text{supp}(f) + \text{supp}(g)}.$$

Since both $\text{supp}(f)$ and $\text{supp}(g)$ are compact sets, we use a theorem from topology: The Minkowski sum of two compact sets is compact. Thus, $\text{supp}(f) + \text{supp}(g)$ is compact. Since it is compact, it is closed in $\mathbb{R}^d$. Therefore,

$$\overline{\text{supp}(f) + \text{supp}(g)} = \text{supp}(f) + \text{supp}(g).$$

This gives us $\text{supp}(f * g) \subseteq \text{supp}(f) + \text{supp}(g)$. Since $\text{supp}(f * g)$ is a closed subset of the compact set $\text{supp}(f) + \text{supp}(g)$, it is compact. Hence, $f * g$ is a continuous function with compact support, so $f * g \in C_c(\mathbb{R}^d)$.

Remark.

1. If a function has compact support, it must be bounded, and here's why: A set in $\mathbb{R}^d$ is compact if and only if it is closed and bounded. So when we say a function f has compact support, it means that the closure of the set $\{x : f(x) \neq 0\}$ is both closed and bounded. For a continuous function f with compact support, we can apply the extreme value theorem, which states that a continuous function on a compact set attains both its maximum and minimum values. Specifically:

   - Since f is continuous and its support is compact
   - By the extreme value theorem, f must attain a maximum value on its support
   - Outside its support, f is zero by definition
   - Therefore, f is bounded by the maximum absolute value it achieves on its support

   This means that there exists some constant $M > 0$ such that $|f(x)| \leq M$ for all $x \in \mathbb{R}^d$.

2. Lebesgue measure on $\mathbb{R}^d$ is indeed sigma-finite. We can express $\mathbb{R}^d$ as a countable union of bounded sets:

   $$\mathbb{R}^d = \bigcup_{n=1}^{\infty} B_n.$$

   where $B_n = \{x \in \mathbb{R}^d : |x| \leq n\}$ is the closed ball of radius n centered at the origin. Each B_n has finite Lebesgue measure. Specifically:

   $$\lambda(B_n) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} \cdot n^d < \infty.$$

This decomposition into a countable union of finite-measure sets is exactly the definition of a sigma-finite measure space.

Proposition 1

Let $p \in [1, \infty]$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in L^p$, $h \in L^q$. Then, $(f * h)(x)$ is defined for all $x \in \mathbb{R}^d$ and

$$\sup_x |(f * h)(x)| \leq \|f\|_p \|h\|_q.$$

Moreover, $f * h$ is continuous. Also, $\lim_{|x| \to \infty} (f * h)(x) = 0$ if $p \in (1, \infty)$.

Proof. For each $x \in \mathbb{R}^d$, by Hölder's inequality,

$$\int |f(x-y)||h(y)| \, dy \leq \|f(x - \cdot)\|_p \|h\|_q = \|f\|_p \|h\|_q < \infty,$$

$$\implies |(f * h)(x)| = \left| \int f(x-y) h(y) \, dy \right| \leq \int |f(x-y)||h(y)| \, dy \leq \|f\|_p \|h\|_q,$$

$$\implies \sup_x |(f * h)(x)| \leq \|f\|_p \|h\|_q.$$

Continuity

(i) Let $p \in (1, \infty)$. There exist $f_n, h_n \in C_c(\mathbb{R}^d)$ such that $\|f_n - f\|_p \to 0$ and $\|h_n - h\|_q \to 0$ as $n \to \infty$. Recall by Lemma 2, $f_n * h_n \in C_c(\mathbb{R}^d)$, and

$$\begin{aligned} \sup_x |(f_n * h_n)(x) - (f * h)(x)| &\leq \sup_x |((f_n - f) * h_n)(x)| + \sup_x |(f * (h_n - h))(x)| \\ &\leq \|f_n - f\|_p \|h_n\|_q + \|f\|_p \|h_n - h\|_q \\ &\leq \left(\sup_k \|h_k\|_q \right) \|f_n - f\|_p + \|f\|_p \|h_n - h\|_q \xrightarrow{n \to \infty} 0. \end{aligned}$$

Thus, $f_n * h_n \to f * h$ uniformly. The uniform convergence of a sequence of continuous functions preserves continuity, so $f * h$ must be continuous. Also,

$$\lim_{|x| \to \infty} (f_n * h_n)(x) = 0 \implies \lim_{|x| \to \infty} (f * h)(x) = 0.$$

(ii) Let $p = 1$, $f \in L^1$, $h \in L^\infty$. There exists $f_n \in C_c(\mathbb{R}^d)$ such that $\|f_n - f\|_1 \xrightarrow{n \to \infty} 0$. Then,

$$\sup_x |(f_n * h)(x) - (f * h)(x)| = \sup_x |((f_n - f) * h)(x)| \leq \|f_n - f\|_1 \|h\|_\infty \xrightarrow{n \to \infty} 0.$$

Thus, $f_n * h$ converges uniformly to $f * h$ as $n \to \infty$. So we just need to show that $(f_n * h)(x) = \int f_n(x-y) h(y) \, dy$ is continuous in x. By the DCT and continuity of f_n:

$$\lim_{x \to x_0} ((f_n * h)(x) - (f_n * h)(x_0)) = \int \lim_{x \to x_0} [f_n(x-y) - f_n(x_0 - y)] h(y) = 0.$$

The uniform convergence of a sequence of continuous functions preserves continuity, so $f * h$ must be continuous.

Remark. The property $\lim_{|x|\to\infty}(f_n * h_n)(x) = 0$ for functions $f_n * h_n \in C_c(\mathbb{R}^d)$ follows directly from the definition of compact support. By definition, a function $g \in C_c(\mathbb{R}^d)$ has compact support, meaning that the set $\mathrm{supp}(g) = \overline{\{x \in \mathbb{R}^d : g(x) \neq 0\}}$ is compact. Since $\mathrm{supp}(g)$ is compact in $\mathbb{R}^d$, it must be bounded. This means there exists some radius $R > 0$ such that:

$$\mathrm{supp}(g) \subseteq B(0, R) = \{x \in \mathbb{R}^d : |x| \leq R\}.$$

In other words, for any $|x| > R$, we have $x \notin \mathrm{supp}(g)$, which means $g(x) = 0$. Applying this to $f_n * h_n \in C_c(\mathbb{R}^d)$, there exists some $R_n > 0$ such that for all $|x| > R_n$, we have $(f_n * h_n)(x) = 0$. Therefore, $\lim_{|x|\to\infty}(f_n * h_n)(x) = 0$ is immediate because the function is exactly zero outside some bounded region.

Proposition 2

Let $p \in [1, \infty)$, $f \in L^p$, $h \in L^1$. Then, $f * h$ is well-defined and $f * h \in L^p$ with

$$\|f * h\|_p \leq \|f\|_p \|h\|_1.$$

Proof. By Minkowski's inequality for integrals,

$$\begin{aligned}
\left(\int \left|\int f(x-y)h(y)\,dy\right|^p dx\right)^{1/p} &\leq \int \left(\int |f(x-y)|^p\,dx\right)^{1/p} |h(y)|\,dy \\
&= \|f\|_p \int |h(y)|\,dy \\
&= \|f\|_p \|h\|_1 < \infty.
\end{aligned}$$

Hence, $\int |f(x-y)||h(y)|\,dy < \infty$ a.e. with respect to x and $\|f * h\|_p \leq \|f\|_p \|h\|_1$.

Continuity of Shifts

For f on $\mathbb{R}^d$ and $z \in \mathbb{R}^d$,

$$\tau_z f(x) = f(x+z), \quad x \in \mathbb{R}^d.$$

Note that $\|\tau_z f\|_p = \|f\|_p$, $p \in (0, \infty]$. Also note that $\tau_0 f = f$.

Lemma 3

Let $p \in [1, \infty)$, $f \in L^p$. Then,

1. $\|\tau_z f - f\|_p \xrightarrow{z\to 0} 0$ or $\int |f(x-z) - f(x)|^p\,dx \xrightarrow{z\to 0} 0$
2. $\|\tau_z f - \tau_{z_0} f\|_p = \|\tau_{z-z_0} f - f\|_p \xrightarrow{z\to z_0} 0$

The second part follows directly from $\|\tau_z f - f\|_p \xrightarrow{z\to 0} 0$.

Proof. Let $g \in \mathcal{C}_c(\mathbb{R}^d)$. Then, by DCT,

$$\int |g(x+z) - g(x)|^p\,dx \xrightarrow{z\to 0} 0.$$

Since if $|z| \leq 1$, $\text{supp}(g) \subseteq B_R(0)$,

$$|g(x+z) - g(x)| \leq |g(x+z)| + |g(x)| \leq \chi_{B_{R+1}}(x) \cdot 2\|g\|_\infty \in L^p.$$

Let $f \in L^p$ with $p \in [1, \infty)$. For any $g \in \mathcal{C}_c(\mathbb{R}^d)$,

$$\begin{aligned}
\|\tau_z f - f\|_p &\leq \|\tau_z f - \tau_z g\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\
&= \|f - g\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\
&= 2\,\|f - g\|_p + \|\tau_z g - g\|_p\,.
\end{aligned}$$

Since g is continuous,

$$\limsup_{z \to 0} \|\tau_z f - f\|_p \leq 2\,\|f - g\|_p + \limsup_{z \to 0} \|\tau_z g - g\|_p = 2\,\|f - g\|_p\,.$$

Functions in $\mathcal{C}_c(\mathbb{R}^d)$ are dense in L^p. The density property states that for any function $f \in L^p$ and any $\epsilon > 0$, there exists a function $g \in \mathcal{C}_c(\mathbb{R}^d)$ such that $\|f - g\|_p < \epsilon$. Using this density property, for any $\epsilon > 0$, we can find a $g \in \mathcal{C}_c(\mathbb{R}^d)$ with $\|f - g\|_p < \epsilon/2$. This means $2\|f - g\|_p < \epsilon$. Therefore,

$$\limsup_{z \to 0} \|\tau_z f - f\|_p \leq 2\|f - g\|_p < \epsilon.$$

Since ϵ is arbitrary,

$$\limsup_{z \to 0} \|\tau_z f - f\|_p = 0.$$

Since $\limsup_{z \to 0} \|\tau_z f - f\|_p \geq \liminf_{z \to 0} \|\tau_z f - f\|_p \geq 0$, we conclude $\lim_{z \to 0} \|\tau_z f - f\|_p = 0$.

Mollifiers

Let $\tilde{w}(x) := \ell(|x|_2^2)$, $x \in \mathbb{R}^d$ where $|x| := \sqrt{x_1^2 + \ldots + x_d^2}$. Then,

$$\tilde{w} \in \mathcal{C}_c^\infty(\mathbb{R}^d), \quad \text{supp}(\tilde{w}) = B_1(0).$$

$\mathcal{C}_c^\infty(\mathbb{R}^d)$ is the space of infinitely differentiable functions with compact support. Let $w(x) = c\tilde{w}(x)$, $x \in \mathbb{R}^d$, with $c > 0$ chosen so that $\int w\,dx = 1$. For $\varepsilon > 0$, set

$$w_\varepsilon(x) := \varepsilon^{-d} w\left(\frac{x}{\varepsilon}\right), \quad x \in \mathbb{R}^d.$$

Note. $w_\varepsilon \in \mathcal{C}_c^\infty(\mathbb{R}^d)$, $\text{supp}(w_\varepsilon) = B_\varepsilon(0)$ with $w_\varepsilon > 0$, and $\int w_\varepsilon(x)\,dx = 1$ for any $\varepsilon > 0$.

Remark. w is a probability density function of a random variable $\xi \in \mathbb{R}^d$, $|\xi| \leq 1$ almost surely. w_ε is a probability density function of $\varepsilon\xi$.

$$\mathbb{E}[f(\varepsilon\xi)] = \int f(\varepsilon x) w(x)\,dx, \quad y = \varepsilon x \implies \int f(y) w_\varepsilon(y)\,dy.$$

Smoothing

Let $p \in [1, \infty)$, $f \in L^p$. Then,

$$f_\varepsilon(x) = (f * w_\varepsilon)(x) = \int f(x-y) w_\varepsilon(y)\, dy = \mathbb{E}[f(x - \varepsilon\xi)], \quad x \in \mathbb{R}^d.$$

1. $f_\varepsilon \in L^p$, $\|f_\varepsilon\|_p \le \|f\|_p$, $\|f_\varepsilon - f\|_p \xrightarrow{\varepsilon \to 0} 0$.
2. $f_\varepsilon \in C_b^\infty$, where C_b^∞ is the space of all infinitely differentiable bounded functions with all derivatives bounded.
3. Let $\gamma = (\gamma_1, \ldots, \gamma_d)$, $|\gamma| = \gamma_1 + \ldots + \gamma_d$, and $D^\gamma = \frac{\partial^{|\gamma|}}{\partial x_1^{\gamma_1} \ldots \partial x_d^{\gamma_d}}$. Then,

$$D^\gamma f_\varepsilon(x) = \int D^\gamma w_\varepsilon(x-y) f(y)\, dy = (D^\gamma w_\varepsilon) * f(x), \quad x \in \mathbb{R}^d.$$

Proof.

1. By Proposition 2,

$$\|f_\varepsilon\|_p = \|f * w_\varepsilon\|_p \le \|f\|_p \cdot \|w_\varepsilon\|_1 = \|f\|_p\,.$$

   Convergence. First, since $\int w_\varepsilon(y)\, dy = 1$,

$$|(f * w_\varepsilon)(x) - f(x)| = \left| \int [f(x-y) - f(x)] w_\varepsilon(y)\, dy \right| \le \int |\tau_{-y} f(x) - f(x)| w_\varepsilon(y)\, dy.$$

   By Minkowski's inequality for integrals,

$$\|f * w_\varepsilon - f\|_p \le \left\| \int |\tau_{-y} f - f| w_\varepsilon(y)\, dy \right\|_p \le \int \|\tau_{-y} f - f\|_p w_\varepsilon(y)\, dy \le \int \sup_{|y| \le \varepsilon} \|\tau_{-y} f - f\|_p w_\varepsilon(y)\, dy.$$

   By Lemma 3,

$$\|f * w_\varepsilon - f\|_p \le \int \sup_{|y| \le \varepsilon} \|\tau_{-y} f - f\|_p w_\varepsilon(y)\, dy = \sup_{|y| \le \varepsilon} \|\tau_y f - f\|_p \xrightarrow{\varepsilon \to 0} 0,$$

$$\implies \|f_\varepsilon - f\|_p \xrightarrow{\varepsilon \to 0} 0.$$

2. Apply Theorem 2.27 and induction.
3. Note that $D^\alpha w_\varepsilon \in C_c^\infty(\mathbb{R}^d) \subset L^q$. $D^\alpha w_\varepsilon$ belongs to L^q because:
   (a) First, note that w_ε is defined as $\varepsilon^{-d} w(x/\varepsilon)$, where $w(x) = C \cdot \ell(|x|/\lambda)$, and ℓ is the standard bump function that is infinitely differentiable and has compact support.
   (b) Since w inherits the properties of ℓ, it is also infinitely differentiable with compact support, meaning $w \in C_c^\infty(\mathbb{R}^d)$.

(c) When we take any partial derivative D^α of w_ε, the result is still a smooth function with compact support. This is because differentiation preserves smoothness, and the rescaling by ε doesn't change the compact support property - it just scales it.

(d) Any function in $C_c^\infty(\mathbb{R}^d)$ automatically belongs to every L^q space for $q \in [1, \infty]$ because: (1) It's bounded since it's continuous on a compact set. (2) It has finite support (zero outside a compact set). Therefore, the integral $\int |D^\alpha w_\varepsilon(x)|^q \, dx < \infty$ for any $q \in [1, \infty)$ and $\sup_x |D^\alpha w_\varepsilon(x)| < \infty$.

By Proposition 1,

$$\sup_x |D^\alpha f_\varepsilon(x)| = \sup_x |(D^\alpha w_\varepsilon) * f(x)| \leq \|f\|_p \|D^\alpha w_\varepsilon\|_q.$$

Duals of Continuous Function Spaces

Let X be a compact metric space (CM), and $C(X)$ be the space of all continuous real-valued functions on X with finite norm:

$$|f|_u = \sup_{x \in X} |f(x)| = \max_{x \in X} |f(x)|.$$

We always consider the Borel σ-algebra $\mathcal{B}(X)$ on X. Note that X is compact if any sequence x_n in X has a converging subsequence.

Proposition 1

Let μ be a finite signed measure on X. Define:

$$\ell_\mu(f) = \int f \, d\mu, \quad f \in C(X).$$

Then, $\ell_\mu \in [C(X)]^*$ and

$$|\ell_\mu|_{[C(X)]^*} = \sup_{|f|_{C(X)} \leq 1} |\ell_\mu(f)| = \|\ell_\mu\| \leq \|\mu\| = |\mu|(X) < \infty,$$

where $|\mu| = \mu^+ + \mu^-$ is the total variation of μ.

Proof. For $f \in C(X)$,

$$|\ell_\mu(f)| = \left| \int f \, d\mu \right| \leq \int |f| \, d|\mu| \leq |f|_u |\mu|(X).$$

Hence $\ell_\mu \in [C(X)]^*$, $\|\ell_\mu\| \leq \|\mu\|$.

Tietze-Urysohn Theorem

Let (E, ρ) be a metric space, $A \subseteq E$ be closed, and $f : A \to \mathbb{R}$ be bounded and continuous. Then, f has a continuous extension $g : E \to \mathbb{R}$ such that:

$$\sup_{x \in E} g(x) = \sup_{x \in A} f(x), \quad \inf_{x \in E} g(x) = \inf_{x \in A} f(x),$$

$$g(x) = \begin{cases} f(x), & x \in A \\ \inf_{y \in A} \frac{f(y)\rho(x,y)}{\rho(x,A)}, & x \notin A \end{cases}.$$

where $\rho(x, A) = \inf_{y \in A} \rho(x, y)$. The ratio $\frac{\rho(x,A)}{\rho(x,y)} \geq 1$. Here is the reason: For a fixed x,

- When $y \in A$ is close to x, this ratio approaches 1.
- When $y \in A$ is far from x, this ratio is smaller than 1.

$$\inf_{y \in A} f(y) \leq \inf_{y \in A} \frac{f(y)\rho(x,y)}{\rho(x,A)}.$$

Corollary 0

Let $A, B \subseteq E$ be closed and disjoint subsets of a metric space (E, ρ). Then, there exists a continuous $g : E \to [0, 1]$ such that

$$g|_A = 1, \quad g|_B = 0.$$

Proof. Applying Tietze-Urysohn Theorem to the closed set $A \cup B$,

$$f(x) = \begin{cases} 1, & x \in A \\ 0, & x \in B \end{cases}.$$

Lemma 1

Let X be a compact metric space, $K \subseteq U \subseteq X$ where K is compact and U is open. Then, there exists open $V \subseteq X$ such that $K \subseteq V \subseteq \overline{V} \subseteq U$ with $\overline{V}$ compact.

Proof. For all $x \in K$, there exists an open ball V_x and closed ball N_x, both centered at x, such that

$$V_x \subseteq N_x \subseteq U, \quad K \subseteq \bigcup_{x \in K} V_x.$$

Since K is compact, every open cover of K has a finite subcover. Then, there exists $x_1, \ldots, x_m \in K$ so that

$$K \subseteq \bigcup_{j=1}^{m} V_{x_j} \subseteq \bigcup_{j=1}^{m} N_{x_j} \subseteq U.$$

Then, we set

$$V = \bigcup_{j=1}^{m} V_{x_j}.$$

The $\overline{V_{x_j}}$ is the smallest closed set containing V_{x_j}. Since N_{x_j} is closed and $V_{x_j} \subseteq N_{x_j}$, it follows that:

$$\overline{V_{x_j}} \subseteq N_{x_j}.$$

In general, the closure of a union satisfies $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Therefore, for $V = \bigcup_{j=1}^{m} V_{x_j}$,

$$\overline{V} = \overline{\bigcup_{j=1}^{m} V_{x_j}} = \bigcup_{j=1}^{m} \overline{V_{x_j}},$$

Since each $\overline{V_{x_j}} \subseteq N_{x_j}$,

$$\overline{V} \subseteq \bigcup_{j=1}^{m} N_{x_j} \implies V \subseteq \overline{V} \subseteq \bigcup_{j=1}^{m} N_{x_j} \subseteq U.$$

$\bigcup_{j=1}^{m} N_{x_j}$ is compact because each closed ball N_{x_j} is bounded and closed. The finite union of compact sets is compact. Since $\overline{V}$ is a closed subset of the compact set $\bigcup_{j=1}^{m} N_{x_j}$, it must also be compact.

Notation. Let $U \subseteq X$ be open, $f \in C(X)$.

$$f \prec U \text{ if } 0 \leq f \leq 1 \text{ and } \operatorname{supp}(f) \subseteq U.$$

Proposition 1

Let X be a compact metric space, $K \subseteq U \subseteq X$ where K is compact and U is open. Then, there exists $f \prec U$ such that $f|_K = 1$.

Proof. By Lemma 1, there exists open V such that

$$K \subseteq V \subseteq \overline{V} \subseteq U.$$

By Corollary 0, there exists $f \in C(X)$ such that

$$f|_K = 1, \quad f|_{V^c} = 0.$$

Note $\operatorname{supp}(f) \subseteq \overline{V} \subseteq U$. By Corollary 0, $0 \leq f \leq 1$ ($g : E \to [0,1]$).

Corollary 1

Let X be a compact metric space, $K \subseteq \bigcup_{j=1}^{n} U_j$ where K is compact and U_j are open for all $j = 1, \ldots, n$. Then, there exists $f_j \prec U_j$, $j = 1, \ldots, n$, such that

$$\left(\sum_{j=1}^{n} f_j\right)\bigg|_K = 1.$$

Proof. For any $x \in K$, there exists an open ball V_x and closed ball N_x, both centered at x such that:

$$V_x \subseteq N_x \subseteq U_j \text{ for some } j \in \{1, \ldots, n\}$$

Hence, $K \subseteq \bigcup_{x \in K} V_x$ and there exists $x_1, \ldots, x_m$ so that

$$K \subseteq \bigcup_{k=1}^{m} V_{x_k} \subseteq \bigcup_{k=1}^{m} N_{x_k}$$

Let K_j be the union of all N_{x_k} s.t. $N_{x_k} \subseteq U_j$. Then $K \subseteq \bigcup_{j=1}^{n} K_j$ and all K_j are compact with $K_j \subseteq U_j$.

By Proposition 1, there exists $g_j \prec U_j, g_j|_{K_j} = 1$. Let $U = \bigcup_{j=1}^n \{x : g_j(x) > 0\}$. Then, $\bigcup_{j=1}^n K_j \subseteq U$ and open since $g_j \in C(X)$. Specifically, the set $\{x : g_j(x) > 0\}$ is open as the preimage of the open set $(0, \infty)$ under the continuous function g_j.

By Proposition 1, there exists $f \prec U$ such that $f|_{\bigcup_{j=1}^n K_j} = 1$. Set:

$$g_{n+1} = 1 - f = \begin{cases} 0 & \text{on } \bigcup_{j=1}^n K_j \\ 1 & \text{on } U^c \end{cases}$$

On K, since $K \subseteq \bigcup_{j=1}^n K_j$, for any $x \in K$, at least one $g_j(x) = 1$, so $f(x) \geq 1$. On U^c, all $g_j = 0$, so $f = 0$. Therefore, on $\bigcup_{j=1}^n K_j$, $f = 1$, so $g_{n+1} = 0$.

$$f_j = \frac{g_j}{\sum_{i=1}^{n+1} g_i}, \quad j = 1, \ldots, n$$

Since $g_{n+1} \geq 0$, we must have $g_{n+1} = 0$ on $\bigcup_{j=1}^n K_j$. On U^c, $f = 0$, so $g_{n+1} = 1$.

Since $g_{n+1} = 0$ on K,

$$\sum_{i=1}^{n+1} g_i = \sum_{j=1}^{n} g_j \text{ on } K \implies f_j = \frac{g_j}{\sum_{k=1}^n g_k} \text{ on } K \implies \sum_{j=1}^{n} f_j = \frac{\sum_{j=1}^n g_j}{\sum_{k=1}^n g_k} = 1 \text{ on } K.$$

$$\implies \left(\sum_{j=1}^{n} f_j\right)\Bigg|_K = 1.$$

Since $g_j \prec U_j$, $\operatorname{supp}(g_j) \subseteq U_j$. The denominator $\sum_{i=1}^{n+1} g_i$ is positive everywhere, so dividing by it does not change the support. Thus, $\operatorname{supp}(f_j) = \operatorname{supp}(g_j) \subseteq U_j$, meaning $f_j \prec U_j$.

Representations of Positive Linear Functionals

A linear functional $\ell : C(X) \to \mathbb{R}$ is **positive** if $\ell(f) \geq 0$ if $f \geq 0$.

Remark. If $f \geq g$, then $\ell(f) \geq \ell(g)$ for positive ℓ. In fact, ℓ is positive iff this holds.

Proposition 2

If ℓ is positive, then $\ell \in [C(X)]^*$.

Proof. Let $f \in C(X)$. Then,

$$-|f|_u \leq f \leq |f|_u \implies -|f|_u \ell(1) \leq \ell(f) \leq |f|_u \ell(1) \implies |\ell(f)| \leq \ell(1)|f|_u.$$

Definition

Let μ be a Borel measure on a metric space X. Let $A \in \mathcal{B}$ (Borel σ-algebra). We say μ is:

(i) **Outer regular** on A if

$$\mu(A) = \inf\{\mu(U) : U \supseteq A, U \text{ is open}\}$$

(ii) **Inner regular** on A if

$$\mu(A) = \sup\{\mu(K) : K \subseteq A, K \text{ is compact}\}$$

(iii) **Regular** if μ is both inner and outer regular for all $E \in \mathcal{B}$.

Theorem 1

Let $\ell : C(X) \to \mathbb{R}$ be a positive linear functional. Then, there exists a unique finite Borel regular measure μ on X such that:

$$\ell(f) = \int f d\mu, \quad f \in C(X).$$

(i) For open $U \subseteq X$,

$$\mu(U) = \sup\{\ell(f) : f \prec U\} \leq \ell(1) = \mu(X)$$
$$\ell(1) = \int_X 1\, d\mu = \mu(X).$$

(ii) For compact $K \subseteq X$,

$$\mu(K) = \inf\{\ell(f) : f \geq \chi_K, f \in C(X)\}.$$

Note. For open $U \subseteq X$,

$$\chi_U(x) = \sup\{f(x) : f \prec U\}, \quad x \in X.$$

Comment. (i) implies (ii): if K is compact,

$$\begin{aligned}
\mu(K) &= \mu(X) - \mu(K^c) \\
&= \ell(1) - \sup\{\ell(f) : f \prec K^c\} \\
&= \ell(1) - \sup\{\ell(f) : 0 \leq f \leq 1, \operatorname{supp}(f) \subseteq K^c\} \\
&= \inf\{\ell(1 - f) : 0 \leq f \leq 1, \operatorname{supp}(f) \subseteq K^c\} \\
&\geq \inf\{\ell(g) : g \geq \chi_K, g \in C(X)\}.
\end{aligned}$$

For f with $\operatorname{supp}(f) \subseteq K^c$, f is zero on K since K and $\operatorname{supp}(f)$ are disjoint. Thus, $1 - f \geq 1$ on K because $f = 0$ on K, and $1 - f \geq 0$ everywhere since $f \leq 1$. Therefore, $1 - f \geq \chi_K$, where χ_K is the indicator function of K.

Proof. *Uniqueness.* Let μ be a finite regular measure and $\ell(f) = \int f d\mu$. Let $U \subseteq X$ be open. By the note above,

$$\mu(U) = \int_U 1\, d\mu = \sup\{\ell(f) : f \prec U\}.$$

Let $K \subseteq U$ be compact. By Proposition 1, there exists $f \prec U$ such that $f|_K = 1$. Then,

$$\mu(K) \leq \int_X f d\mu = \ell(f)$$

$$\implies \mu(U) = \sup\{\mu(K') : K' \subseteq U, K' \text{ compact}\} \leq \sup\{\ell(f) : f \prec U\}$$
$$\implies \mu(U) = \sup\{\ell(f) : f \prec U\} \text{ which is (i)}$$

Why does it imply uniqueness? Suppose there were two measures μ_1 and μ_2 representing ℓ:

$$\ell(f) = \int f\, d\mu_1 = \int f\, d\mu_2 \quad \forall f \in C(X).$$

For any open set U, both $\mu_1(U)$ and $\mu_2(U)$ must satisfy:

$$\mu_i(U) = \sup\{\ell(f) : f \prec U\}, \quad i = 1, 2.$$

Since the right-hand side depends only on ℓ, we must have:

$$\mu_1(U) = \mu_2(U).$$

Thus, μ_1 and μ_2 agree on all open sets. The measure μ is outer regular, meaning for any Borel set $E \subseteq X$,

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}.$$

Since μ_1 and μ_2 agree on open sets, they must assign the same measure to any Borel set E:

$$\mu_1(E) = \inf\{\mu_1(U) : U \supseteq E\} = \inf\{\mu_2(U) : U \supseteq E\} = \mu_2(E).$$

Hence, $\mu_1 = \mu_2$ on the entire Borel σ-algebra. In short, open sets generate the topology of X, and their measures are uniquely specified by ℓ. Since μ is outer regular, its values on arbitrary sets are determined by its values on open sets.

Existence.

$$\mu(U) := \sup\{\ell(f) : f \prec U\} \text{ for } U \subseteq X \text{ open.}$$
$$\mu^*(E) := \inf\{\mu(U) : U \supseteq E, U \text{ open}\} \text{ for } E \subseteq X.$$

Claim 1. μ^* is an outer measure.

An outer measure must satisfy:

1. $\mu^*(\emptyset) = 0$.
2. $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$.
3. $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$.

$\mu^*(\emptyset) = 0$ because $\emptyset$ is covered by itself, and $\mu(\emptyset) = 0$. If $A \subseteq B$, any open cover of B also covers A, so the infimum for A is over a larger set of covers, so $\mu^*(A) \leq \mu^*(B)$. Thus, it is enough to prove that

$$\mu^*\left(\bigcup_{n=1}^{\infty} U_n\right) \leq \sum_{n=1}^{\infty} \mu^*(U_n) \text{ for } U_n \text{ open.}$$

The key observation is that open sets can be used to approximate arbitrary sets when defining the outer measure. There exists $f \prec \bigcup_{n=1}^{\infty} U_n = U$ such that $\ell(f) > \mu\left(\bigcup_{n=1}^{\infty} U_n\right) - \epsilon$ for a given $\epsilon > 0$ since $\mu(U) = \sup\{\ell(f) : f \prec U\}$. Since every open cover of a compact set has a finite subcover, there exists N such that $\operatorname{supp}(f) \subseteq \bigcup_{n=1}^{N} U_n$. By Corollary 1, there exists $f_n \prec U_n$ such that

$$\left(\sum_{n=1}^{N} f_n\right)\Bigg|_{\operatorname{supp}(f)} = 1, \quad f = \sum_{n=1}^{N} (f_n f),$$

$$\mu\left(\bigcup_{n=1}^{\infty} U_n\right) \leq \ell(f) + \epsilon = \left[\sum_{n=1}^{N} \ell(f f_n)\right] + \epsilon \leq \left[\sum_{n=1}^{N} \mu(U_n)\right] + \epsilon.$$

Hence μ^* is an outer measure.

Claim 2. μ has a finite regular Borel extension.

By Caratheodory's theorem, μ^* is a measure on the σ-algebra $\mathcal{F}$ of all $A \subseteq X$ such that

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A) \text{ for all } E \subseteq X \tag{2}$$

We now show $\mathcal{F}$ contains all the open sets of X, i.e., $U \in \mathcal{F}$ if $U \subseteq X$ is open. If this is the case, $\mathcal{F}$ contains $\mathcal{B}$ and μ can be extended to a measure on $\mathcal{B}$.

Let $U \subseteq X$ be open. Equation 2 holds if E is open. In this case, $E \cap U$ is open so there exists $f \prec E \cap U$ such that $\ell(f) > \mu(E \cap U) - \frac{\epsilon}{2}$ where $\epsilon > 0$. There exists $g \prec E \setminus \operatorname{supp}(f)$ such that

$$\ell(g) > \mu(E \setminus \operatorname{supp}(f)) - \frac{\epsilon}{2}.$$

Then, $f + g \prec E$ since $\operatorname{supp}(f), \operatorname{supp}(g) \subseteq E$. Then,

$$\begin{aligned}\mu(E) \geq \ell(f+g) &= \ell(f) + \ell(g) \\ &> \mu(E \cap U) - \frac{\epsilon}{2} + \mu(E \setminus \operatorname{supp}(f)) - \frac{\epsilon}{2} \\ &\geq \mu(E \cap U) + \mu^*(E \setminus U) - \epsilon.\end{aligned}$$

Now, we prove Equation 2 holds for arbitrary $E \subseteq X$. For all $\epsilon > 0$, there exists open $\tilde{E} \supset E$ such that

$$\mu^*(E) \geq \mu(\tilde{E}) - \epsilon$$

$$\implies \mu^*(E) \geq \mu(\tilde{E}) - \epsilon \geq \mu^*(\tilde{E} \cap U) + \mu^*(\tilde{E} \setminus U) - \epsilon \geq \mu^*(E \cap U) + \mu^*(E \setminus U) - \epsilon.$$

Since $\mu(U) \leq \ell(1) < \infty$ for all open $U \subseteq X$, μ is finite.

We define for any subset $E \subseteq X$ by:

$$\mu^*(E) = \inf\{\mu(U) : U \supseteq E,\ U \text{ open}\}.$$

This ensures μ^* is outer regular. Why Outer Regularity on the Borel σ-Algebra Implies Inner Regularity (in a Compact Space)? Let $A \in \mathcal{B}$ be a Borel set. Since μ is outer regular, for the complement A^c, there exists an open set $U \supseteq A^c$ such that:

$$\mu(U) \leq \mu(A^c) + \epsilon \text{ for any } \epsilon > 0.$$

Since $U \supseteq A^c$, its complement $K = U^c$ satisfies $K \subseteq A$. In a compact space, closed subsets are compact, so K is compact. From $\mu(X) = \mu(A) + \mu(A^c)$ and $\mu(U) \leq \mu(A^c) + \epsilon$,

$$\mu(K) = \mu(U^c) = \mu(X) - \mu(U) \geq \mu(X) - (\mu(A^c) + \epsilon) = \mu(A) - \epsilon.$$

Since $K \subseteq A$, this shows:

$$\mu(A) \leq \mu(K) + \epsilon \leq \sup\{\mu(K') : K' \subseteq A,\ K' \text{ compact}\} + \epsilon.$$

Taking $\epsilon \to 0$ gives:

$$\mu(A) \leq \sup\{\mu(K) : K \subseteq A,\ K \text{ compact}\}.$$

The reverse inequality ($\geq$) is trivial since $\mu(K) \leq \mu(A)$ for $K \subseteq A$. Thus,

$$\mu(A) = \sup\{\mu(K) : K \subseteq A,\ K \text{ compact}\}.$$

We conclude μ is inner regular. Since μ is both outer and inner regular, μ is regular.

Claim 3. $\int f d\mu = \ell(f), f \in C(X)$.

Let $0 \leq f \leq 1, f \in C(X)$. Let $N > 1$. Then,

$$\begin{aligned} f &= \sum_{n=1}^{N} f\chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} \\ &= \sum_{n=1}^{N} \left[\left(f - \frac{n-1}{N}\right)\chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} + \frac{n-1}{N}\chi_{\frac{n-1}{N} \leq f < \frac{n}{N}}\right] \end{aligned}$$

$$\implies f = \sum_{n=1}^{N} f\chi_{\frac{n-1}{N} \leq f < \frac{n}{N}} = \sum_{n=1}^{N} \left[\left(f - \frac{n-1}{N}\right)\chi_{\frac{n-1}{N} \leq f \leq \frac{n}{N}} + \frac{1}{N}\chi_{\frac{n}{N} \leq f}\right] = \sum_{n=1}^{N} f_n$$

where $f_n \in C(X)$.

$$\begin{aligned} &\sum_{n=1}^{N} \frac{n-1}{N}\left(\chi_{\frac{n-1}{N} \leq f} - \chi_{\frac{n}{N} \leq f}\right) \\ &= \frac{1}{N}\sum_{n=1}^{N}(n-1)\left(\chi_{\frac{n-1}{N} \leq f} - \chi_{\frac{n}{N} \leq f}\right) \\ &= \frac{1}{N}\left[1 \cdot \left(\chi_{\frac{1}{N} \leq f} - \chi_{\frac{2}{N} \leq f}\right) + 2 \cdot \left(\chi_{\frac{2}{N} \leq f} - \chi_{\frac{3}{N} \leq f}\right) + \ldots + (N-1) \cdot \left(\chi_{\frac{N-1}{N} \leq f} - \chi_{\frac{N}{N} \leq f}\right)\right] \\ &= \frac{1}{N}\left[\chi_{\frac{1}{N} \leq f} - \chi_{\frac{2}{N} \leq f} + 2\chi_{\frac{2}{N} \leq f} - 2\chi_{\frac{3}{N} \leq f} + \ldots - (N-1)\chi_{\frac{N}{N} \leq f}\right] \\ &= \frac{1}{N}\left[\chi_{\frac{1}{N} \leq f} + \chi_{\frac{2}{N} \leq f} + \chi_{\frac{3}{N} \leq f} + \ldots + \chi_{\frac{N-1}{N} \leq f} - (N-1) \cdot \chi_{\frac{N}{N} \leq f}\right]. \end{aligned}$$

Since $(N-1)\cdot\chi_{\frac{N}{N}\le f}=0$,

$$\sum_{n=1}^{N}\frac{n-1}{N}\chi_{\frac{n-1}{N}\le f<\frac{n}{N}}=\sum_{n=1}^{N}\frac{n-1}{N}\left(\chi_{\frac{n-1}{N}\le f}-\chi_{\frac{n}{N}\le f}\right)=\frac{1}{N}\sum_{n=1}^{N}\chi_{\frac{n}{N}\le f}.$$

For each n,

$$f_n=\left(f-\frac{n-1}{N}\right)\chi_{\frac{n-1}{N}\le f\le\frac{n}{N}}+\frac{1}{N}\chi_{\frac{n}{N}\le f}\implies\frac{1}{N}\chi_{\frac{n}{N}\le f}\le f_n\le\frac{1}{N}\chi_{\frac{n-1}{N}\le f}$$

By (ii),

$$\mu\left(\frac{n}{N}\le f\right)\le N\ell(f_n)\le\mu\left(\frac{n-1}{N}\le f\right)$$

$$\implies\frac{1}{N}\sum_{n=1}^{N}\mu\left(\frac{n}{N}\le f\right)\le\sum_{n=1}^{N}\ell(f_n)\le\frac{1}{N}\sum_{n=1}^{N}\mu\left(\frac{n-1}{N}\le f\right)$$

Focus on the right-hand side (RHS):

$$\ell(f)\le\sum_{n=1}^{N}\frac{1}{N}\mu\left(\frac{n-1}{N}\le f\right).$$

We can express $\mu\left(\frac{n-1}{N}\le f\right)$ as:

$$\mu\left(\frac{n-1}{N}\le f\right)=\sum_{k=n}^{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right)+\mu(f\ge 1).$$

Since $f\le 1$,

$$\mu\left(\frac{n-1}{N}\le f\right)=\sum_{k=n}^{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right).$$

Substitute back into the RHS:

$$\ell(f)\le\sum_{n=1}^{N}\frac{1}{N}\sum_{k=n}^{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right).$$

Interchange the order of summation:

$$\sum_{n=1}^{N}\sum_{k=n}^{N}\frac{1}{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right)=\sum_{k=1}^{N}\sum_{n=1}^{k}\frac{1}{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right),$$

$$\implies\sum_{n=1}^{k}\frac{1}{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right)=\frac{k}{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right),$$

$$\implies\ell(f)\le\sum_{k=1}^{N}\frac{k}{N}\mu\left(\frac{k-1}{N}\le f<\frac{k}{N}\right),$$

Write $\frac{k}{N} = \frac{k-1}{N} + \frac{1}{N}$:

$$\ell(f) \leq \sum_{k=1}^{N} \left(\frac{k-1}{N} + \frac{1}{N} \right) \mu \left(\frac{k-1}{N} \leq f < \frac{k}{N} \right),$$

$$\implies \ell(f) \leq \sum_{k=1}^{N} \frac{k-1}{N} \mu \left(\frac{k-1}{N} \leq f < \frac{k}{N} \right) + \frac{1}{N} \sum_{k=1}^{N} \mu \left(\frac{k-1}{N} \leq f < \frac{k}{N} \right),$$

The second sum is:

$$\frac{1}{N} \sum_{k=1}^{N} \mu \left(\frac{k-1}{N} \leq f < \frac{k}{N} \right) = \frac{1}{N} \mu(f > 0).$$

because the sets $\frac{k-1}{N} \leq f < \frac{k}{N}$ partition $\{f > 0\}$. Then,

$$\sum_{n=1}^{N} \frac{n-1}{N} \mu \left(\frac{n-1}{N} \leq f < \frac{n}{N} \right) \leq \ell(f) \leq \sum_{n=1}^{N} \frac{n-1}{N} \mu \left(\frac{n-1}{N} \leq f < \frac{n}{N} \right) + \frac{1}{N} \mu(f > 0)$$

The left-hand side is the lower Riemann sum and the right-hand side is the upper Riemann sum. As we integrate (Riemann sum) and let $N \to \infty$,

$$\int f d\mu = \ell(f) = \int f d\mu$$

Therefore, $\int f d\mu = \ell(f)$ for all $f \in C(X)$.

Corollary 2

Let X be a compact metric space. Then, any finite Borel measure γ is regular.

Proof. Let $\ell(f) = \int f \, d\gamma$, $f \in C(X)$. Then, ℓ is positive. By Theorem 1, there exists a regular finite μ so that:

$$\ell(f) = \int f \, d\mu = \int f \, d\gamma \text{ for any } f \in C(X).$$

Let $K \subset X$ be compact and define:

$$F_n = \{x \in X : \rho(x, K) \geq \frac{1}{n}\} \text{ for all } n \geq 1.$$

We show that the complement $F_n^c = \{x \in X : \rho(x, K) < 1/n\}$ is open: Take any $x \in F_n^c$, meaning $\rho(x, K) = d < 1/n$. By definition of infimum, there exists $y \in K$ such that $\rho(x, y) < (1/n + d)/2$. Now, consider the open ball $B(x, r)$ where $r = (1/n - d)/2 > 0$. For any $z \in B(x, r)$, by the triangle inequality:

$$\rho(z, K) \leq \rho(z, y) \leq \rho(z, x) + \rho(x, y) < r + \frac{1/n + d}{2} = \frac{1/n - d}{2} + \frac{1/n + d}{2} = 1/n.$$

Thus, $\rho(z, K) < 1/n$, so $z \in F_n^c$. Since every $x \in F_n^c$ has an open neighborhood entirely contained in F_n^c, F_n^c is open, and hence F_n is closed.

By Corollary 0, there exists continuous $f_n : X \to [0,1]$ such that $f_n|_K = 1$, $f_n|_{F_n} = 0$. Note that $f_n(x) \to \chi_K(x)$ for all $x \in X$ and:

$$\int f_n \, d\mu = \int f_n \, d\gamma \implies \mu(K) = \gamma(K)$$

Therefore, γ is regular, and in fact $\gamma = \mu$, since μ is regular. For open U,

$$\gamma(U) = \gamma(X) - \gamma(U^c) = \mu(X) - \mu(U^c) = \mu(U)$$

Let $A \in \mathcal{B}$, $\epsilon > 0$. Since μ is regular, there exists $U_\epsilon \supseteq A \supseteq K_\epsilon$ so that:

$$\mu(U_\epsilon \setminus K_\epsilon) < \epsilon \implies \gamma(U_\epsilon) \leq \gamma(K_\epsilon) + \epsilon \leq \gamma(A) + \epsilon.$$

Hence, γ is outer regular on A. Similarly,

$$\mu(K_\epsilon) \geq \mu(A) - \epsilon \implies \gamma(K_\epsilon) \geq \gamma(A) - \epsilon.$$

Thus, γ is inner regular. Then, γ is regular and $\gamma = \mu$ since γ and μ are equal on compact and open sets.

Lemma 2

Let $\ell \in [C(X)]^*$. Then, $\ell = \ell_1 - \ell_2$ with ℓ_1, ℓ_2 positive.

Proof. Define $P = \{f \in C(X) : f \geq 0\}$. For any $f \in P$,

$$\ell_1(f) := \sup\{\ell(g) : 0 \leq g \leq f, g \in C(X).\}.$$

Note that $\ell_1(f) \geq 0$ since we can take $g = 0$ in the supremum. We first show that ℓ_1 is linear on P. For any $f, f' \in P$ and $c \geq 0$, we need to verify:

1. *Additivity.* $\ell_1(f + f') = \ell_1(f) + \ell_1(f')$.

   For any $0 \leq g \leq f$ and $0 \leq g' \leq f'$, we have $0 \leq g + g' \leq f + f'$, and thus:

   $$\ell(g) + \ell(g') = \ell(g + g') \leq \ell_1(f + f').$$

   Taking suprema over g and g' gives $\ell_1(f) + \ell_1(f') \leq \ell_1(f + f')$. For the reverse inequality, given any $0 \leq h \leq f + f'$, we can write $h = u + u'$ where $0 \leq u \leq f$ and $0 \leq u' \leq f'$ by taking $u = \min(h, f)$ and $u' = h - u$. Then,

   $$\ell(h) = \ell(u) + \ell(u') \leq \ell_1(f) + \ell_1(f').$$

   Taking supremum over h gives $\ell_1(f + f') \leq \ell_1(f) + \ell_1(f')$.

2. *Homogeneity.* $\ell_1(cf) = c\ell_1(f)$.

   $$\ell_1(cf) = \sup\{\ell(g) : 0 \leq g \leq cf\} = \sup\{\ell(ch) : 0 \leq h \leq f\} = c \sup\{\ell(h) : 0 \leq h \leq f\}.$$

   $$\implies \ell_1(cf) = c\ell_1(f)$$

We now extend ℓ_1 to all of $C(X)$. Any $f \in C(X)$ can be written as $f = f_1 - f_2$ where $f_1, f_2 \in P$, using positive and negative parts. Define:

$$\ell_1(f) := \ell_1(f_1) - \ell_1(f_2).$$

If $f = f_1 - f_2 = f_1' - f_2'$, then $f_1 + f_2' = f_1' + f_2$, so by additivity on P:

$$\ell_1(f_1) + \ell_1(f_2') = \ell_1(f_1') + \ell_1(f_2).$$

$$\implies \ell_1(f_1) - \ell_1(f_2) = \ell_1(f_1') - \ell_1(f_2')$$

Thus, $\ell_1(f) = \ell_1(f_1) - \ell_1(f_2)$ is well-defined. Finally, define $\ell_2 := \ell_1 - \ell$. To see that ℓ_2 is positive, note that for any $f \geq 0$:

$$\ell_2(f) = \ell_1(f) - \ell(f) = \sup\{\ell(g) : 0 \leq g \leq f\} - \ell(f) \geq \ell(f) - \ell(f) = 0.$$

Thus, ℓ_1 and ℓ_2 are both positive linear functionals with $\ell = \ell_1 - \ell_2$.

Proposition 4

Let X be a compact metric space, μ be a finite regular Borel measure, $p \in [1, \infty)$. Then, $C(X)$ is dense in L^p.

Proof. $\Sigma = \mathcal{E}$ as $\mu(X) < \infty$. Since $\Sigma = \mathcal{E}$ is dense in L^p, it is enough to approximate χ_E in L^p by continuous functions. Let $E \in \mathcal{B}$, $\epsilon > 0$. Since μ is regular, for any Borel set E and $\epsilon > 0$, there exist a compact set $K_\epsilon \subseteq E$ and an open set $U_\epsilon \supseteq E$, such that:

$$\mu(U_\epsilon) < \mu(E) + \frac{\epsilon}{2}, \quad \mu(K_\epsilon) > \mu(E) - \frac{\epsilon}{2}.$$

Subtracting these inequalities gives:

$$\mu(U_\epsilon \setminus K_\epsilon) = \mu(U_\epsilon) - \mu(K_\epsilon) < \epsilon.$$

Since $K_\epsilon \subseteq E \subseteq U_\epsilon$, by Corollary 0, we can construct a continuous function f such that: $f = 1$ on K_ϵ, $f = 0$ outside U_ϵ, and $0 \leq f \leq 1$ everywhere. The difference $|\chi_E - f|$ is nonzero only on $U_\epsilon \setminus K_\epsilon$, because:

- On K_ϵ, $\chi_E = 1$ and $f = 1$, so $|\chi_E - f| = 0$.
- Outside U_ϵ, $\chi_E = 0$ and $f = 0$, so $|\chi_E - f| = 0$.
- On $U_\epsilon \setminus K_\epsilon$, $|\chi_E - f| \leq 1$ since $0 \leq f \leq 1$ and $\chi_E \in \{0, 1\}$.

Therefore:

$$\int_X |\chi_E - f|^p \, d\mu \leq \int_{U_\epsilon \setminus K_\epsilon} 1 \, d\mu = \mu(U_\epsilon \setminus K_\epsilon) < \epsilon.$$

Since simple functions are dense in L^p, and we can approximate any simple function by continuous functions, we can approximate any L^p-function by continuous functions. Thus, $C(X)$ is dense in $L^p(X, \mu)$.

Lusin's theorem

Let X be a compact metric space and μ be finite. Let f be a bounded measurable function on X ($|f|_\infty \leq C$). Then, for all $\epsilon > 0$, there exists $g \in C(X)$ such that $\mu(\{f \neq g\}) \leq \epsilon$. Moreover, $|g|_\infty \leq \sup_x |f(x)|$, and $0 \leq g \leq \sup_x |f(x)|$ if $f \geq 0$.

Proof. By Proposition 4, since $C(X)$ is dense in $L^1(X, \mu)$, there exists a sequence $\{f_n\} \subset C(X)$ such that $f_n \to f$ in L^1, and hence $f_n \to f$ almost everywhere. Since $f_n \to f$ almost everywhere and $\mu(X) < \infty$, Egoroff's Theorem guarantees that for any $\epsilon > 0$, there exists a measurable set $A \subset X$ such that:

$$\mu(X \setminus A) \leq \frac{\epsilon}{2},$$

and $f_n \to f$ uniformly on A. Since μ is regular from Corollary 2, for the set A, there exists a compact subset $K \subseteq A$ such that:

$$\mu(A \setminus K) \leq \frac{\epsilon}{2}.$$

$$\implies \mu(X \setminus K) = \mu(X \setminus A) + \mu(A \setminus K) \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $f_n \to f$ uniformly on A, and $K \subseteq A$, the restriction $f|_K$ is continuous as a uniform limit of continuous functions. By the Tietze-Urysohn Theorem, since K is closed in X and X is a compact metric space, there exists a continuous function $g \in C(X)$ such that:

$$g|_K = f|_K, \quad \sup_X g = \sup_K f, \quad \inf_X g = \inf_K f.$$

By construction, $f = g$ on K and $\mu(X \setminus K) \leq \epsilon$. Therefore:

$$\{x \in X : f(x) \neq g(x)\} \subseteq X \setminus K, \implies \mu(\{f \neq g\}) \leq \mu(X \setminus K) \leq \epsilon.$$

Since g is an extension of $f|_K$, and f is bounded by C,

$$|g(x)| \leq \sup_K |f| \leq \sup_X |f| = C.$$

If $f \geq 0$, then $g \geq 0$ since $\inf_X g \geq \inf_K f \geq 0$.

Proposition 5

Let X be a compact metric space, μ be a finite signed measure. Let $\ell_\mu(f) = \int f d\mu$, $f \in C(X)$. Then, $\ell_\mu \in [C(X)]^*$ and $\|\ell_\mu\| = |\ell_\mu|_{[C(X)]^*} = |\mu|(X) = \|\mu\|$.

Proof. By Proposition 1, since μ is a finite signed measure, $\ell_\mu \in [C(X)]^*$ and $\|\ell_\mu\| \leq \|\mu\|$. Let D^+, D^- be the disjoint sets of Hahn's decomposition such that $d|\mu| = \chi_{D^+} d\mu - \chi_{D^-} d\mu$ and $d\mu = (\chi_{D^+} - \chi_{D^-}) d|\mu|$. Note that $|\rho| = \rho^2 = 1$ where $\rho = \chi_{D^+} - \chi_{D^-}$. Let $\epsilon > 0$. By

Lusin's theorem, there exists $g \in C(X)$ such that $|\mu|(\{\rho \neq g\}) \leq \epsilon$ and $|g|_\infty \leq 1$. Then:

$$\begin{aligned}
\ell_\mu(g) &= \int g\, d\mu \\
&= \int g\rho\, d|\mu| \\
&= \int_{g=\rho} g\rho\, d|\mu| + \int_{g\neq\rho} g\rho\, d|\mu| \\
&= \int_X \rho^2\, d|\mu| - \int_{g\neq\rho} \rho^2\, d|\mu| + \int_{g\neq\rho} g\rho\, d|\mu| \\
&\geq |\mu|(X) - \epsilon - \epsilon \\
&= |\mu|(X) - 2\epsilon.
\end{aligned}$$

Since:

$$0 \leq \int_{g\neq\rho} \rho^2\, d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon,$$

$$\left|\int_{g\neq\rho} g\rho\, d|\mu|\right| \leq \int_{g\neq\rho} |g\rho|\, d|\mu| \leq \int_{g\neq\rho} d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon \implies \int_{g\neq\rho} g\rho\, d|\mu| \geq -\epsilon.$$

Therefore,

$$\|\mu\| \geq |\ell_\mu(g)| \geq \ell_\mu(g) \geq |\mu|(X) - 2\epsilon$$

Taking $\epsilon \to 0$, we have $\|\ell_\mu\| = \|\mu\|$.

Notation. Let $M(x)$ be the space of finite signed measures with $\|\mu\| = |\mu|(X)$ (total variation norm).

Theorem 3

Let X be a compact metric space. Then, for all $\ell \in [C(X)]^*$, there exists a unique $\mu \in M(X)$ such that $\ell(f) = \int f d\mu, \quad f \in C(X)$. Moreover, $\|\ell\| = \|\mu\|$.

Proof. Let $\ell \in [C(X)]^*$. By Lemma 2, $\ell = \ell_1 - \ell_2$ with ℓ_1, ℓ_2 positive. By Theorem 1, there exists finite regular measures μ_1, μ_2 such that for all $f \in C(X)$:

$$\ell_1(f) = \int f\, d\mu_1, \quad \ell_2(f) = \int f\, d\mu_2$$

$$\implies \ell(f) = \int f\, d(\mu_1 - \mu_2) = \int f\, d\mu, \quad f \in C(X)$$

Then, we could prove μ is a finite regular signed measure and is unique following a similar argument as Corollary 2 proof. Moreover, by Proposition 5, $\|\ell\| = \|\mu\|$.

Schwartz Space of Functions with Rapid Decrease

From Fourier Series to Fourier Transform

1. Let $T > 0$. Then, $\{\frac{1}{\sqrt{2\pi}} e^{\frac{in\pi}{T}x},\, n = 0, \pm 1, \ldots\}$ is $2T$-periodic.
   This forms a CONS in $L^2([-T, T])$ (complex-valued functions).

2. If $h \in L^2([-T,T])$, then

$$h(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x} \text{ in } L^2([-T,T]) \text{ with } c_n = \frac{1}{2T}\int_{-T}^{T} h(x)e^{-\frac{in\pi}{T}x}\,dx.$$

Also, $\displaystyle\int_{-T}^{T} |h(x)|^2\,dx = 2T \sum_{n=-\infty}^{\infty} |c_n|^2.$

Proof. First, let's compute the L^2 norm squared of h:

$$\int_{-T}^{T} |h(x)|^2\,dx = \int_{-T}^{T} h(x)\overline{h(x)}\,dx$$

Substitute the Fourier series for $h(x)$ and $\overline{h(x)}$:

$$\int_{-T}^{T} \left(\sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x}\right)\left(\sum_{m=-\infty}^{\infty} \overline{c_m} e^{-\frac{im\pi}{T}x}\right)\,dx$$

Rearranging, we get:

$$\int_{-T}^{T} \sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} c_n\overline{c_m} e^{\frac{i(n-m)\pi}{T}x}\,dx$$

By Fubini's theorem,

$$\sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} c_n\overline{c_m}\int_{-T}^{T} e^{\frac{i(n-m)\pi}{T}x}\,dx$$

The exponential functions form an orthogonal system. When $n \neq m$,

$$\int_{-T}^{T} e^{\frac{i(n-m)\pi}{T}x}\,dx = \left[\frac{T}{i\pi(n-m)}e^{\frac{i(n-m)\pi}{T}x}\right]_{-T}^{T} = 0$$

When $n = m$,

$$\int_{-T}^{T} e^0\,dx = \int_{-T}^{T} 1\,dx = 2T$$

Therefore,

$$\sum_{n=-\infty}^{\infty} c_n\overline{c_n}\cdot 2T = 2T\sum_{n=-\infty}^{\infty} |c_n|^2$$

3. Let $f \in C_c^\infty(\mathbb{R})$, $f(x) = 0$ if $|x| > R > 0$. Then, for $T > R+1$, $|x| < T$,

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T}\int_{-T}^{T} f(y)e^{-\frac{in\pi}{T}y}\,dy\,e^{\frac{in\pi}{T}x} = \sum_{n=-\infty}^{\infty} \frac{1}{2T}\int_{-\infty}^{\infty} f(y)e^{-i\frac{n}{2T}2\pi y}\,dy\,e^{i\frac{n}{2T}2\pi x}.$$

Let $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(y)e^{-i2\pi\xi y}\,dy,\ \xi \in \mathbb{R}$. Note that $\hat{f} \in C_c^\infty(\mathbb{R})$. Hence,

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T}\hat{f}\left(\frac{n}{2T}\right)e^{i\frac{n}{2T}2\pi x}, \quad |x| < T, \text{ for all } T > R+1$$

$$\int_{-\infty}^{\infty} |f(x)|^2 \, dx = \int_{-T}^{T} |f(x)|^2 \, dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \text{ for } T > R+1.$$

For any $x \in \mathbb{R}$,

$$f(x) = \lim_{T\to\infty} \sum_{n=-\infty}^{\infty} \frac{1}{2T} \hat{f}\left(\frac{n}{2T}\right) e^{i2\pi \frac{n}{2T} x} = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi\xi x} \, d\xi.$$

Also,

$$\int_{-\infty}^{\infty} |f(x)|^2 \, dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 \, d\xi \text{ for all } T > R+1.$$

For finite $T > R+1$, we established:

$$\int_{-T}^{T} |f(x)|^2 \, dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2$$

Since f has compact support within $[-R, R]$, when $T > R+1$,

$$\int_{-\infty}^{\infty} |f(x)|^2 \, dx = \int_{-T}^{T} |f(x)|^2 \, dx$$

$$\sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \cdot \frac{1}{2T} = \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi$$

This is precisely the form of a Riemann sum for the function $|\hat{f}(\xi)|^2$ over the entire real line:

$$\sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi$$

As $T \to \infty$, the spacing $\Delta\xi = \frac{1}{2T} \to 0$, making the partition finer. Under appropriate conditions (which $\hat{f}$ satisfies since it comes from a compactly supported $f \in C_c^\infty(\mathbb{R})$), the Riemann sum converges to the integral:

$$\lim_{T\to\infty} \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 \, d\xi$$

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi\xi x} \, d\xi, \quad x \in \mathbb{R} \text{ and } \int_{-\infty}^{\infty} |f(x)|^2 \, dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 \, d\xi$$

Fourier transform of $f \in C_c^\infty(\mathbb{R})$:

$$\hat{f}(\xi) = (\mathcal{F}f)(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi\xi x} \, dx, \quad \xi \in \mathbb{R}$$

The inverse Fourier transform of $g \in C_c^\infty(\mathbb{R})$:

$$(\mathcal{F}^{-1}g)(x) = \int_{-\infty}^{\infty} g(\xi)e^{i2\pi\xi x}\, d\xi, \quad x \in \mathbb{R}.$$

We also have:

$$\mathcal{F}^{-1}\mathcal{F}f = f.$$

Note that $(\mathcal{F}^{-1}f)(x) = \hat{f}(-x)$, for any x. Also, $\mathcal{F}$ is "unitary". It turns out $F(C_c^\infty(\mathbb{R})) \subseteq S(\mathbb{R})$, the space of functions of rapid decrease. Moreover, $F : S(\mathbb{R}) \to S(\mathbb{R})$ is a bijection.

Space of functions of Rapid Decrease

$$S(\mathbb{R}) = \{f \in C^\infty(\mathbb{R}) : \sup_x (1+|x|)^N |f^{(n)}(x)| < \infty \text{ for any } N, n \in \mathbb{Z} \text{ and } N, n \geq 0\}.$$

Functions of Rapid Decrease

Let $f, g : \mathbb{R}^d \to \mathbb{C}$. For $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}_0^d$, where $\mathbb{N}_0 = \{0, 1, 2 \cdots\}$,

$$|\alpha| = \alpha_1 + \cdots + \alpha_d,$$

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}, \text{ total derivative of order } |\alpha|,$$

$$\alpha! = \alpha_1! \cdots \alpha_d!, \quad 0! = 1.$$

Product Rule

$$\partial^\alpha(fg) = \sum_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} \partial^\beta f \partial^{\alpha-\beta} g,$$

where $\beta \leq \alpha \iff \beta_k \leq \alpha_k$ for any k.

$$x^\alpha := x_1^{\alpha_1} \cdots x_d^{\alpha_d}, \quad \text{where } \alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}_0^d,$$

$$x = (x_1, \ldots, x_d) \in \mathbb{R}^d.$$

Definition of Schwartz Space

For $\alpha \in \mathbb{N}_0^d$ and $N \in \mathbb{N}_0$,

$$|f|_{N,\alpha} = \sup_{x \in \mathbb{R}^d} (1+|x|)^N |\partial^\alpha f(x)|.$$

The Schwartz space of functions of rapid decrease is defined as:

$$S = S(\mathbb{R}^d) = \{f \in C^\infty(\mathbb{R}^d) : |f|_{N,\alpha} < \infty \text{ for any } N, \alpha\}.$$

Properties

1. If $|f|_{N,\alpha} < \infty$ if and only if $(1+|x|)^N |f(x)| \le C_{\alpha,N}$ for all $x \in \mathbb{R}^d$, then
$$|\partial^\alpha f(x)| \le \frac{C_{\alpha,N}}{(1+|x|)^N} \text{ for any } x \in \mathbb{R}^d.$$
2. If $f \in S$, then
$$|\partial^\alpha f(x)| \le \frac{C_{\alpha,N}}{(1+|x|)^N} \text{ for any } x \in \mathbb{R}^d.$$
3. $C_c^\infty(\mathbb{R}^d) \subseteq S(\mathbb{R}^d)$. Example: $e^{-|x|^2} \in S(\mathbb{R}^d)$ but $e^{-|x|} \notin C_c^\infty(\mathbb{R}^d)$ because it is non-zero everywhere.

Seminorms and Convergence

1. $|f|_{N,\alpha}$ with $f \in S$ is a seminorm.
$$|f+g|_{N,\alpha} \le |f|_{N,\alpha} + |g|_{N,\alpha},$$
$$|af|_{N,\alpha} = |a||f|_{N,\alpha},$$
$$|f|_{N,\alpha} = 0 \text{ not implies } f = 0.$$
2. Let $f_k, f \in S$. We say $f_k \to f$ in S if
$$|f_k - f|_{N,\alpha} \to 0 \text{ for any } N, \alpha.$$
3. For $N \in \mathbb{N}_0$ and $f \in S$,
$$|f|_{(N)} = \max_{|\alpha| \le N} |f|_{N,\alpha} = \max_{|\alpha|\le N} \sup_x (1+|x|)^N |\partial^\alpha f(x)|$$
$|f|_{(N)}$ are norms on S, increasing in N, $|f|_{(N)} \le |f|_{(N+1)}$.
4. For $f, g \in S$,
$$\rho(f,g) = \sum_{N=1}^\infty 2^{-N} \min(|f-g|_{(N)}, 1)$$

Properties of $S(\mathbb{R}^d)$

1. If $f \in S$ then $\partial^\alpha f \in S$.
2. If $f, g \in S$ then $fg \in S$.
3. If $f \in S$ then $Pf \in S$ for any polynomial P.
4. $f \in S \Rightarrow f \in L^p(\mathbb{R}^d)$ for any $p > 0$.

   Proof. For $p = \infty$ this is easy. For $0 < p < \infty$,
$$\int_{\mathbb{R}^d} |f|^p \, dx = \int_{\mathbb{R}^d} (|f|(1+|x|)^{\frac{d+1}{p}})^p \frac{dx}{(1+|x|)^{d+1}} \le |f|^p_{\lceil \frac{d+1}{p} \rceil, 0} \int_{\mathbb{R}^d} \frac{dx}{(1+|x|)^{d+1}} < \infty.$$

5. If $f, g \in S$ then $f * g \in S$.

Proof. By definition of convolution,

$$\partial^\alpha (f * g)(x) = \int_{\mathbb{R}^d} \partial^\alpha f(x-y) g(y)\, dy,$$

$$1 + |x| \le 1 + |x-y| + |y| \le (1 + |x-y|)(1+|y|).$$

$$\begin{aligned}
(1+|x|)^N |\partial^\alpha (f*g)(x)| &\le \int (1+|x-y|)^N |\partial^\alpha f(x-y)| (1+|y|)^N |g(y)|\, dy \\
&\le |f|_{N,\alpha} \int (1+|y|)^{N+d+1} |g(y)| \frac{dy}{(1+|y|)^{d+1}} \\
&\le |f|_{N,\alpha} |g|_{N+d+1,0} \int \frac{dy}{(1+|y|)^{d+1}} < \infty
\end{aligned}$$

Continuous Linear Functionals on S

Let $T : S(\mathbb{R}^d) \to \mathbb{C}$ be linear. Then, T is continuous with respect to the metric ρ on S if and only if

There exists $C > 0$ and $N \in \mathbb{N}_0$ such that $|T(f)| \le C |f|_{(N)}$ for all $f \in S$.

Proof. $(\Rightarrow)$ Let T be continuous. Since T is continuous at zero, $|T(f)| = |T(f) - T(0)| \le 1$ if $\rho(f, 0) \le 2\delta$ for some $\delta > 0$.

$$\begin{aligned}
\rho(f, 0) &= \sum_{N=1}^{\infty} 2^{-N} \min\{1, |f|_{(N)}\} \\
&\le \sum_{N=1}^{N_0} 2^{-N} |f|_{(N_0)} + \sum_{N=N_0+1}^{\infty} 2^{-N} \\
&\le |f|_{(N_0)} + 2^{-N_0}
\end{aligned}$$

Choose N_0 so that $2^{-N_0} \le \delta$. Let $g \in S$ and $g \ne 0$. Then, $f = \frac{\delta}{|g|_{(N_0)}} g$ has $|f|_{(N_0)} = \delta$.

$$\implies |f|_{(N_0)} \le \delta \Rightarrow \rho(f,0) \le 2\delta \Rightarrow |T(f)| \le 1$$

$$\implies \sum_{N=1}^{\infty} 2^{-N} \min\{1, |f|_{(N)}\} \le |f|_{(N_0)} + 2^{-N_0} \le 2\delta \text{ for all } N_0 \in \mathbb{N},$$

$$|T(f)| = |T(g)| \cdot \frac{\delta}{|g|_{(N_0)}} \le 1 \Rightarrow |T(g)| \le \frac{1}{\delta} |g|_{(N_0)}.$$

$(\Leftarrow)$ Let $|T(f)| \le C|f|_{(N_0)}, f \in S$. Let $f_n \to f$ in S. Then, when $|f_n - f|_{(N_0)} \to 0$,

$$|T(f_n) - T(f)| = |T(f_n - f)| \le C |f_n - f|_{(N_0)} \to 0.$$

Corollary 0

A linear functional $T : S \to \mathbb{C}$ is continuous if and only if there exist $C > 0$, $\alpha_1, \ldots, \alpha_m \in \mathbb{N}_0^d$, and $N_1, \ldots, N_m \in \mathbb{N}_0$ such that:

$$|T(f)| \leq C \sum_{k=1}^{m} |f|_{N_k,\alpha_k}.$$

Fourier Transform on $S(\mathbb{R}^d)$

(i) Given $f \in L^1(\mathbb{R}^d)$, its Fourier transform is:

$$\mathcal{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-i2\pi x\cdot\xi}\, dx, \quad \xi \in \mathbb{R}^d.$$

where $x \cdot \xi = x_1\xi_1 + \cdots + x_d\xi_d$.

(ii) Given $g \in L^1(\mathbb{R}^d)$, its inverse Fourier transform is:

$$\mathcal{F}^{-1}g(x) = \int_{\mathbb{R}^d} g(\xi) e^{i2\pi x\cdot\xi}\, d\xi, \quad x \in \mathbb{R}^d.$$

Note.

(a) $\mathcal{F}f(\xi)$ and $\mathcal{F}^{-1}g(x)$ are bounded, continuous functions.

(b) $\mathcal{F}$ and $\mathcal{F}^{-1}$ are linear operators.

(c) $|\mathcal{F}f(\xi)| \leq \int_{\mathbb{R}^d} |f|\, dx = |f|_{L^1}$.

(d) $|\mathcal{F}^{-1}g(x)| \leq |g|_{L^1}$.

(e) $\mathcal{F}^{-1}g(x) = \mathcal{F}g(-x)$, $x \in \mathbb{R}^d$.

(f) $\widehat{\partial^\alpha f} = (i2\pi\xi)^\alpha \hat{f}$.

Simple Properties of Fourier Transform

Property 1

If $f \in S$, then for $\alpha \in \mathbb{N}_0^d$,

$$\widehat{\partial^\alpha f}(\xi) = (i2\pi\xi)^\alpha \hat{f}(\xi) = (i2\pi)^{|\alpha|} \xi_1^{\alpha_1} \cdots \xi_d^{\alpha_d} \hat{f}(\xi)$$

where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d) \in \mathbb{N}_0^d$, $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d}$, and $x = (x_1, x_2, \ldots, x_d) \in \mathbb{R}^d$.

Proof. For $d = 1$ and $\alpha = 1$,

$$\widehat{f^{(1)}}(\xi) = i2\pi\xi \hat{f}(\xi), \quad \xi \in \mathbb{R}.$$

By definition,

$$\begin{aligned}\widehat{f^{(1)}}(\xi) &= \int_{-\infty}^{\infty} f'(x)e^{-i2\pi x\xi}\,dx = \lim_{n\to\infty}\int_{-n}^{n} f'(x)e^{-i2\pi x\xi}\,dx \\ &= \lim_{n\to\infty}\left[f(x)e^{-i2\pi x\xi}\right]_{-n}^{n} - (-i2\pi\xi)\int_{-n}^{n} f(x)e^{-i2\pi x\xi}\,dx \\ &= 0 + i2\pi\xi\hat{f}(\xi) = i2\pi\xi\hat{f}(\xi)\end{aligned}$$

Example 1

Let $\Delta u(x) = \frac{\partial^2}{\partial x_1^2}u(x) + \cdots + \frac{\partial^2}{\partial x_d^2}u(x)$, where $u \in S$. Find $\widehat{\Delta u}(\xi)$.

$\frac{\partial^2}{\partial x_1^2} = \partial^\alpha$ with $\alpha = [2, 0, 0, \cdots 0]$. By property 1,

$$\begin{aligned}\widehat{\Delta u}(\xi) &= \widehat{\frac{\partial^2}{\partial x_1^2}u}(\xi) + \cdots + \widehat{\frac{\partial^2}{\partial x_d^2}u}(\xi) \\ &= (i2\pi\xi_1)^2\hat{u}(\xi) + \cdots + (i2\pi\xi_d)^2\hat{u}(\xi) \\ &= \hat{u}(\xi)\left(-4\pi^2(\xi_1^2 + \xi_2^2 + \cdots + \xi_d^2)\right) \\ &= -4\pi^2|\xi|^2\hat{u}(\xi)\end{aligned}$$

Property 2

1. Let $f \in S$, $y \in \mathbb{R}^d$. Then:

$$\widehat{f(\cdot + y)}(\xi) = e^{i2\pi\xi\cdot y}\hat{f}(\xi)$$

   Proof.

$$\begin{aligned}\widehat{f(\cdot + y)}(\xi) &= \int_{\mathbb{R}^d} f(x+y)e^{-i2\pi x\cdot\xi}\,dx \\ &= \int_{\mathbb{R}^d} f(z)e^{-i2\pi(z-y)\cdot\xi}\,dz \quad \text{(where } z = x + y \text{ and } dz = dx\text{)} \\ &= e^{i2\pi y\cdot\xi}\hat{f}(\xi)\end{aligned}$$

2. Let $f \in S$, $\delta \in \mathbb{R}$ and $\delta \neq 0$. Then,

$$\widehat{f(\delta\cdot)}(\xi) = |\delta|^{-d}\hat{f}\left(\frac{\xi}{\delta}\right)$$

   Proof.

$$\begin{aligned}\widehat{f(\delta\cdot)}(\xi) &= \int_{\mathbb{R}^d} f(\delta x)e^{-i2\pi x\cdot\xi}\,dx \\ &= \int_{\mathbb{R}^d} f(z)e^{-i2\pi\frac{z\cdot\xi}{\delta}}\delta^{-d}\,dz \quad \text{(where } z = \delta x \text{ and } dz = \delta^d dx\text{)} \\ &= \delta^{-d}\hat{f}\left(\frac{\xi}{\delta}\right)\end{aligned}$$

Example 2

Let $f \in S$, $\int f\,dx \neq 0$. Show that:

$$\lim_{\delta\to 0} |\widehat{f(\delta\cdot)}(\xi)| = \begin{cases} 0, & \xi \neq 0 \\ +\infty, & \xi = 0 \end{cases}.$$

Assume $\delta \neq 0$.

$$|\widehat{f(\delta\cdot)}(\xi)| \overset{\delta\neq 0}{=} |\delta|^{-d} \left| \hat{f}\left(\frac{\xi}{\delta}\right) \right|$$

Since $\hat{f} \in S$, it satisfies:

$$|\hat{f}(\xi)| \leq \frac{C}{(1+|\xi|)^{d+1}},$$

where $C = |\hat{f}|_{d+1,0}$ is a seminorm. When $\xi \neq 0$,

$$\begin{aligned} |\widehat{f(\delta\cdot)}(\xi)| &= \frac{1}{|\xi|^d} \cdot \left(\frac{|\xi|}{|\delta|}\right)^d \cdot |\hat{f}(\xi)| \\ &\leq \frac{1}{|\xi|^d} \cdot \left(\frac{|\xi|}{|\delta|}\right)^d \cdot \frac{|\hat{f}|_{d+1,0}}{(1+|\xi|/|\delta|)^{d+1}} \to 0 \quad \text{as} \quad \delta \to 0 \end{aligned}$$

When $\xi = 0$,

$$|\widehat{f(\delta\cdot)}(0)| = |\delta|^{-d} |\hat{f}(0)| \to \infty \quad \text{as} \quad \delta \to 0$$

Note that $\hat{f}(0) = \int f(x)\,dx \neq 0$, so $|\hat{f}(0)| \neq 0$. On the other hand, $\lim_{\delta\to 0} f(\delta x) = f(0)$ for $f \in S$. Note the opposite tendencies of $f(\delta x)$ and $\widehat{f(\delta\cdot)}(\xi)$ as $\delta \to 0$.

Property 3

If $f \in S$, then $\hat{f} \in C^\infty$ with all derivatives bounded, and:

$$\partial^\beta \hat{f}(\xi) = \hat{g}(\xi) \quad \text{with} \quad g(x) = (-i2\pi x)^\beta f(x) \in S, \quad \beta \in \mathbb{N}_0^d.$$

Proof. (See Folland 2-27)

$$\partial^\beta \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) \frac{\partial^\beta}{\partial \xi} e^{-i2\pi x\cdot\xi}\,dx = \int_{\mathbb{R}^d} \underbrace{f(x)(-i2\pi x)^\beta}_{g(x)} e^{-i2\pi x\cdot\xi}\,dx = \hat{g}(\xi)$$

Proposition 1

The Fourier transform F maps S to S. We write $F : S \to S$.

Proof. Let $f \in S$. By Property 3 above, $\hat{f} \in C^\infty$. Then, $\hat{f} \in S$ if and only if $\sup_{\xi\in\mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| < \infty$ for any $\alpha, \beta \in \mathbb{N}_0^d$.

$$\begin{aligned} \xi^\alpha \partial^\beta \hat{f}(\xi) &= (i2\pi)^{-|\alpha|} (i2\pi\xi)^\alpha \partial^\beta \hat{f}(\xi) \\ &= (i2\pi)^{-|\alpha|} (i2\pi\xi)^\alpha \hat{g}(\xi) \text{ with } g(x) = (-i2\pi x)^\beta f(x) \in S \end{aligned}$$

Hence,

$$\sup_{\xi\in\mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| = (2\pi)^{-|\alpha|} \sup_{\xi\in\mathbb{R}^d} \widehat{\partial^\alpha g}(\xi) \leq (2\pi)^{-|\alpha|} |\partial^\alpha g|_{L_1} < \infty$$

Proposition 2

Let $f, g \in S$. Then, $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Proof. By Fubini's Theorem,

$$\begin{aligned}
&\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} f(x-y)g(y)\,dy\,e^{-i2\pi x\cdot\xi}\,dx\\
&= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} f(x-y)\,e^{-i2\pi(x-y)\cdot\xi}\,dx\right)\cdot g(y)\,e^{-i2\pi y\cdot\xi}\,dy\,, \quad \text{set } z = x-y \text{ and } dz = dx\\
&= \int_{\mathbb{R}^d}\left(\hat{f}(\xi)\right)\cdot g(y)\,e^{-i2\pi y\cdot\xi}\,dy = \hat{f}(\xi)\cdot\hat{g}(\xi)
\end{aligned}$$

We can swap integrals because $\iint |f(x-y)|\cdot|g(y)|\,dy\,dx = |f|_{L^1}\cdot|g|_{L^1} < \infty$.

Proposition 3

Let $f, g \in S$. Then,

$$\int_{\mathbb{R}^d} \hat{f}(z)g(z)\,dz = \int_{\mathbb{R}^d} f(y)\hat{g}(y)\,dy$$

$$\int_{\mathbb{R}^d} \mathcal{F}^{-1}(f)(z)g(z)\,dz = \int_{\mathbb{R}^d} f(y)\mathcal{F}^{-1}(g)(y)\,dy$$

.

Proof. The proof will use Fubini's Theorem. Let $f, g \in S$. Then,

$$\begin{aligned}
\int_{\mathbb{R}^d} \hat{f}(z)g(z)\,dz &= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} f(y)\,e^{-i2\pi y\cdot z}\,dy\right)g(z)\,dz\\
&= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} g(z)\,e^{-i2\pi y\cdot z}\,dz\right)f(y)\,dy\\
&= \int_{\mathbb{R}^d} \hat{g}(y)f(y)\,dy
\end{aligned}$$

We can use Fubini's Theorem because:

$$\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} |f(y)|\cdot|g(z)|\cdot|e^{i2\pi y\cdot z}|\,dy\,dz = |f|_{L^1}\cdot|g|_{L^1} < \infty$$

since $f, g \in L^1$ since $f, g \in S$. Analogous argument proves the $\mathcal{F}^{-1}$ case.

Inverse of $\mathcal{F}$

Theorem 1

Let $f \in S$. Then, $\mathcal{F}^{-1}\mathcal{F}(f)(x) = \mathcal{F}\mathcal{F}^{-1}(f)(x)$ for $x \in \mathbb{R}^d$. In particular,

$$\mathcal{F}^{-1}(\hat{f})(x) = \int_{\mathbb{R}^d} \hat{f}(\xi)e^{i2\pi x\cdot\xi}\,d\xi = f(x) \text{ for } x \in \mathbb{R}^d.$$

This means both $\mathcal{F}, \mathcal{F}^{-1} : S \to S$ are bijections.

Proof. Let $f \in S$. We first prove that inversion holds at $x = 0$.

$$f(0) = \int_{\mathbb{R}^d} \hat{f}(\xi)\, d\xi.$$

For $\delta > 0$ and $k(x) = e^{-\pi|x|^2}$,

$$\int_{\mathbb{R}^d} f(\delta y)\hat{k}(y)\, dy \longrightarrow f(0) \text{ as } \delta \to 0 \text{ by dominant convergence theorem.}$$

By Proposition 3,

$$\begin{aligned}
\int_{\mathbb{R}^d} f(\delta y)\hat{k}(y)\, dy &= \int \widehat{f(\delta \cdot)}(\xi)k(\xi)\, d\xi \\
&= \int \delta^{-d}\hat{f}\left(\frac{\xi}{\delta}\right) k(\xi)\, d\xi \text{ (by Property 2(b))} \\
&= \int \delta^{-d}\hat{f}(z)k(\delta z)\delta^d\, dz \text{ where } z = \frac{\xi}{\delta} \text{ and } d\xi = \delta^d dz \\
&= \int \hat{f}(z)k(\delta z)\, dz
\end{aligned}$$

By dominant convergence theorem, when $\delta \to 0$, $\int \hat{f}(z)k(\delta z)\, dz \to \int \hat{f}(z)k(0)\, dz = \int \hat{f}(z)\, dz$ as $k(0) = 1$ and $\int \hat{f}(z)\, dz = f(0)$. In general case, let $y \in \mathbb{R}^d$ and consider $f(x+y)$ for $x \in \mathbb{R}^d$. Note that $f(y) = f(0+y)$. Using the result that inversion holds at 0,

$$\begin{aligned}
f(y) = f(0+y) &= \int \mathcal{F}(f(\cdot + y))(\xi)\, d\xi \\
&= \int \hat{f}(\xi)e^{i2\pi\xi\cdot y}\, d\xi \text{ (by property 2(a))} \\
&= \mathcal{F}^{-1}(\hat{f})(y)
\end{aligned}$$

We have shown $\mathcal{F}^{-1}\mathcal{F}(f) = f$ so we now show $\mathcal{F}\mathcal{F}^{-1}(f) = f$. Recall that $\mathcal{F}(g)(x) = \mathcal{F}^{-1}(g)(-x)$. Now $\mathcal{F}\mathcal{F}^{-1}(f)(x) = \mathcal{F}^{-1}(\mathcal{F}^{-1}(f))(-x)$ and:

$$\mathcal{F}^{-1}f(x) = \int f(\xi)e^{i2\pi x\cdot\xi}\, d\xi = \int f(-z)e^{-i2\pi z\cdot x}\, dz = \mathcal{F}\tilde{f}(x)$$

By the change of variable, $\xi = -z$ and $\tilde{f}(x) = f(-x)$ for $x \in \mathbb{R}^d$. Therefore,

$$\mathcal{F}(\mathcal{F}^{-1}(f))(x) = \mathcal{F}^{-1}(\mathcal{F}^{-1}(f))(-x) = \mathcal{F}^{-1}(\mathcal{F}(\tilde{f}))(-x) = \tilde{f}(-x) = f(x).$$

Thus, we prove that $\mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x) = \mathcal{F}^{-1}\mathcal{F}(f)(x)$.

Remark.

1. First Use of DCT in the Proof.

$$\int_{\mathbb{R}^d} f(\delta y)\hat{k}(y)\, dy \longrightarrow f(0) \quad \text{as } \delta \to 0.$$

- *Pointwise Convergence.* Since f is continuous (because $f \in S$, the Schwartz space), $f(\delta y) \to f(0)$ as $\delta \to 0$ for each y.
- *Dominating Function.* Since $\hat{f} \in S$, it is rapidly decreasing, meaning $|\hat{f}(\xi)| \le \frac{C}{(1+|\xi|)^{d+1}}$ for some $C > 0$. Meanwhile, $|k(\delta\xi)| \le 1$ (since k is a Gaussian). Thus, $|\hat{f}(\xi)k(\delta\xi)| \le |\hat{f}(\xi)|$, and $\hat{f}(\xi)$ is integrable (because $\hat{f} \in S$).

DCT allows us to move the limit inside:

$$\lim_{\delta\to 0}\int f(\delta y)\hat{k}(y)\,dy = \int \lim_{\delta\to 0} f(\delta y)\hat{k}(y)\,dy = \int f(0)\hat{k}(y)\,dy = f(0).$$

2. Second Use of DCT in the Proof.

$$\int \hat{f}(\xi)k(\delta\xi)\,d\xi \longrightarrow \int \hat{f}(\xi)\,d\xi \quad \text{as } \delta \to 0.$$

- *Pointwise Convergence.* $k(\delta\xi) = e^{-\pi\delta^2|\xi|^2} \to 1$ as $\delta \to 0$ for each ξ.
- *Dominating Function.* Since $\hat{f} \in S$, it is rapidly decreasing, meaning $|\hat{f}(\xi)| \le \frac{C}{(1+|\xi|)^{d+1}}$ for some $C > 0$. Meanwhile, $|k(\delta\xi)| \le 1$ since k is a Gaussian. Thus, $|\hat{f}(\xi)k(\delta\xi)| \le |\hat{f}(\xi)|$, and $\hat{f}(\xi)$ is integrable because $\hat{f} \in S$.

DCT allows us to move the limit inside:

$$\lim_{\delta\to 0}\int \hat{f}(\xi)k(\delta\xi)\,d\xi = \int \hat{f}(\xi)\lim_{\delta\to 0} k(\delta\xi)\,d\xi = \int \hat{f}(\xi)\,d\xi.$$

3. Why the Schwartz Space S is Important?

- Functions in S decay rapidly: Both f and $\hat{f}$ decrease faster than any polynomial, ensuring integrability.
- Gaussians are nice dominating functions: $k(x) = e^{-\pi|x|^2}$ is smooth, integrable, and dominates the integrals involved.

If f were not in S, we might not have a suitable dominating function.

Plancherel's Equality

For $f \in S$, $\int |f(x)|^2\,dx = \int |\hat{f}(\xi)|^2\,d\xi$, so $|f|_{L^2} = |\hat{f}|_{L^2}$ where $L^2 = L^2(\mathbb{R}^d)$. *Idea of the proof:* Use $f(0) = \int \hat{f}(\xi)\,d\xi$ and the Fourier transform of the convolution: Let $(f*g)(x) = \int f(x-y)g(y)\,dy$ and $\mathcal{F}(f*g)(x) = \hat{f}(\xi)\hat{g}(\xi)$. Applying Proposition 2 and $f(0) = \int \hat{f}(\xi)\,d\xi$,

$$(f*g)(0) = \int g(y)f(-y)\,dy = \int \mathcal{F}(f*g)(\xi)\,d\xi = \int \hat{f}(\xi)\hat{g}(\xi)\,d\xi \quad (\#)$$

In (#) above, we want to take $f = g$ and deal with the negative y part.

Proposition 4 (Plancherel's Equality)

For any $f, g \in S$, $\int g(y)\overline{f(y)}\,dy = \int \hat{g}(\xi)\overline{\hat{f}(\xi)}\,d\xi$. Thus, $\langle g, f\rangle_{L^2} = \langle \hat{g}, \hat{f}\rangle_{L^2}$. In particular, $|f|_{L^2} = |\hat{f}|_{L^2}$ in the case $f = g$.

Proof. Let $f, g \in S$. Set $\tilde{f}(y) = \overline{f(-y)}$ for $y \in \mathbb{R}^d$. Note that $\tilde{f} \in S$. By (#),

$$\int g(y)\overline{f(y)}\,dy = \int g(y)\tilde{f}(-y)\,dy = \int \hat{g}(\xi)\mathcal{F}(\tilde{f})(\xi)\,d\xi$$

Now, we compute $\mathcal{F}(\tilde{f})(\xi)$.

$$\begin{aligned}\mathcal{F}(\tilde{f})(\xi) &= \int \overline{f(-y)}e^{-i2\pi y\cdot\xi}\,dy = \int \overline{f(x)}e^{i2\pi x\cdot\xi}\,dx \quad \text{where } x = -y,\ dx = dy\,|J| = dy \\ &= \overline{\int f(x)e^{-i2\pi x\cdot\xi}\,dx} = \overline{\hat{f}(\xi)}\end{aligned}$$

Hence, $\int g(y)\overline{f(y)}\,dy = \int \hat{g}(\xi)\overline{\hat{f}(\xi)}\,d\xi \iff \langle g, f\rangle_{L^2} = \langle \hat{g}, \hat{f}\rangle_{L^2}$.

Remark. $\mathcal{F}$ and $\mathcal{F}^{-1}$ can be extended to unitary maps on L^2. For $f \in L^2$, there exists $f_n \in S$ with $f_n \to f$ in L^2 since $C_c^\infty \subseteq S$ and C_c^∞ is dense in L^2. Then, by Proposition 4, if $|f_n - f_m|_{L^2} \to 0$ as $m, n \to \infty$, then

$$|f_n - f_m|_{L^2} = |\hat{f}_n - \hat{f}_m|_{L^2} = |\mathcal{F}^{-1}(f_n) - \mathcal{F}^{-1}(f_m)|_{L^2}$$

We define $\mathcal{F}f = \lim_{n\to\infty}\mathcal{F}(f_n)$ and $\mathcal{F}^{-1}f = \lim_{n\to\infty}\mathcal{F}^{-1}(f_n)$. This definition does not depend on the choice of the sequence f_n. Indeed, let $g_n \in S$ such that $g_n \to f$ in L^2. Then, $|g_n - f_n|_{L^2} \to 0$ as $n \to \infty$. Therefore, by proposition 4,

$$|g_n - f_n|_{L^2} = |\hat{g}_n - \hat{f}_n|_{L^2} = |\mathcal{F}^{-1}g_n - \mathcal{F}^{-1}f_n|_{L^2}$$

$$\implies \hat{g}_n \to \mathcal{F}f \text{ and } \mathcal{F}^{-1}g_n \to \mathcal{F}^{-1}f_n \text{ in } L^2$$

Hence, $\mathcal{F} : L^2 \to L^2$ is unitary with $\mathcal{F}^* = \mathcal{F}^{-1}$.

Corollary 1

$\mathcal{F}, \mathcal{F}^{-1} : S \to S$ can be extended to $\mathcal{F}, \mathcal{F}^{-1} : L^2 \to L^2$ as

$$\mathcal{F}f = \hat{f} = \lim_{n\to\infty}\mathcal{F}f_n,\ \mathcal{F}^{-1}f = \lim_{n\to\infty}\mathcal{F}^{-1}f_n \text{ in } L^2 \text{ with } f_n \in S,\ f_n \to f \text{ in } L^2.$$

We have $\mathcal{F}\mathcal{F}^{-1}f = \mathcal{F}^{-1}\mathcal{F}f = f$ for any $f \in L^2$. Also,

$$|\mathcal{F}f|_{L^2} = |\mathcal{F}^{-1}f|_{L^2} = |f|_{L^2}, \quad f \in L^2.$$

Proposition 6

Let $f \in L^1 \cap L^2$. Let $\hat{f} = \lim_{n\to\infty} \hat{f}_n, \mathcal{F}^{-1}f = \lim_{n\to\infty} \mathcal{F}^{-1}f_n$ in L^2 with $f_n \in S$ and $f_n \to f$ in L^2. Then, $\hat{f}(\xi) = \int f(x)e^{-i2\pi x\cdot\xi}\,dx$ with $\xi \in \mathbb{R}^d$. $\mathcal{F}^{-1}f(x) = \int f(\xi)e^{i2\pi x\cdot\xi}\,d\xi$ with $x \in \mathbb{R}^d$.

Proof. By Proposition 3, for any $h \in S$,

$$\int \hat{f}_n h = \int f_n \hat{h} \quad \text{and} \quad \int \mathcal{F}^{-1}(f_n)h = \int f_n \mathcal{F}^{-1}(h)$$

As $n \to \infty$,

$$\int \hat{f}_n h \to \int \hat{f} h, \int f_n \hat{h} \to \int f\hat{h}, \int \mathcal{F}^{-1}(f_n)h \to \int \mathcal{F}^{-1}(f)h, \int f_n\mathcal{F}^{-1}(h) \to \int f\mathcal{F}^{-1}(h)$$

$$\implies \int \hat{f}h = \int f\hat{h}, \quad \int \mathcal{F}^{-1}(f)h = \int f\mathcal{F}^{-1}(h)$$

Now, for all $h \in S$,

$$\begin{aligned}
\int \hat{f}(y)h(y)\,dy &= \int f(x)\hat{h}(x)\,dx \\
&= \int f(x)\left(\int h(y)e^{-i2\pi x\cdot y}\,dy\right)\,dx \\
&= \int \left(f(x)e^{-i2\pi x\cdot y}\,dx\right)h(y)\,dy \quad \text{(By Fubini's Theorem)}
\end{aligned}$$

Thus,

$$\hat{f}(y) = \int f(x)e^{-i2\pi x\cdot y}\,dx \quad \text{a.e.}$$

Fubini holds since $\iint |f(x)|\cdot|h(y)|\cdot|e^{-i2\pi x\cdot y}|\,dx\,dy = |f|_{L^1}\cdot|h|_{L^1} < \infty$. An analogous proof holds for $\mathcal{F}^{-1}$.

Corollary 2

Let $f \in L^2$. Then,

$$\mathcal{F}f(\xi) = \hat{f}(\xi) = \lim_{n\to\infty} \int_{|x|\le n} f(x)e^{-i2\pi x\cdot\xi}\,dx \text{ in } L^2,$$

$$\mathcal{F}^{-1}f(x) = \lim_{n\to\infty} \int_{|\xi|\le n} f(\xi)e^{i2\pi x\cdot\xi}\,d\xi \text{ in } L^2.$$

Proof. Since $f \in L^2$, we can approximate it by truncating it to the ball $|x| \le n$. Define:

$$f_n(x) = f(x)\cdot\chi_{|x|\le n},$$

where $\chi_{|x|\le n}$ is the indicator function of the ball of radius n. Then, f_n has compact support since it vanishes outside $|x| \le n$. By Hölder's inequality, and $f_n \in L^2$ since $f \in L^2$,

$$\int |f_n| \le \left(\int |f_n|^2\right)^{1/2} \cdot \left(\int \chi_{|x|\le n}\right)^{1/2} < \infty$$

Therefore, $f_n \in L^1 \cap L^2$.

$f_n \to f$ in L^2 as $n \to \infty$ by the dominated convergence theorem, since $|f_n| \le |f|$ and $|f|^2$ is integrable. For $f_n \in L^1 \cap L^2$,

$$\mathcal{F}f(\xi) = \hat{f}_n(\xi) = \int_{\mathbb{R}^d} f_n(x) e^{-i2\pi x\cdot\xi}\, dx = \int_{|x|\le n} f(x) e^{-i2\pi x\cdot\xi}\, dx.$$

This is well-defined pointwise since $f_n \in L^1$. Since $f_n \to f$ in L^2, and the Fourier transform $\mathcal{F}$ is a unitary operator on L^2, we have:

$$\hat{f}_n \to \hat{f} \quad \text{in } L^2 \text{ as } n \to \infty.$$

This is because $|\hat{f}_n - \hat{f}|_{L^2} = |f_n - f|_{L^2} \to 0$ by Plancherel's Equality. Now, define:

$$g_n(\xi) = f(\xi) \cdot \chi_{|\xi|\le n}.$$

Then $g_n \to f$ in L^2. By the continuity of $\mathcal{F}^{-1}$ on L^2, we have $\mathcal{F}^{-1} g_n \to \mathcal{F}^{-1} f \quad$ in L^2, which gives the second formula:

$$\mathcal{F}^{-1} f(x) = \lim_{n\to\infty} \int_{|\xi|\le n} f(\xi) e^{i2\pi x\cdot\xi}\, d\xi \quad (\text{in } L^2).$$

Recall. $f \in S$. Then, $\mathcal{F}^{-1}\mathcal{F}(f)(x) = \mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x)$, $x \in \mathbb{R}^d$. $\mathcal{F}, \mathcal{F}^{-1} : S \to S$ are bijective.

Plancherel's Equality

For $f \in S$, $\int |f(x)|^2\, dx = \int |\hat{f}(\xi)|^2\, d\xi$, so $|f|_{L^2} = |\hat{f}|_{L^2}$ where $L^2 = L^2(\mathbb{R}^d)$.

Idea of the proof: Use $f(0) = \int \hat{f}(\xi)\, d\xi$ and the Fourier transform of the convolution: Let $(f * g)(x) = \int f(x-y) g(y)\, dy$ and $\mathcal{F}(f * g)(x) = \hat{f}(\xi)\hat{g}(\xi)$. Applying Proposition 2 and $f(0) = \int \hat{f}(\xi)\, d\xi$,

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For any $f, g \in S$, $\int g(y)\overline{f(y)}\, dy = \int \hat{g}(\xi)\overline{\hat{f}(\xi)}\, d\xi$. Thus, $\langle g, f\rangle_{L^2} = \langle \hat{g}, \hat{f}\rangle_{L^2}$. In particular, $|f|_{L^2} = |\hat{f}|_{L^2}$ in the case $f = g$.

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$$\mathcal{F}(\tilde{f})(\xi) = \int \overline{f(-y)} e^{-i2\pi y\cdot\xi}\, dy = \int \overline{f(x)} e^{i2\pi x\cdot\xi}\, dx \quad \text{where } x=-y,\, dx = dy\,|J| = dy$$
$$= \overline{\int f(x) e^{-i2\pi x\cdot\xi}\, dx} = \overline{\hat{f}(\xi)}$$

Hence, $\int g(y)\overline{f(y)}\, dy = \int \hat{g}(\xi)\overline{\hat{f}(\xi)}\, d\xi \iff \langle g, f\rangle_{L^2} = \langle \hat{g}, \hat{f}\rangle_{L^2}$.

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$$|g_n - f_n|_{L^2} = |\hat{g}_n - \hat{f}_n|_{L^2} = |\mathcal{F}^{-1}g_n - \mathcal{F}^{-1}f_n|_{L^2},$$
$$\implies \hat{g}_n \to \mathcal{F}f \text{ and } \mathcal{F}^{-1}g_n \to \mathcal{F}^{-1}f_n \text{ in } L^2.$$

Hence, $\mathcal{F} : L^2 \to L^2$ is unitary with $\mathcal{F}^* = \mathcal{F}^{-1}$.

Corollary 1

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We have $\mathcal{F}\mathcal{F}^{-1}f = \mathcal{F}^{-1}\mathcal{F}f = f$ for any $f \in L^2$. Also,

$$|\mathcal{F}f|_{L^2} = |\mathcal{F}^{-1}f|_{L^2} = |f|_{L^2}, \quad f \in L^2.$$

Proposition 6

Let $f \in L^1 \cap L^2$. Let $\hat{f} = \lim_{n\to\infty} \hat{f}_n, \mathcal{F}^{-1}f = \lim_{n\to\infty}\mathcal{F}^{-1}f_n$ in L^2 with $f_n \in S$ and $f_n \to f$ in L^2. Then, $\hat{f}(\xi) = \int f(x)e^{-i2\pi x\cdot\xi}\, dx$ with $\xi \in \mathbb{R}^d$. $\mathcal{F}^{-1}f(x) = \int f(\xi)e^{i2\pi x\cdot\xi}\, d\xi$ with $x \in \mathbb{R}^d$.

Proof. By Proposition 3, for any $h \in S$,

$$\int \hat{f}_n h = \int f_n \hat{h} \quad \text{and} \quad \int \mathcal{F}^{-1}(f_n)h = \int f_n \mathcal{F}^{-1}(h)$$

As $n \to \infty$,

$$\int \hat{f}_n h \to \int \hat{f}h, \int f_n\hat{h} \to \int f\hat{h}, \int \mathcal{F}^{-1}(f_n)h \to \int \mathcal{F}^{-1}(f)h, \int f_n\mathcal{F}^{-1}(h) \to \int f\mathcal{F}^{-1}(h),$$

$$\implies \int \hat{f} h = \int f \hat{h}, \quad \int \mathcal{F}^{-1}(f) h = \int f \mathcal{F}^{-1}(h).$$

Now, for all $h \in S$,

$$\begin{aligned}
\int \hat{f}(y) h(y)\, dy &= \int f(x) \hat{h}(x)\, dx \\
&= \int f(x) \left(\int h(y) e^{-i2\pi x \cdot y}\, dy \right) dx \\
&= \int \left(\int f(x) e^{-i2\pi x \cdot y}\, dx \right) h(y)\, dy \quad \text{(By Fubini's Theorem)}
\end{aligned}$$

Thus,

$$\hat{f}(y) = \int f(x) e^{-i2\pi x \cdot y}\, dx \quad \text{a.e.}$$

Fubini holds since $\iint |f(x)| \cdot |h(y)| \cdot |e^{-i2\pi x \cdot y}|\, dx\, dy = |f|_{L^1} \cdot |h|_{L^1} < \infty$. An analogous proof holds for $\mathcal{F}^{-1}$.

Corollary 2

Let $f \in L^2$. Then,

$$\mathcal{F} f(\xi) = \hat{f}(\xi) = \lim_{n \to \infty} \int_{|x| \le n} f(x) e^{-i2\pi x \cdot \xi}\, dx \text{ in } L^2,$$

$$\mathcal{F}^{-1} f(x) = \lim_{n \to \infty} \int_{|\xi| \le n} f(\xi) e^{i2\pi x \cdot \xi}\, d\xi \text{ in } L^2.$$

Proof. Let $f \in L^2$. Set $f_n = \chi_{|x| \le n} f$. Then $f_n \in L^1 \cap L^2$ (by Hölder, $\int |f_n| \le |f_n|_{L^2} |\chi_{|x| \le n}|_{L^2} < \infty$) and $f_n \to f$ in L^2 by the dominated convergence theorem, since $|f_n| \le |f|$ and $|f|^2$ is integrable. By Proposition 6, since $f_n \in L^1 \cap L^2$,

$$\hat{f}_n(\xi) = \int_{\mathbb{R}^d} f_n(x) e^{-i2\pi x \cdot \xi}\, dx = \int_{|x| \le n} f(x) e^{-i2\pi x \cdot \xi}\, dx.$$

Since $\mathcal{F}$ is unitary on L^2, $|\hat{f}_n - \hat{f}|_{L^2} = |f_n - f|_{L^2} \to 0$, so $\hat{f}_n \to \hat{f}$ in L^2. Hence

$$\hat{f}(\xi) = \lim_{n \to \infty} \int_{|x| \le n} f(x) e^{-i2\pi x \cdot \xi}\, dx \quad \text{in } L^2.$$

An analogous argument with $g_n = \chi_{|\xi| \le n} f$ proves the formula for $\mathcal{F}^{-1}$.

Some Applications of the Fourier Transform

1. Heisenberg Principle

Let $\psi \in S(\mathbb{R})$ be a wave function, $\int_{\mathbb{R}} |\psi(x)|^2\, dx = 1$. Then, in quantum mechanics, $|\psi(x)|^2$ is the pdf of a particle's location and

$$\int |\hat{\psi}(\xi)|^2\, d\xi = \int |\psi(x)|^2\, dx = 1.$$

$|\hat{\psi}(\xi)|^2$ is the pdf of a particle's momentum. If x_0, ξ_0 are the measured location and momentum, then their mean square errors are

$$(\Delta x)^2 = \int_{\mathbb{R}} |x - x_0|^2 \, |\psi(x)|^2 \, dx, \qquad (\Delta \xi)^2 = \int_{\mathbb{R}} |\xi - \xi_0|^2 \, |\hat{\psi}(\xi)|^2 \, d\xi.$$

Theorem 2 (Heisenberg Uncertainty Principle). Let $\psi \in S(\mathbb{R})$ with $\int_{\mathbb{R}} |\psi(x)|^2 \, dx = 1$. Then

$$(\Delta x)^2 (\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

Proof. First take $x_0 = \xi_0 = 0$, so that $(\Delta x)^2 = \int x^2 |\psi(x)|^2 \, dx$ and $(\Delta \xi)^2 = \int \xi^2 |\hat{\psi}(\xi)|^2 \, d\xi$. Since $\psi \in S(\mathbb{R})$, it decays faster than any polynomial, so $x|\psi(x)|^2 \to 0$ as $|x| \to \infty$. Integrating by parts,

$$\begin{aligned} 1 = \int_{-\infty}^{\infty} |\psi(x)|^2 \, dx &= \Big[x |\psi(x)|^2 \Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x \big(\psi'(x) \overline{\psi(x)} + \psi(x) \overline{\psi'(x)} \big) \, dx \\ &= -2 \operatorname{Re} \int_{-\infty}^{\infty} x \, \psi'(x) \, \overline{\psi(x)} \, dx. \end{aligned}$$

Hence, by the triangle inequality and then Cauchy–Schwarz,

$$1 \leq 2 \int_{-\infty}^{\infty} |x \, \psi(x)| \, |\psi'(x)| \, dx \leq 2 \left(\int x^2 |\psi(x)|^2 \, dx \right)^{1/2} \left(\int |\psi'(x)|^2 \, dx \right)^{1/2} = 2 \, (\Delta x) \left(\int |\psi'|^2 \, dx \right)^{1/2}.$$

By Plancherel's equality and Property 1, $\widehat{\psi'}(\xi) = i 2\pi \xi \, \hat{\psi}(\xi)$, so

$$\int |\psi'(x)|^2 \, dx = \int |\widehat{\psi'}(\xi)|^2 \, d\xi = \int (2\pi |\xi|)^2 |\hat{\psi}(\xi)|^2 \, d\xi = 4\pi^2 (\Delta \xi)^2.$$

Therefore $1 \leq 2(\Delta x) \cdot 2\pi (\Delta \xi) = 4\pi (\Delta x)(\Delta \xi)$, which gives

$$(\Delta x)^2 (\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

For general x_0, ξ_0, apply the result to $\varphi(x) = e^{-i2\pi\xi_0 x} \psi(x + x_0)$. Then $\int |\varphi|^2 = 1$, $|\hat{\varphi}(\xi)|^2 = |\hat{\psi}(\xi + \xi_0)|^2$, and a change of variables gives $\int x^2 |\varphi(x)|^2 dx = (\Delta x)^2$ and $\int \xi^2 |\hat{\varphi}(\xi)|^2 d\xi = (\Delta \xi)^2$. Applying the case above to φ yields the general inequality.

2. Helmholtz Equation in $S = S(\mathbb{R}^d)$

Consider

$$u(x) - \Delta u(x) = f(x), \quad x \in \mathbb{R}^d, \ f \in S, , \tag{H}$$

where $\Delta u(x) = \frac{\partial^2}{\partial x_1^2} u(x) + \cdots + \frac{\partial^2}{\partial x_d^2} u(x)$. We found (Example 1, Property 1)

$$\widehat{\Delta u}(\xi) = -4\pi^2 |\xi|^2 \, \hat{u}(\xi).$$

Proposition 7

For any $f \in S$, there exists a unique $u \in S$ solving (H); i.e., $I - \Delta : S \to S$ is bijective.

Proof. Let $f \in S$. Note that $\hat{f} \in S$ and, since $1 + 4\pi^2|\xi|^2$ is a polynomial that is bounded below by 1 with $\frac{1}{1+4\pi^2|\xi|^2} \in C^\infty$ having all derivatives bounded, the product $\frac{1}{1+4\pi^2|\xi|^2}\,\hat{f}(\xi) \in S$.

(1) Uniqueness. Let $u \in S$ with $u - \Delta u = f$. Taking Fourier transforms,

$$\widehat{u - \Delta u}(\xi) = \hat{f}(\xi) \implies (1 + 4\pi^2|\xi|^2)\,\hat{u}(\xi) = \hat{f}(\xi) \implies \hat{u}(\xi) = \frac{\hat{f}(\xi)}{1 + 4\pi^2|\xi|^2}.$$

Thus $\hat{u}$ is uniquely determined, and since $\mathcal{F} : S \to S$ is a bijection, u is unique.

(2) Existence. Let $f \in S$ and set $g(\xi) = \dfrac{\hat{f}(\xi)}{1 + 4\pi^2|\xi|^2} \in S$. Then $u(x) = (\mathcal{F}^{-1}g)(x) \in S$ solves (H): indeed $\hat{u} = g$, so

$$\widehat{u - \Delta u}(\xi) = (1 + 4\pi^2|\xi|^2)\,\hat{u}(\xi) = (1 + 4\pi^2|\xi|^2)\,g(\xi) = \hat{f}(\xi),$$

and injectivity of $\mathcal{F}$ gives $u - \Delta u = f$.

Tempered Distributions

Notation. Let $S' = S'(\mathbb{R}^d)$ be the space of all linear continuous maps $T : S \to \mathbb{C}$.

Recall. A linear map $T : S \to \mathbb{C}$ is continuous if and only if there exist $m \geq 1$, $C \geq 0$, $\alpha_1, \alpha_2, \ldots, \alpha_m \in \mathbb{N}_0^d$, and $N_1, N_2, \ldots, N_m \in \mathbb{N}_0$ such that

$$|T(f)| \leq C \sum_{k=1}^{m} |f|_{N_k, \alpha_k}, \quad f \in S.$$

Equivalently, a linear map $T : S \to \mathbb{C}$ is continuous if and only if there exist $N \in \mathbb{N}_0$ and $C \geq 0$ such that $|T(f)| \leq C|f|_{(N)}$.

$T \in S'$ is called a *tempered distribution*, and $S' = S'(\mathbb{R}^d)$ is called the space of tempered distributions.

Example 1 ($\mathcal{O}_M$)

Let $\mathcal{O}_M$ be the space of all $f \in C^\infty$ such that for every $\alpha \in \mathbb{N}_0^d$ there exist $C = C(\alpha) \geq 0$ and $N = N(\alpha) \in \mathbb{N}_0$ with

$$|\partial^\alpha f(x)| \leq C(1 + |x|)^N, \quad x \in \mathbb{R}^d.$$

(a) Every $g \in \mathcal{O}_M$ defines $T_g \in S'$ by $T_g(f) = \displaystyle\int_{\mathbb{R}^d} g(x)f(x)\,dx$, $f \in S$.

Proof. Suppose $|g(x)| \le C(0)(1+|x|)^N$ for $x \in \mathbb{R}^d$ (the bound for $\alpha = 0$). Then

$$\begin{aligned}\int |g(x)| \cdot |f(x)|\, dx &\le C(0) \int (1+|x|)^N |f(x)|\, dx \\ &= C(0) \int (1+|x|)^{N+d+1} |f(x)| \cdot \frac{dx}{(1+|x|)^{d+1}} \\ &\le C(0)\, |f|_{N+d+1,\,0} \int \frac{dx}{(1+|x|)^{d+1}}.\end{aligned}$$

Hence

$$|T_g(f)| \le \left| \int g(x) f(x)\, dx \right| \le C\, |f|_{N+d+1,\,0}, \qquad C = C(0) \int \frac{dx}{(1+|x|)^{d+1}} < \infty,$$

so T_g is continuous, i.e. $T_g \in S'$.

(b) The map $\mathcal{O}_M \ni g \mapsto T_g \in S'$ is injective.

Proof. Let $g_1, g_2 \in \mathcal{O}_M$ with $T_{g_1}(f) = T_{g_2}(f)$ for all $f \in S$. Then $\int g_1(x) f(x)\, dx = \int g_2(x) f(x)\, dx$ for all $f \in S$. Since g_1, g_2 are locally integrable and $\int g_1 f = \int g_2 f$ for all $f \in C_c^\infty$, it follows that $g_1 = g_2$ a.e.

(c) Let $\alpha \in \mathbb{N}_0^d$. Then $\partial^\alpha g \in \mathcal{O}_M$ for $g \in \mathcal{O}_M$, and since $f \in S$ decays rapidly, integration by parts gives

$$\langle \partial^\alpha g, f \rangle = \int_{\mathbb{R}^d} \partial^\alpha g \cdot f\, dx = (-1)^{|\alpha|} \int g\, \partial^\alpha f\, dx = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} T_g(\partial^\alpha f), \quad f \in S.$$

Notation. $C_c^\infty \subseteq S \subseteq \mathcal{O}_M \subseteq S'$. We also write $T_g = g$ and $T_g(f) = \langle g, f \rangle$ for $f \in S$.

Definition. Let $T_n, T \in S'$. We say $T_n \to T$ in S' if $T_n(f) \to T(f)$ for every $f \in S$.

Example 2 (Polynomial Growth)

We say a measurable function $g : \mathbb{R}^d \to \mathbb{C}$ is of *polynomial growth* if

$$|g(x)| \le C(1+|x|)^N, \quad x \in \mathbb{R}^d, \text{ for some } C > 0,\ N \ge 1.$$

Remark. If g is of polynomial growth, then g defines $T_g \in S'$ by $T_g(f) = \int gf$, $f \in S$ (equivalently $\langle g, f \rangle = \int gf$), and $|T_g(f)| \le C|f|_{N+d+1,\,0}$, $f \in S$. Again we write $T_g = g$.

For $\alpha \in \mathbb{N}_0^d$ we define the distributional derivative $\partial^\alpha g \in S'$ by

$$\langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} T_g(\partial^\alpha f), \quad f \in S., \tag{\#}$$

Formally this reads $\langle \partial^\alpha g, f \rangle = \int \partial^\alpha g(x) f(x)\, dx = (-1)^{|\alpha|} \int g(x) \partial^\alpha f(x)\, dx$. As an operator,

$$\partial^\alpha T_g = (-1)^{|\alpha|} T_g \circ \partial^\alpha, \qquad S \xrightarrow{\partial^\alpha} S \xrightarrow{T_g} \mathbb{C}.$$

More generally, (#) defines $\partial^\alpha T$ for any $T \in S'$ by $(\partial^\alpha T)(f) = (-1)^{|\alpha|} T(\partial^\alpha f)$, $f \in S$.

Definition. $T \in S'$ is *regular* if $T = T_g$ where g is of polynomial growth.

E.g. Let

$$g(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \text{(the ramp function).}$$

(a) Find $\frac{d}{dx} g = g'$. For $f \in S$,

$$\begin{aligned}\left\langle \tfrac{d}{dx} g, f \right\rangle &= -\int_{-\infty}^{\infty} g(x) f'(x)\, dx = -\int_0^{\infty} x f'(x)\, dx \\ &= -\Big[x f(x)\Big]_0^{\infty} + \int_0^{\infty} f(x)\, dx = \int_0^{\infty} f(x)\, dx = \int_{-\infty}^{\infty} H(x) f(x)\, dx,\end{aligned}$$

where the boundary term vanishes since $f \in S$. Hence $g' = H$, the Heaviside function,

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}, \qquad H = g' = \frac{d}{dx} g.$$

(b) Find $\frac{dH}{dx}$. For $f \in S$,

$$\begin{aligned}\left\langle \tfrac{dH}{dx}, f \right\rangle &= -\int_{-\infty}^{\infty} H(x) f'(x)\, dx \\ &= -\int_0^{\infty} f'(x)\, dx \\ &= -\Big[f(x)\Big]_0^{\infty} = f(0) \\ &= \int_{-\infty}^{\infty} f(x)\, \delta_0(dx) = \int_{-\infty}^{\infty} f(x)\, dH(x).\end{aligned}$$

Thus $\frac{dH}{dx} = \delta_0$, the Dirac measure, where (informally)

$$\delta_0(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \qquad \text{(Dirac measure),} \qquad \langle \delta_0, f \rangle = f(0).$$

Formally, $\displaystyle\int_{-\infty}^{\infty} f(x) H'(x)\, dx = \int_{-\infty}^{\infty} f(x)\, dH(x).$

Example 3 **(L^p)**

Let $g \in L^p$ with $p \geq 1$. Then g defines $T_g \in S'$ by $T_g(f) = \displaystyle\int g(x) f(x)\, dx$, $f \in S$.

Proof. By Hölder's inequality, with $\frac{1}{p} + \frac{1}{q} = 1$,

$$\int |g(x)| \cdot |f(x)|\, dx \leq |g|_{L^p}\, |f|_{L^q}.$$

Now, for $f \in S$, writing $m = \lceil \frac{d+1}{q} \rceil$,

$$|f|_{L^q}^q = \int |f(x)|^q \, dx = \int \left(|f(x)|(1+|x|)^{\frac{d+1}{q}} \right)^q \cdot \frac{dx}{(1+|x|)^{d+1}} \leq |f|_{m,0}^q \int \frac{dx}{(1+|x|)^{d+1}},$$

where $\int \frac{dx}{(1+|x|)^{d+1}} = C < \infty$. Then $|f|_{L^q} \leq C^{1/q} |f|_{m,0}$, so

$$|T_g(f)| \leq |g|_{L^p} \, |f|_{L^q} \leq C^{1/q} \, |g|_{L^p} \, |f|_{m,0} < \infty,$$

showing $T_g \in S'$. The map $g \in L^p \mapsto T_g \in S'$ is injective (similar proof to $\mathcal{O}_M$ for injectivity).

Recall. We define $L^p \subseteq S'$ and often write $g = T_g$, where

$$T_g(f) = \langle g, f \rangle = \int g(x) f(x) \, dx, \quad f \in S.$$

We define $\partial^\alpha g$ by

$$\langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle, \quad f \in S.$$

In particular $L^2 \subseteq S'$. Recall that if $g \in L^2$, then $\mathcal{F}g = \hat{g}$ and $\mathcal{F}^{-1}g \in L^2$. By Proposition 3 in simple properties of fourier transform section,

$$\int \hat{g} \, f = \int g \, \hat{f}, \quad \int \mathcal{F}^{-1} g \, f = \int g \, \mathcal{F}^{-1} f, \quad f \in S,$$

and this even holds for $f \in L^2$:

$$\langle \hat{g}, f \rangle = \langle g, \hat{f} \rangle, \quad \langle \mathcal{F}^{-1} g, f \rangle = \langle g, \mathcal{F}^{-1} f \rangle, \quad f \in S.$$

For $g \in L^p$ with $p \geq 1$, we define $\mathcal{F}g = \hat{g}$, $\mathcal{F}^{-1}g \in S'$ by

$$\langle \mathcal{F}g, f \rangle = \langle \hat{g}, f \rangle = \langle g, \mathcal{F}f \rangle = \langle g, \hat{f} \rangle, \quad \langle \mathcal{F}^{-1} g, f \rangle = \langle g, \mathcal{F}^{-1} f \rangle, \quad f \in S.$$

Note. $S \xrightarrow{\mathcal{F}} S \xrightarrow{T_g} \mathbb{C}$, i.e. $\mathcal{F}g = T_g \circ \mathcal{F}$.

Example 4 ($\mathcal{M}_{\text{mod}}$)

Let $\mathcal{M}_{\text{mod}}$ (moderate) be the space of signed Borel measures μ on $\mathbb{R}^d$ of *moderate growth*:

$$\int (1+|x|)^{-N} \, d|\mu| < \infty \quad \text{for some integer } N \geq 1.. \tag{\#}$$

E.g. Lebesgue measure on $\mathbb{R}^d$: $N = d+1$ works (indeed any $N = d + \epsilon$, $\epsilon > 0$).

Every $\mu \in \mathcal{M}_{\text{mod}}$ defines $T_\mu \in S'$ by $T_\mu(f) = \displaystyle\int f \, d\mu$, $f \in S$.

Proof (continuity).

$$\int |f| \, d|\mu| = \int |f(x)|(1+|x|)^N \cdot \frac{1}{(1+|x|)^N} \, d|\mu| \leq |f|_{N,0} \int \frac{d|\mu|}{(1+|x|)^N} < \infty \quad \text{by (\#)},$$

for $f \in S$. Hence $|T_\mu(f)| \le C\,|f|_{N,0}$, so $T_\mu \in S'$.

The map $\mu \in \mathcal{M}_{\mathrm{mod}} \mapsto T_\mu \in S'$ is injective; we write $\mathcal{M}_{\mathrm{mod}} \subseteq S'$, $T_\mu(f) = \langle \mu, f\rangle$ for $f \in S$, and $\mu = T_\mu$. We define

$$\langle \partial^\alpha \mu, f\rangle = (-1)^{|\alpha|}\langle \mu, \partial^\alpha f\rangle, \quad \langle \mathcal{F}\mu, f\rangle = \langle \mu, \mathcal{F}f\rangle, \quad \langle \mathcal{F}^{-1}\mu, f\rangle = \langle \mu, \mathcal{F}^{-1}f\rangle, \quad f \in S.$$

Let μ be a finite signed Borel measure on $\mathbb{R}^d$. Then, by Fubini,

$$\begin{aligned}\langle \mathcal{F}^{-1}\mu, f\rangle = \langle \mu, \mathcal{F}^{-1}f\rangle = \int \mathcal{F}^{-1}f\,d\mu = \int \left(\int_{\mathbb{R}^d} f(\xi)e^{i2\pi x\cdot\xi}\,d\xi\right) d\mu(x)\\ = \int_{\mathbb{R}^d} f(\xi)\left(\int e^{i2\pi x\cdot\xi}\,d\mu(x)\right) d\xi = \int_{\mathbb{R}^d} f(\xi)\,\phi(\xi)\,d\xi,\end{aligned}$$

where

$$\phi(\xi) = \int e^{i2\pi x\cdot\xi}\,d\mu(x) = E\big[e^{i2\pi x\cdot\xi}\big], \quad \xi \in \mathbb{R}^d,$$

(the characteristic function of μ). Thus $\mathcal{F}^{-1}\mu = \phi$ in S', where $\phi = T_\phi$ is defined by $T_\phi(f) = \int f\phi$, $f \in S$.

Main Operations on $S' = S'(\mathbb{R}^d)$

Differentiation. Given $T \in S'$ and $\alpha \in \mathbb{N}_0^d$, we define $\partial^\alpha T \in S'$ by

$$\partial^\alpha T(f) = (-1)^{|\alpha|}T(\partial^\alpha f), \quad f \in S, \quad \text{i.e.} \quad \langle \partial^\alpha T, f\rangle = (-1)^{|\alpha|}\langle T, \partial^\alpha f\rangle, \quad f \in S,$$

where, for $R \in S'$, $R(f) = \langle R, f\rangle$.

Fourier Transform. Given $T \in S'$, we define $\mathcal{F}T = \hat{T}$, $\mathcal{F}^{-1}T \in S'$ by $\hat{T}(f) = T(\hat{f})$, i.e.

$$\mathcal{F}T(f) = T(\mathcal{F}f), \quad \mathcal{F}^{-1}T(f) = T(\mathcal{F}^{-1}f), \quad f \in S.$$

Notation. $\langle \hat{T}, f\rangle = \langle T, \hat{f}\rangle$, $\quad \langle \mathcal{F}^{-1}T, f\rangle = \langle T, \mathcal{F}^{-1}f\rangle$.

Multiplication by $F \in \mathcal{O}_M$***.*** Given $T \in S'$ and $F \in \mathcal{O}_M$, we define $FT \in S'$ by

$$FT(f) = T(Ff), \quad f \in S, \quad \text{i.e.} \quad \langle FT, f\rangle = \langle T, Ff\rangle, \quad f \in S,$$

which is well defined since $f \in S \mapsto Ff \in S$ is continuous.

Proposition 1

$\mathcal{F}, \mathcal{F}^{-1} : S' \to S'$ are bijections (recall $S \subseteq L^2 \subseteq S'$).

Proof. Let $f \in S$. Then

$\mathcal{F}^{-1}\mathcal{F}T(f) = \mathcal{F}T(\mathcal{F}^{-1}f) = T(\mathcal{F}\mathcal{F}^{-1}f) = T(f)$, $\mathcal{F}\mathcal{F}^{-1}T(f) = \mathcal{F}^{-1}T(\mathcal{F}f) = T(\mathcal{F}^{-1}\mathcal{F}f) = T(f)$.

Thus $\mathcal{F}^{-1}\mathcal{F}T = T = \mathcal{F}\mathcal{F}^{-1}T$.

Example

Let $T \in S'$, $\alpha \in \mathbb{N}_0^d$. Show that $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$.

We know $(i2\pi\xi)^\alpha \in \mathcal{O}_M$. Let $f \in S$. Then

$$\begin{aligned}\widehat{\partial^\alpha T}(f) &= \partial^\alpha T(\hat{f}) = (-1)^{|\alpha|} T(\partial^\alpha \hat{f}) = (-1)^{|\alpha|} T(\hat{g}) \quad \text{with } g(x) = (-i2\pi x)^\alpha f(x) \in S \\ &= (-1)^{|\alpha|} T\big(\widehat{(-i2\pi\cdot)^\alpha f}\big) = (-1)^{|\alpha|} \hat{T}\big((-i2\pi\cdot)^\alpha f\big) \quad \text{with } (-i2\pi\cdot)^\alpha \in \mathcal{O}_M \\ &= (-1)^{|\alpha|} (-i2\pi\cdot)^\alpha \hat{T}(f) = (i2\pi\cdot)^\alpha \hat{T}(f),\end{aligned}$$

using Property 3, $\partial^\alpha \hat{f} = \hat{g}$ with $g = (-i2\pi\cdot)^\alpha f$. Hence $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$.

Recall. Let $g(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$, $g \in S'$. We found $\frac{d^2}{dx^2} g = \delta_0$, and $\delta_0 \in S'$ with $\int_{\mathbb{R}^d} f(x)\, d\delta_0(x) = f(0)$. Also recall that $g : \mathbb{R}^d \to \mathbb{C}$ is of *polynomial growth* if $|g(x)| \leq C(1 + |x|)^N$, $x \in \mathbb{R}^d$, for some $C > 0$ and integer $N \geq 1$.

Theorem 1 (Structure Theorem)

For each $T \in S'$, there exist a continuous function g of polynomial growth and $\alpha \in \mathbb{N}_0^d$ such that $T = \partial^\alpha g$, i.e.

$$T(f) = \langle T, f \rangle = \langle \partial^\alpha g, f \rangle = (-1)^{|\alpha|} \langle g, \partial^\alpha f \rangle = (-1)^{|\alpha|} \int_{\mathbb{R}^d} g(x)\, \partial^\alpha f(x)\, dx.$$

Proof. We prove it for $d = 1$. In particular, we prove the following first.

Lemma 1

Let $h : \mathbb{R} \to \mathbb{C}$ be of polynomial growth. Then there exists a continuous function H of polynomial growth so that

$$\langle h, f^{(n)} \rangle = \langle H, f^{(n+1)} \rangle, \quad f \in S,\ n \in \mathbb{N}_0, \quad \text{i.e.} \quad \int_{-\infty}^{\infty} h(x) f^{(n)}(x)\, dx = \int_{-\infty}^{\infty} H(x) f^{(n+1)}(x)\, dx.$$

Proof. Set $G(x) = \int_0^x h(t)\, dt$, $x \in \mathbb{R}$. Suppose $|h(x)| \leq C_1(1 + |x|)^N$. Then, using $|x| \leq \frac{|x|^2+1}{2}$,

$$|h(x)| \leq C_2(1 + |x|^2)^N, \quad \text{and also} \quad |h(x)| \leq C_3(1 + |x|^{2N}), \quad x \in \mathbb{R},$$

(since $|x|^k \leq \frac{|x|^{2N}}{C_4} + C_5$). Then G is a continuous function of polynomial growth: for $x \geq 0$,

$$|G(x)| \leq \int_0^x |h(t)|\, dt \leq C_6 \int_0^x (1 + t^{2N})\, dt = C_6 x + C_6 \frac{x^{2N+1}}{2N+1},$$

and for $x < 0$, $|G(x)| \le \int_x^0 |h(t)|\,dt \le C_6|x| + C_6\frac{|x|^{2N+1}}{2N+1}$; hence $|G(x)| \le C_7(1+|x|)^{2N+1}$.

$$\begin{aligned}
\langle G, f^{(n+1)}\rangle &= \int_{-\infty}^{\infty} G(x) f^{(n+1)}(x)\,dx \\
&= \Big[G(x) f^{(n)}(x)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f^{(n)}(x) h(x)\,dx \\
&= 0 - \langle h, f^{(n)}\rangle \quad (f^{(n)} \in S,\ h \in S').
\end{aligned}$$

Take $H = -G$, which finishes the proof of the lemma.

Now we return to Theorem 1. Let $T \in S'$. There exist $C > 0$, $N \in \mathbb{N}_0$ such that

$$|T(f)| \le C|f|_{(N)} \le C\sum_{k=0}^{N} \sup_x (1+|x|)^N |f^{(k)}(x)| \le C_1 \sum_{k=0}^{N} \sup_x (1+|x|^2)^N |f^{(k)}(x)|, \quad f \in S.. \tag{1}$$

Let $\varphi(x) = (1+|x|^2)^{-N}$. Note $\varphi \in \mathcal{O}_M$ and $|\varphi^{(\ell)}(x)| \le C_\ell\,\varphi(x)$ for all $\ell \in \mathbb{N}_0$, $x \in \mathbb{R}$. Consider $\varphi T(f) = T(\varphi f)$, $f \in S$.

Claim 1. $|\varphi T(f)| \le C_2 \sum_{k=0}^{N} \sup_x |f^{(k)}(x)|$, $f \in S$.

Proof. By (1), $|\varphi T(f)| = |T(\varphi f)| \le C_1 \sum_{k=0}^{N} \sup_x (1+|x|^2)^N |(\varphi f)^{(k)}(x)|$. By Leibniz,

$$|(\varphi f)^{(k)}(x)| \le \sum_{j=0}^{k} \frac{k!}{j!(k-j)!} |f^{(j)}(x)|\,|\varphi^{(k-j)}(x)| \le \sum_{j=0}^{k} C'_{k,j}\,|f^{(j)}(x)|\,\varphi(x), \quad C'_{k,j} = \tbinom{k}{j} C_{k-j}.$$

Since $(1+|x|^2)^N \varphi(x) = 1$,

$$|\varphi T(f)| \le C_1 \sum_{k=0}^{N}\sum_{j=0}^{k} C'_{k,j} \sup_x |f^{(j)}(x)| \le C_2 \sum_{k=0}^{N} \sup_x |f^{(k)}(x)|.$$

Claim 2. There exist complex (finite Borel) measures μ_k, $k = 0, 1, \ldots, N$, on $\mathbb{R}$ such that

$$\varphi T(f) = \sum_{k=0}^{N} \int_{\mathbb{R}} f^{(k)}(x)\,d\mu_k, \quad f \in S.$$

Proof. Let $C_0(\mathbb{R})$ be the Banach space of continuous functions on $\mathbb{R}$ vanishing at $\pm\infty$, with norm $|g|_u = \sup_x |g(x)|$. Let $E = \underbrace{C_0(\mathbb{R}) \times \cdots \times C_0(\mathbb{R})}_{N+1}$ with norm $|(g_0, \ldots, g_N)|_u = |g_0|_u + \cdots + |g_N|_u$; E is a Banach space. Let

$$M = \Big\{(f^{(0)}, f^{(1)}, \ldots, f^{(N)}) : f \in S\Big\} \subseteq E,$$

and consider $\Lambda : M \to \mathbb{C}$ defined by $\Lambda(f^{(0)}, f^{(1)}, \dots, f^{(N)}) = \varphi T(f)$. By Claim 1,

$$|\Lambda(f^{(0)}, \dots, f^{(N)})| = |\varphi T(f)| \le C_2 \, |(f^{(0)}, \dots, f^{(N)})|_u,$$

so Λ is bounded on M. By a corollary of the Hahn–Banach theorem, Λ extends to $\Lambda \in E^*$ with $\|\Lambda\|_{E^*} \le C_2$. Now $|\Lambda(g_0, 0, \dots, 0)| \le C_2 |g_0|_u$ for $g_0 \in C_0(\mathbb{R})$, so by the Riesz representation theorem there exists a finite complex Borel measure μ_0 on $\mathbb{R}$ with $\Lambda(g_0, 0, \dots, 0) = \int_{\mathbb{R}} g_0 \, d\mu_0$. Continuing slot by slot, we obtain $\mu_1, \dots, \mu_N$ with

$$\Lambda(g_0, g_1, \dots, g_N) = \sum_{k=0}^{N} \int_{\mathbb{R}} g_k(x) \, d\mu_k, \quad (g_0, \dots, g_N) \in E.$$

(for example, $\Lambda(g_0, g_1) = \Lambda(g_0, 0) + \Lambda(0, g_1)$). Hence $\varphi T(f) = \Lambda(f^{(0)}, \dots, f^{(N)}) = \sum_{k=0}^{N} \int_{\mathbb{R}} f^{(k)} \, d\mu_k$.

Claim 3. There exist bounded measurable functions F_k, $k = 0, 1, \dots, N$, such that

$$\varphi T(f) = \sum_{k=0}^{N} \int_{\mathbb{R}} f^{(k+1)}(x) \, F_k(x) \, dx, \quad f \in S.$$

Proof. Let $G_k(x) = \mu_k\big((-\infty, x]\big)$, $x \in \mathbb{R}$; then G_k is bounded (as μ_k is finite) and of bounded variation. Then

$$\begin{aligned}
\varphi T(f) &= \sum_{k=0}^{N} \int_{-\infty}^{\infty} f^{(k)}(x) \, d\mu_k(x) = \sum_{k=0}^{N} \int_{-\infty}^{\infty} f^{(k)}(x) \, dG_k(x) \\
&= \sum_{k=0}^{N} \Big[f^{(k)} G_k \Big]_{-\infty}^{\infty} - \sum_{k=0}^{N} \int_{-\infty}^{\infty} f^{(k+1)}(x) G_k(x) \, dx.
\end{aligned}$$

The boundary terms vanish ($f^{(k)} \in S$, G_k bounded). Set $F_k = -G_k$.

Finishing the proof. Since $\frac{1}{\varphi} = (1 + |x|^2)^N \in \mathcal{O}_M$ and $\frac{1}{\varphi} f \in S$,

$$T(f) = \frac{1}{\varphi} \cdot \varphi T(f) = \varphi T\Big(\tfrac{1}{\varphi} f\Big) = \sum_{k=0}^{N} \int_{\mathbb{R}} F_k(x) \left(\tfrac{1}{\varphi} f\right)^{(k+1)}(x) \, dx.$$

By Leibniz, $\left(\tfrac{1}{\varphi} f\right)^{(k+1)} = \sum_{\ell=0}^{k+1} \binom{k+1}{\ell} f^{(\ell)} \left(\tfrac{1}{\varphi}\right)^{(k+1-\ell)}$, so

$$T(f) = \sum_{k=0}^{N} \sum_{\ell=0}^{k+1} \int_{\mathbb{R}} f^{(\ell)}(x) \, H_{k,\ell}(x) \, dx, \quad H_{k,\ell} = \tbinom{k+1}{\ell} F_k \left(\tfrac{1}{\varphi}\right)^{(k+1-\ell)},$$

where each $H_{k,\ell}$ is measurable of polynomial growth (bounded $\times$ polynomial growth). Applying Lemma 1 repeatedly to raise every term to order $N+2$,

$$T(f) = \sum_{k=0}^{N} \sum_{\ell=0}^{k+1} \int_{\mathbb{R}} f^{(N+2)}(x) \, \widetilde{H}_{k,\ell}(x) \, dx,$$

where the $\widetilde{H}_{k,\ell}$ are continuous of polynomial growth. Set

$$g(x) = (-1)^{N+2} \sum_{k=0}^{N} \sum_{\ell=0}^{k+1} \widetilde{H}_{k,\ell}(x), \quad x \in \mathbb{R},$$

a continuous function of polynomial growth. Then $T(f) = (-1)^{N+2} \int g(x) f^{(N+2)}(x)\, dx = \langle \partial^{N+2} g, f \rangle$, i.e. $T = \partial^{N+2} g$ (so $\alpha = N+2$). This concludes the proof of Theorem 1.

Some Applications of Tempered Distributions

Helmholtz Equation in $S' = S'(\mathbb{R}^d)$

Consider

$$u - \Delta u = g \text{ in } S'., \tag{H}$$

Given $g \in S'$, find $u \in S'$ solving (H).

Proposition 2

For any $g \in S'$, there exists a unique $u \in S'$ solving (H); i.e. $I - \Delta : S' \to S'$ is bijective. Moreover

$$u = \mathcal{F}^{-1}\Big(\frac{1}{1 + 4\pi^2 |\xi|^2}\, \hat{g} \Big), \quad \text{where } \frac{1}{1 + 4\pi^2 |\xi|^2} \in \mathcal{O}_M.$$

Proof. Note that both $1 + 4\pi^2|\xi|^2$ and $\frac{1}{1+4\pi^2|\xi|^2}$ are in $\mathcal{O}_M$. Using $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$ (so $\widehat{\Delta u} = -4\pi^2 |\xi|^2 \hat{u}$),

$$\begin{aligned} u - \Delta u = g \text{ in } S' \iff \widehat{u - \Delta u} = \hat{g} \text{ in } S' &\iff (1 + 4\pi^2|\xi|^2)\hat{u} = \hat{g} \text{ in } S' \\ \iff \hat{u} = \frac{1}{1+4\pi^2|\xi|^2}\hat{g} \text{ in } S' &\iff u = \mathcal{F}^{-1}\Big(\frac{1}{1+4\pi^2|\xi|^2}\hat{g} \Big) \text{ in } S'. \end{aligned}$$

This determines u uniquely and provides a solution, so $I - \Delta : S' \to S'$ is bijective.

(H) with $g \in L^2$

$H_2^2 = \big\{ u \in S' : u,\, \partial^\alpha u \in L^2 \text{ for all } |\alpha| \le 2 \big\}$ with norm

$$|u|_{2,2} = \left(\sum_{|\alpha| \le 2} |\partial^\alpha u|_{L^2}^2 \right)^{1/2}.$$

In fact H_2^2 is a Hilbert space with inner product $\langle u, v \rangle = \sum_{|\alpha|\le 2} \int_{\mathbb{R}^d} \partial^\alpha u\, \overline{\partial^\alpha v}\, dx$.

Note. $u \in H_2^2$ means $u \in L^2$ and there exists $\partial^\alpha u \in L^2$ so that $\int \partial^\alpha u\, f\, dx = (-1)^{|\alpha|} \int u\, \partial^\alpha f\, dx$, $f \in S$, for all $|\alpha| \le 2$.

Proposition 3

For all $g \in L^2$, there exists a unique $u \in H_2^2$ solving (H). Moreover, $u - \Delta u = g$ a.e., and $|u|_{2,2} \leq C|g|_{L^2}$ with C independent of g. Also $I - \Delta : H_2^2 \to L^2$ is bijective.

Proof. Let $g \in L^2$ and $u = \mathcal{F}^{-1}\Big(\frac{1}{1+4\pi^2|\xi|^2}\hat{g}\Big) \in L^2$. Then for all $|\alpha| \leq 2$,

$$\widehat{\partial^\alpha u} = (i2\pi\xi)^\alpha \hat{u} = \frac{(i2\pi\xi)^\alpha}{1+4\pi^2|\xi|^2}\hat{g} \implies |\widehat{\partial^\alpha u}(\xi)| \leq |\hat{g}(\xi)|,$$

since for $|\alpha| \leq 2$, $|(i2\pi\xi)^\alpha| \leq (2\pi|\xi|)^{|\alpha|} \leq 1 + 4\pi^2|\xi|^2$. Hence $\partial^\alpha u \in L^2$ and, by Plancherel,

$$|\partial^\alpha u|_{L^2} = |\widehat{\partial^\alpha u}|_{L^2} \leq |\hat{g}|_{L^2} = |g|_{L^2}.$$

Summing over $|\alpha| \leq 2$ gives $|u|_{2,2} \leq C|g|_{L^2}$. Uniqueness follows from Proposition 2 (uniqueness in S'). Hence $I - \Delta : H_2^2 \to L^2$ is bijective.

Solution of (H) as a limit with $g \in L^2$***.*** Let $g \in L^2$. There exist $g_n \in S$ with $g_n \to g$ in L^2. For each $g_n \in S$ there is a unique $u_n \in S$ solving (H) (since $I - \Delta : S \to S$ is bijective, $S \subseteq L^2$, and S is dense in L^2). Then $(u_n - u_m) - \Delta(u_n - u_m) = g_n - g_m$, and by Proposition 3, for all $|\alpha| \leq 2$,

$$|\partial^\alpha u_n - \partial^\alpha u_m|_{L^2} \leq C|g_n - g_m|_{L^2} \to 0 \quad \text{as } n, m \to \infty.$$

Hence there exist $u_\alpha \in L^2$ with $\partial^\alpha u_n \to u_\alpha$ in L^2 for $|\alpha| \leq 2$; write $u = u_0$. For any $f \in S$ and $|\alpha| \leq 2$, $\int \partial^\alpha u_n\, f = (-1)^{|\alpha|} \int u_n\, \partial^\alpha f$, and passing to the limit,

$$\int u_\alpha\, f = (-1)^{|\alpha|} \int u\, \partial^\alpha f, \quad f \in S,$$

which means $\partial^\alpha u = u_\alpha$ in S'. Thus $u \in H_2^2$ and $u - \Delta u = g$ a.e.

(H) with $g \in H_2^s$

Let $s \in \mathbb{N}_0$. Define

$$H_2^s = \big\{u \in S' : u,\, \partial^\alpha u \in L^2,\ |\alpha| \leq s\big\}, \quad |u|_{s,2} = \left(\sum_{|\alpha|\leq s} |\partial^\alpha u|_{L^2}^2\right)^{1/2}.$$

Note. $H_2^s \subseteq L^2$.

Proposition 4

For any $g \in H_2^s$, there exists a unique $u \in H_2^{s+2}$ solving (H); i.e. $I - \Delta : H_2^{s+2} \to H_2^s$ is bijective.

Proof. $u - \Delta u = g$ implies $\partial^\alpha u - \Delta\partial^\alpha u = \partial^\alpha g$ in S'. Since $\partial^\alpha g \in L^2$ for all $|\alpha| \leq s$, by Proposition 3 we have, for all $|\beta| \leq 2$,

$$|\partial^{\beta+\alpha} u|_{L^2} = |\partial^\beta(\partial^\alpha u)|_{L^2} \leq C|\partial^\alpha g|_{L^2}.$$

Every γ with $|\gamma| \le s+2$ can be written $\gamma = \beta + \alpha$ with $|\alpha| \le s$, $|\beta| \le 2$, so

$$|\partial^\gamma u|_{L^2} \le C|g|_{s,2} \quad \text{for all } |\gamma| \le s+2.$$

Hence $u \in H_2^{s+2}$ solves (H), and uniqueness follows from Proposition 2. Therefore $I - \Delta : H_2^{s+2} \to H_2^s$ is bijective.

Weak Convergence

Let E be a normed vector space and $E^* = \mathcal{L}(E, \mathbb{R})$ be its dual. Let $x \in E$ and let $x_n \in E$ be a sequence "approximating" x. In what sense do we mean "approximating"? By the Hahn–Banach theorem,

$$|x_n - x|_E = \sup_{|\ell|_{E^*} \le 1} |\ell(x_n - x)| \to 0 \ \text{ as } n \to \infty,$$

if and only if for any $R > 0$, $\sup_{|\ell|_{E^*} \le R} |\ell(x_n) - \ell(x)| \to 0$ as $n \to \infty$. That is, norm convergence asks for $\ell(x_n) \to \ell(x)$ *uniformly* over bounded sets of functionals. Weak convergence relaxes this to pointwise.

We say $x_n \to x$ *weakly*, written $x_n \xrightarrow{w} x$, if

$$\ell(x_n) \to \ell(x) \ \text{ as } n \to \infty \ \text{ for all } \ell \in E^*, \quad \text{i.e.} \quad \langle \ell, x_n \rangle \to \langle \ell, x \rangle \ \text{ as } n \to \infty \ \text{ for all } \ell \in E^*.$$

Remark.

(a) Let $E = L^p$, $p \in [1, \infty)$. For the measure spaces considered above, $(L^p)^* = L^q$ with $\frac{1}{p} + \frac{1}{q} = 1$. Therefore $f_n \xrightarrow{w} f$ in L^p if and only if $\int f_n g \to \int fg$ for all $g \in L^q$.

(b) Let X be a compact metric space and $E = C(X)$ with the $|f|_u$ (sup) norm. Then $[C(X)]^* = M(X)$, the space of finite signed Borel measures, and $f_n \xrightarrow{w} f$ if and only if $\int f_n \, d\mu \to \int f \, d\mu$ for all $\mu \in M(X)$.

(c) Let $E = H$ be a Hilbert space. Then $H^* = H$ and $x_n \xrightarrow{w} x$ if and only if $\langle z, x_n \rangle \to \langle z, x \rangle$ as $n \to \infty$ for all $z \in H$. (Recall Problem 3 of HW4: $x_n \to x$ in H if and only if $x_n \xrightarrow{w} x$ and $\limsup_n |x_n|_H \le |x|_H$.)

Proposition 1

Let E be finite dimensional (e.g. $E = \mathbb{R}^d$). Then

$$x_n \to x \quad \text{if and only if} \quad x_n \xrightarrow{w} x.$$

Proof. ($\Rightarrow$) In any normed space, if $x_n \to x$ then for each $\ell \in E^*$, $|\ell(x_n) - \ell(x)| \le |\ell|_{E^*} |x_n - x|_E \to 0$, so $x_n \xrightarrow{w} x$.

($\Leftarrow$) Suppose $\dim E = d$ and $x_n \xrightarrow{w} x$. Fix a basis $v_1, \dots, v_d$ of E with coordinate functionals $\ell_1, \dots, \ell_d \in E^*$ ($\ell_i(v_j) = \delta_{ij}$), so $y = \sum_{i=1}^d \ell_i(y) v_i$ for every $y \in E$. Weak convergence gives

$\ell_i(x_n) \to \ell_i(x)$ for each i. Since all norms on a finite-dimensional space are equivalent,

$$|x_n - x|_E \le C \sum_{i=1}^{d} |\ell_i(x_n) - \ell_i(x)| \to 0,$$

so $x_n \to x$ in norm.

E.g. For $x^{(n)}, x \in \mathbb{R}^d$ with $\{x^{(n)}\}$ a sequence, $x^{(n)} \to x$ iff $x_i^{(n)} \to x_i$ for every $i = 1, 2, \ldots, d$.

Remark. In general, $x_n \xrightarrow{w} x$ does *not* imply $x_n \to x$ in norm. For example, let H be an infinite-dimensional Hilbert space with a CONS $\{e_n : n \ge 1\}$. Then for $n \ne m$,

$$|e_n - e_m|_H^2 = |e_n|^2 - 2\langle e_n, e_m\rangle + |e_m|^2 = 2,$$

so $\{e_n\}$ has no norm-convergent subsequence. But $e_n \xrightarrow{w} 0$: for every $x \in H$, by Parseval $\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 = |x|^2 < \infty$, hence $|\langle x, e_n\rangle|^2 \to 0$, i.e. $\langle x, e_n\rangle \to 0 = \langle x, 0\rangle$ as $n \to \infty$.

Recall the canonical embedding $E \subseteq (E^*)^* = E^{**}$: every $x \in E$ defines a functional $T_x \in (E^*)^*$ by $T_x(\ell) = \ell(x)$, $\ell \in E^*$. By the Hahn–Banach theorem,

$$\|T_x\| = \sup_{|\ell|_{E^*} \le 1} |T_x(\ell)| = \sup_{|\ell|_{E^*} \le 1} |\ell(x)| = |x|_E,$$

so $x \mapsto T_x$ is an isometry.

Definition. A sequence $\{x_n\}$ in E is *weakly Cauchy* if $\{\ell(x_n)\}$ is Cauchy for all $\ell \in E^*$.

Note. $x_n \xrightarrow{w} x \implies \{x_n\}$ is weakly Cauchy.

Proposition 2

Let E be a normed vector space. Then:

(a) $x_n \xrightarrow{w} x \implies |x|_E \le \liminf_n |x_n|_E$.

(b) If $\{x_n\}$ is weakly Cauchy, then $\{x_n\}$ is bounded: $\sup_n |x_n|_E < \infty$.

(c) Suppose E is a reflexive Banach space. Then every weakly Cauchy sequence in E has a weak limit in E.

Note on reflexive spaces. $E \subseteq (E^*)^* = E^{**}$. The space is *reflexive* when $E = E^{**}$: for all $T \in (E^*)^*$ there exists $x \in E$ such that $T(\ell) = \ell(x)$ for all $\ell \in E^*$.

Proof. (a) Let $x_n \xrightarrow{w} x$. By a corollary of the Hahn–Banach theorem, there exists $\ell \in E^*$ with $|\ell|_{E^*} = 1$ and $\ell(x) = |x|_E$. Then

$$|x|_E = \ell(x) = \lim_n \ell(x_n), \qquad |\ell(x_n)| \le |\ell|_{E^*} |x_n|_E = |x_n|_E \text{ for any } n.$$

Hence $|x|_E = \lim_n \ell(x_n) = \liminf_n \ell(x_n) \le \liminf_n |x_n|_E$.

(b) Let $\{x_n\}$ be weakly Cauchy. Then $\sup_n |\ell(x_n)| < \infty$ for all $\ell \in E^*$, because $\{\ell(x_n)\}$ is Cauchy in $\mathbb{R}$, hence convergent and bounded. Consider $T_{x_n} \in (E^*)^*$ defined by $T_{x_n}(\ell) =$

$\ell(x_n)$, $\ell \in E^*$. Then $\sup_n |T_{x_n}(\ell)| = \sup_n |\ell(x_n)| < \infty$ for all $\ell \in E^*$. Since E^* is Banach, by the Banach–Steinhaus theorem (uniform boundedness),

$$\sup_n \|T_{x_n}\| = \sup_n \sup_{|\ell|_{E^*} \le 1} |T_{x_n}(\ell)| = \sup_n \sup_{|\ell|_{E^*} \le 1} |\ell(x_n)| = \sup_n |x_n|_E < \infty.$$

(c) Let $\{x_n\}$ be a weakly Cauchy sequence in a reflexive Banach space E. By (b), $M := \sup_n |x_n|_E < \infty$. Set $T(\ell) = \lim_n \ell(x_n)$; the limit exists since $\{x_n\}$ is weakly Cauchy, and $T : E^* \to \mathbb{R}$ is linear. Now $|\ell(x_n)| \to |T(\ell)|$ and

$$|\ell(x_n)| \le |\ell|_{E^*} |x_n|_E \le M|\ell|_{E^*} \implies |T(\ell)| \le M|\ell|_{E^*}, \quad \ell \in E^*,$$

so $T \in (E^*)^*$. By reflexivity $(E^*)^* = E$, so there exists $x \in E$ with $T(\ell) = \ell(x)$ for all $\ell \in E^*$. Then $\ell(x) = \lim_n \ell(x_n)$ for all $\ell \in E^*$, i.e. $x_n \xrightarrow{w} x$.

Note. "$\lim_n x_n = x$" means (a) $\lim_n x_n$ exists and (b) it equals x. For (a), it helps to know there is a convergent subsequence x_{n_k} — equivalently, that $\{x_n : n \ge 1\}$ is relatively compact.

Fact. If E is infinite dimensional, the weak topology on E is not metrizable; in particular, weak convergence cannot be defined by any metric.

Proof. Suppose, for contradiction, that the weak topology is induced by a metric. Then 0 has a countable weak-neighborhood basis $\{V_k\}_{k \ge 1}$. Every weak neighborhood of 0 contains a basic neighborhood

$$W(\ell_1, \dots, \ell_m; \epsilon) = \{x \in E : |\ell_i(x)| < \epsilon,\ i = 1, \dots, m\}, \quad \ell_i \in E^*,\ \epsilon > 0,$$

so each V_k contains such a W_k built from a finite set $F_k \subseteq E^*$. Let $F = \bigcup_k F_k$, a countable subset of E^*.

Given any $\ell \in E^*$, the set $\{x : |\ell(x)| < 1\}$ is a weak neighborhood of 0, hence contains some $V_k \supseteq W_k = W(\phi_1, \dots, \phi_m; \epsilon)$ with $\phi_j \in F_k$. In particular $\bigcap_j \ker \phi_j \subseteq \{x : |\ell(x)| < 1\}$, and by scaling $\bigcap_j \ker \phi_j \subseteq \ker \ell$. By the elementary lemma (if $\bigcap_j \ker \phi_j \subseteq \ker \ell$ then $\ell \in \operatorname{span}\{\phi_1, \dots, \phi_m\}$), we get $\ell \in \operatorname{span}(F_k) \subseteq \operatorname{span}(F)$.

Thus $E^* = \operatorname{span}(F)$ has a countable spanning set, i.e. countable Hamel dimension. But E^* is a Banach space, and by the Baire category theorem an infinite-dimensional Banach space cannot have countable Hamel dimension. Since E is infinite dimensional, so is E^* — a contradiction. Hence the weak topology is not metrizable.